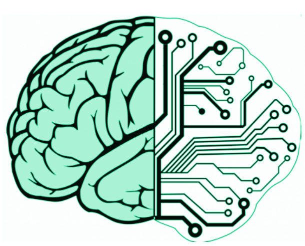
Quotation:

”Though this be madness, yet there is method in’t.”
Shakespeare: Hamlet: Act 2 Scene 2

The Hodgkin-Huxley neuron model V2.0

The true physics and physiology behind neuronal operation



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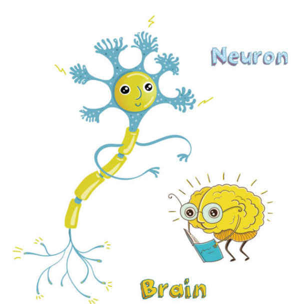

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Foreword

The importance of Hodgkin and Huxley's classic (V1.0) "model", which fundamentally changed our ideas about the physiology of neurons and which has served as the solid fundamentum for describing the functioning of neurons for several decades, can hardly be overestimated; it is truly the crown jewel of neuroscience. However, its authors, quite realistically, assessed that the physical picture behind their observations they mathematically formulated could be partially incomplete and incorrect. Already in their original communication, they drew attention to the fact that their equations did not fully describe their observations. Later they added that there exists observed (thermodynamic) evidence that does not fit into their picture at all. When making the first (and really enormous) step on terra incognita, lacking most of the required technical apparatus, structural and operational knowledge, they were forced to formulate ad hoc assumptions and they also admitted that the formalism of the equations could describe more than one physical process. Most of their assumptions proved to be a brilliant vision, but in quite a few cases it turned out that they did not always put the right physical background behind their observations which were generally correctly described mathematically, and that their model designed for mechanical computers had obvious limitations. Already at the time of the introduction of the model, doubts arose as to whether the purely electrical model could fully describe the observations known up to that time, and later, as the accuracy, directionality and detailedness of the dedicated observations increased, it became necessary to introduce more and more ad hoc assumptions in order to maintain the consistency of the model assuming purely electrical operation, although in quite a few cases the original hypothesis had to be corrected.

In the past decades, numerous (V1.0x) models have been created, modifying and supplementing this purely electrical, disciplinary, original model with more or less success (which ultimately became a model due to the activities of Hodgkin and Huxley (HH)'s followers). There have also been (V1.y) models that try to model the neuron within the framework of an entirely different discipline, mainly interpreting its thermodynamic, mechanical, optical, etc. phenomena. The common feature of all these models is that they cannot provide a complete model (describing all observations without contradiction). There are also models that do not even strive to describe a physical operating mechanism; they are satisfied with providing a mathematical formalism that shows a behavior that is consistent with experience in some context; without even trying to answer the questions 'why' and 'by what mechanism'.

Due to the lack of the needed expertize in physics, biophysics introduced the idea of the "whatnot" [1] protein mechanisms for the unexplainable forces that totally misguided physiology and neuroscience. The speculative hypotheses hyde that science can describe living matter, but needs a different approach. Biophysics (and consequently, biology) has no laws of motion in the sense as physics has Newton's laws. Instead, it is attempted to apply the more-or-less similar laws of inanimate science to the living matter. Claims such as "pumps [by using protein mechanisms] ...transport ions *against their electrical and chemical gradients*" [2], page 101, or "the flux of ions through ion channels is passive, requiring *no expenditure of metabolic energy* by the channels." [2], page 107, are as scientific as in the pre-Newtonian age, there was a notion related to Aristotelian philosophy about the celectial mechanics. That concept gave credence to the "theory" that the celestial spheres and, later on, the planets were pushed or moved by celestial movers, celestial intelligences, or angels against physical forces, without investing energy. Inventing the universal gravitational law made the complex interplay of those spheres obsolete, although, for the first look, it seemed to be unrelated.

Quotation:

"*We stand on the verge of a great journey into the unknown—the interior terrain of thinking, feeling, perceiving, learning, deciding, and acting to achieve our goals—that is the special province of the human brain* " — from the preamble to "BRAIN 2025: A Scientific Vision" [3] @2015

Describing life science, from many points of view, is still in the state similar to the pre-Newtonian stae of physics: it has a lot of "spheres" described by complicated theories that contradict each other, and does not have its laws of motion, in the sense as Newton's Laws in physics. What is worse, in some fields it did not invent and test the laws describing living matter. Instead, it borrowed the disciplinary laws of physics, without recognizing that its different discipline has different conditions and requires different approximations; that is, similar, but differently used laws. Without questioning the merits of the V1.0 model, the experimental experience accumulated in recent decades, the dizzying development of experimental technology and instrumental measurements, and the ever-increasing demand for the results of neuroscience have made it necessary to develop the (V2.0) model. The model fuzes and unifies the concepts of the former V1.x models, purifies the wrong conceps and synthesizes the rigth ones. The change in the major version number symbolizes that it is a leap-like development (a drastic change in the physics concepts) instead of an incremental one (a slight improvement by introducing a new ad-hoc hypothesis).

The V2.0 model is necessarily cross-disciplinary; no single physical discipline alone allows for a complete and consistent interpretation of the phenomena of the living organism. The reader who grew up on the 'old school' doctrines will find our approach strange: the professors' professors also grew up on the old school doctrines. In physics it is not strange at all that the concepts that are valid for the observations made in everyday life became invalid in the world of too small or too large masses, too small or too large distances, and so on. Up to now, invalid scientific concepts have been applied to the living matter, and because of the wrong concepts, many wrong conclusions have been made. The statical view of biology worked fine for example, for creating the descriptive biology and anatomy, but handling the dynamicity of life as a perturbation of the static anatomy leads to flawed results. To fix those mistakes, one must strictly define the concepts, borrowed from science, for the case of living matter, because for today the wrong concepts have grown into major obstacles toward understanding the extremely complex operation of the brain, from its elementary physical processes up to its information processing and higher-level functionality.

For the same reasons, we follow von Neumann's method when he described principles of technical computing [4]. "The ideal procedure would be, to take up the specific parts in some definite order, to treat each one of them exhaustively, and go on to the next one only after the predecessor is completely disposed of. However, this seems hardly feasible. The desirable features of the various parts, and the decisions based on them, emerge only after a somewhat zigzagging discussion. It is, therefore, necessary to take up one part first, pass after an incomplete discussion to a second part, return after an equally incomplete discussion of the latter with the combined results to the first part, extend the discussion of the first part without yet concluding it, then possibly go on to a third part, etc. Furthermore, these discussions of specific parts will be mixed with discussions of general principles, of the elements to be used, etc." For example, basic concepts such as the membrane potential and its regulation in resting and transiens states, belong to chapter Physics, and their detailed scientific, physics-based, discussion is given there, but their corresponding aspects must be discussed in chapters 'Abstract neuron' and less abstract 'Physiology' as well. The contents are inherently intertwinned, and the form must follow the contents. We make this zigzag reading more accessible by using hyperlinks and cross-references throughout the document.

Chapter 1

Introduction

Contents

Quotation:

"The Human Brain Project should lay the technical foundation for a **new model** of ICT-based brain research, driving *integration between data and knowledge from different disciplines*, and catalysing a community effort to achieve a **new understanding of the brain**... and *new brain-like computing technologies*."
the Human Brain Project, summarised its goal @2012

The *dynamic* operation of individual neurons, their connections, higher-level organizations, connections, the brain with its information processing capability, and finally, the mind with its conscience and behavior, are still among the big mysteries of science: *at which point the non-living matter becomes a living one*, *at which point the living matter becomes intelligent* and conscious; whether and how science can handle all this stuff. We really need a new understanding that we provide here. The hype in the newly launched projects is excessive and seems to lose the hope to build brain research on a firm science base. For example, the newly (at the end of 2025) launched "Brain/MINDS 2.0" program in Japan was launched using only mathematical models, *without targeting the understanding of the underlying physical processes.* In neuron models, 'dynamics' refers to how the state variables of the neuron (primarily membrane potential and ion channel properties) change and evolve over time in response to internal processes and external inputs. However, unlike in our approach, the usual processes simply use a time parameter in empirical or mathematical functions, see for an example [5]. We use a real dynamics, based on physics, in the sense as Newton introduced his laws of motion to describe response of neurons.

1.1 Disciplinary layers

The material has multiple layers. In addition to the sections intended to be understandable by everyone (i.e. primarily readers with a suitable physiological and neuroscience background), there are also panels (typically in another chapters) for those with the relevant background, containing the necessary **mathematical 1.1.1** and **physical 1.1.1** knowledge, which require at least the appropriate level of BSc knowledge to understand, but sometimes knowledge that goes beyond this, and sometimes knowledge and explanations that make you delve deeper into the unusualness of the ideas. This method naturally leads to some redundancy, but enables the audience to read only the text his/her background qualifies for. We strive to make the material understandable without those supplementary material, and with them, the reader with the appropriate qualifications will be able to gain knowledge that encourages further thoughts.

Physics panel 1.1.1:
An example physics panel

During the specialization of sciences, physics specialized in the description of inanimate nature and (in the interest of reasonable description) created fields of limited validity. In the study of living nature, its results, methods and tools were already used in a natural way. In doing so, they started from the correct point of view that the description of objects in nature and their interactions should be governed by the same laws. However, they did not take into account that due to the complexity of nature, approximations and neglects must be made in order to be able to describe them. Disciplinary laws create abstract concepts for a certain range of phenomena, and the laws describe these abstract concepts precisely, not the phenomena that occur in nature. In most cases these laws can be used in their entirety, due to the significant differences in the conditions of the living and non-living environment often they can only be used with significant modification or even complete rethinking; see the interpretation of Erwin Schrödinger's insightful questions in **Physics Panel B.2.1** Physics panels contain the strict physical background.

Mathematics panel 1.1.1:
An example mathematics panel

Mathematics panels typically contain equations and their explanations. The statements in the main text are understandable without these panels, they solely underpin those statements in the language of mathematics; for the audience having the required mathematical background.

Numerics panel 1.1.1:
An example numerics panel

The complexity of the phenomena requires many approximations, but by using the appropriate approximations and abstractions, numerical results (sometimes only reasonable estimates) are provided for the phanomenon in question in the corresponding numeric panels.

Quotation:

"Biological laws are much more difficult to find than physical ones" [6] @1995

1.2 Abstract discussion

Nature uses an infinite variety of implementing neurons. Our goal is not to discuss all infinitely complex molecular, biochemical and physiological details of neuronal operation In the Central Nervous System (CNS) they can cooperate with each other, 'despite the extraordinary diversity and complexity of neuronal morphology and synaptic connectivity, *the nervous systems adopt* ***a number of basic principles*** for all neurons and synapses', [7] independently from the infinitely complex molecular, biochemical and physiological details. We *will base our holistic discussion on those general basic principles and create an 'abstract physical neuron'*; a hypothetical element, skipping the 'implementation details' nature uses. We essentially follow von Neumann's method in describing the foundations of computing science [4]: "we will base our considerations on a hypothetical element, which functions essentially like a neuron". We need different abstractions and approximations (furthermore, 'non-ordinary' laws of science) for describing biological processes. However, an abstraction is usable in practice only when paired with a generalization: the more abstract the assumption, the more general and widely applicable the concept or conclusion.

By abstractions, we can reduce the unbelievably detailed world into manageable pieces, and we can learn anything general. We must show which approximations are oversimplifications and which phenomena are misunderstood; measured, or interpreted in a wrong approach. Abstraction is, in fact, everywhere, including inside each of us. It is a core element of science and cognition.

Chapter 2

The V2.0 model in a nutshell

Contents

Numerics panel 2.0.1:
Constants from Johnston&Wu [7], page 12

Neuron's surface 8×10^{-9} $[m^2]$
Neuron's volume 6.5×10^{-14} $[m^3]$
Number of uncompensated ions 4.7×10^7
Fractional uncompensated ions $2.35 \times 10^{-5} = 0.00235\%$

2.1 Modeling principles

Quotation:

”Everything should be made as simple as possible, but not simpler.” Attributed to Albert Einstein.

The principle, suggested by A. Einstein, means that while one should strip away unnecessary clutter, extra features, or overly complex jargon to achieve clarity, one must be careful not to remove so much that the core message or solution becomes meaningless or inaccurate. For neurons (living matter), on the one side, it actually means that one should not reject quantities from the set of observed quantities (the mass of the charge carrier and the pressure they cause) just because they are not present in a classical discipline (electricity) of the science, which once upon the time someone hypothesized, or decided in advance, or willingly described the first case that some aspect of the functioning of a neuron reminded him of, which they considered suitable for describing neurons.

However, one must include the proven new discoveries (effects and structures) made in the past eight decades (Axon Initial Segment (AIS)) and the intentionally omitted observations (heat emission and absorption; mechanical changes) which do not fit into the pre-authorized disciplinary theory. On the other side, one must remove the foreign and fake ad hoc mechanisms and hypotheses (which contradict fundamental principles of science and elementary logics). All those issues were required to obscure that the current understanding does not match the experimental facts. The discussion does not include ’whatnot’ interactions (such as protein mechanisms that operate contrary to the laws of physics) or ion-specific ion channels operating under synchronized control (for which there is no structural evidence). These are discussed as hypotheses in physiology and biophysics, but cannot be scientifically substantiated. A detailed and precise description of the correct functioning is usually beyond the typical knowledge level of physiologists, so we give only a popular discussion of them here; the physically correct discussion is placed in the appendices. Similarly, additional knowledge about modeling is discussed in the A appendix.

We must also have Schrödinger’s comments in mind when constructing the physical picture about neurons, that we must be extremely careful. *We must not use the ’ordinary’ laws of physics derived for non-living matter without revisiting them* because ”*the construction is different from anything we have yet tested in the physical laboratory*” [1]. The discussion requires deep knowledge in science but has severe consequences for physiology, in this chapter the discussion remains on the surface and provides a holistic picture about the model. The details and deeper layers are explained in the next chapters and in the appendices. We hope the essence can be understood with knowing only the elements of science and physiology. However, the open mind is required already in this section. The fundamental principles are the same, but we must use different approximations and abstractions for the different disciplines of science and for living matter. For this reason, maybe it is more correct to call those laws for living matter 'non-disciplinary' instead of 'non-ordinary' laws.

We must also include the discoveries from the past seven decades, among others, that a component AIS can be found between the neuron's membrane and the axon. It is observed that AIS has a crucial role in creating the Action Potential (AP), but that role has not been integrated into the existing theories. As a consequence, in the transient state, a neuron shall be modeled as a serial (i.e., differentiator-type) electrical oscillator, which has gated current inputs (instead of a parallel oscillator without gating its inputs, which is a good model only in the static view of the resting state) and provides a timed output. We explicitly add that the charge carriers, the neuron works with, are *ions* (instead of electrons) and that the signal propagation mechanism is *bioelectric* (instead of purely an electrical or mechanical wave or "protein mechanism"), in many cases even without having free carriers in the respective volume. (We do not consider the ideas that the associated mechanical, optical, etc. changes cause electric phenomena. They reverse the true causality: they do not have a triggering cause; the view inherited from the narrative biology.) Electrochemical operation, especially that of semipermeable membranes, requires special attention: charge carriers are "created" inside the biological matter. Omitting purposefully the clear physical background results in performing wrong measurements and introducing wrong concepts into biology.

2.1.1 The non-disciplinary behavior

Since *the same physical action generates the voltage gradient and the concentration gradient* (and, consequently, the pressure), they are inseparable and proportional to each other. Generating a voltage change (clamping, direct current, magnetic pulse) or a pressure change (ultrasound, mechanical, shock waves) on the membrane mutually cause the other to change. Inputting ions (synaptic input) generates both changes. The control circuit tries to restore the resting state, during which the mentioned changes decrease proportionally. The transient state of the neuron can, in principle, be described by any of the mentioned quantities, but, as a practical matter, *the effects of changes in the other entities must also be taken into account*: *no single discipline, alone, can describe the changes*. The elastic membrane implements an almost critically damped vibration, but the "vibrating mass" and the "spring force" are constantly changing. In an electric resonant circuit, the capacitor's voltage and capacitance change due to the finite amount of charge. *The pressure wave caused by a force shock and the voltage wave caused by a voltage shock describe the same effect.* In both cases, the other effect must be taken into account to some extent: an infinitesimal movement of the ion also changes both the pressure and the voltage. Changes in pressure and charge alter the membrane's thickness (and therefore its capacity), and both voltage and pressure can cause a structural change (phase change, "melting") in the lipid chains that make up the membrane.

Here, the difference comes to light: the carrier of the electromagnetic (EM) interaction has no mass, while the thermodynamic one does. In the case of the pressure wave, the speed is naturally finite (mass must be moved). For the electric wave, there are laws regarding only instantaneous interaction; so, they must be adapted to *slow ion current*s based on physical assumptions. It is simpler to describe the generation of the action potential as an electrical oscillation, and its propagation as a pressure wave (although this only represents the wave front well: the AP lasts as long as there is a non-equilibrium charge in the neuron: the repulsion between the excess ions pushes the ions out of the closed space, so the intensity of the wave continuously decreases). Hyperpolarization is described in the electrical view as a capacitive current in the sequential oscillator circuit, in the thermodynamic view as a suction effect occurring in the opposite phase of the membrane. In both cases, however, it must be taken into account that charge and mass are inseparable in ions, and that their cross-disciplinary effects significantly reduce the applicability of the disciplinary laws. Furthermore, as our force analysis suggests, the ions' motion results an interference between the longitudinal thermodynamic oscillation of elastic origin and a discharge process of electrical origin.

2.1.2 History of modeling

When ions interact, we always have to work with two types of fields: the deterministic 'instantaneous' electric field and the primary forces from a stochastic slow force field, which can vary by orders of magnitude over nanometer distances, and also depend on the chemical nature of the ions, and biological structures can also induce counterforces of varying direction and magnitude. To take all of this into account, very detailed numerical simulations or very good approximations are required. Our approach divides biological events into spatial and temporal sections in which –to a good approximation– a dominant and a corrective interaction occurs.

Today, two major branches of disciplinary theories compete for being applied to describe the neuronal operation [8, 9, 10, 11, 12, 13]. Those cited references also compare those theories. The 2-decade-old Heimburg-Jackson thermodynamic theory is trying to break in alongside or displace the 7-decade-old all-electric Hodgkin-Huxley theory. The two theories apply one of the scientific disciplines developed for inanimate nature, unchanged, to living nature. There are unexplained observations for each of the two theories, and neither can describe the biological neuron's functioning without contradictions. In a disciplinary view, there is no chance of the two theories fusing. Although they can describe a wider range of phenomena together, the conditions under which they are applicable exclude their joint use. Moreover, to conceal contradictions and fill gaps, biophysics assumes (typically unspecified protein) mechanisms that contradict the fundamental principles of physics, and some of them even the disciplinary laws.

The electrical view essentially stems from the (Nobel Prize-winning) work of Hodgkin and Huxley [14]. They could not make a perfect job, mainly due to the lack of discoveries made several decades later, including ours, about handling the finite speed of the biological ionic currents (and, due to that, the dynamic features), which drastically change their conclusions. In contrast with their *empirical description*, which delivered mathematically formulated measured observations, our discussion, although it follows essentially the same principles, reinterprets and pinpoints the used fundamental terms, sets up a physical model, and *explains* the physically underpinned processes. “However, the theory could not explain the physical phenomena such as reversible heat changes, density changes, and geometrical changes observed in the experiments” [11]. The thermodynamic view roots in the (possibly Nobel-prize-winning) idea of Heimburg and Jackson [15]. They proposed that the action potential is essentially a sound wave (a soliton). ”However, there are several other questions that this has to answer like ion flow involvement in nerve signal propagation as stated by the HH model and also the faster propagation in myelinated nerves than in unmyelinated” [11]. If we consider that the ion flow means charge propagation in a membrane tube where the specific capacity depends on the thickness (of the myelin sheath) of the wall [16], the faster propagation is not mysterious anymore. Those apparent contradictions are simply consequences of ions' non-disciplinary features. By using the correct (non-disciplinary) discussion of the phenomena, we can explain all those observations in more detail in chapter 7. Given that our model natively connects charge and mass of ions [17], it has a good chance of explaining the missing phenomena.

The excellent textbook's statement that ”the resting membrane potential results from the separation of charge across the cell membrane” [2] is only half the truth; we tell the second half in section **??**. We refute their statement that ”resting ion channels establish and maintain the resting potential” [2], page 126. The ion channels are passive players. The interplay of the lipid condenser and the electrolytes on both sides of the permeable membrane establishes the resting potential, and ionic gradients maintain it. The textbook separates the neuron membrane's states into resting and transient states. It introduces resting and transient ion channels, but does not introduce that the different physical processes in those states need different models. As we show in chapter 7, *two different physical mechanisms operate in the resting and the transient states*, so two different models are needed for describing the same components.

A primary reason neurophysiology has been misguided is that it has in mind a static picture of the dynamic life of neurons. It started from the static inanimate anatomic microscopic pictures, continued with the static clamping investigations (where the negative feedback of an external electric circuit eliminates the natural gradients) and underpinning the theory by analyzing frozen (inanimate) samples by electron microscopes. Those investigations revealed a tremendous amount of crucial details, but they obscured the idea that those samples are just snapshots of a permanently changing system. Neuroscience projected the mechanisms of the static resting state to the dynamic transient state and attempted to understand the transient state as perturbations [18] to the (entirely different) resting state. Of course, that attempt required more and more ad hoc non-scientific assumptions. We agree that "Understanding and predicting molecular responses in single cells upon chemical, genetic or mechanical perturbations is a core question in biology." [19] However, it is not possible without understanding that the transient state is a state on its own right, with different physical mechanisms, rather than a perturbation of the entirely different resting state.

Due to the lack of the needed expertize in physics, biophysics introduced the idea of the "whatnot" [1] protein mechanisms. Claims such as "pumps [by using protein mechanisms] . . . transport ions *against their electrical and chemical gradients*" [2], page 101, or "the flux of ions through ion channels is passive, requiring *no expenditure of metabolic energy* by the channels." [2], page 107, are as scientific as in the pre-Newtonian age, there was a notion related to Aristotelian philosophy. This concept gave credence to the "theory" that the celestial spheres and, later on, the planets were pushed or moved by celestial movers, celestial intelligences, or angels against physical forces. Life science does not have its laws of motion, in the sense as Newton's Laws in physics, at least at cellular level.

The book [2] asks the central questions, "How do ionic [i.e., electrical and chemical] gradients contribute to the resting membrane potential? What prevents the ionic gradients from dissipating by diffusion of ions across the membrane through the resting channels?" However, it leaves them essentially open by giving only a qualitative answer, discussing only membrane permeability, without explaining how the resting potential is created. We demonstrate in section **??** that classical methods of electricity enable us to calculate the membrane's potential resulting from charge separation and polarization, why and how its resting value is created, furthermore, how it is regulated. We make some physically plausible assumptions to provide numerical figures.

Our results underpin that "the crucial system in biology isn't a molecule or a molecular class whatsoever, but the interface created by biomolecules in water" [20]. We add that mainly physical processes at various speeds (instead of molecular classes such as proteins) and ionic gradients (instead of protein mechanisms) control the processes. More precisely, inseparable thermodynamic and electrical processes set up and maintain the potential established by the electrical and thermodynamic backbone defined by the lipid/protein structure of the cell and the ionic solutions it contains. It is one of the frequent cases when one effect implements a functionality (the backbone closely defines the frames of the operation), and the other (in this case, combined thermodynamics and electricity) corrects it when it implements a complex operational (dynamical) functionality: "stimulated phase transitions enable the phase-dependent processes to replace each other ... one process to build and the other to correct" [21].

2.2 The abstract neuron

Nature uses an infinite variety of implementing neurons. In the CNS they can cooperate with each other, 'despite the extraordinary diversity and complexity of neuronal morphology and synaptic connectivity, *the nervous systems adopt* ***a number of basic principles*** for all neurons and synapses', [7] independently from the infinitely complex molecular, biochemical and physiological details. Nature focuses only on functionality, the design of an element is irrelevant as long as the element performs its task. Malfunctions are of course tied to the specific implementation. We *will base our holistic discussion on those general basic principles and create an 'abstract physical neuron'; a hypothetical element, which functions essentially like a neuron*, skipping the 'implementation details' nature uses. We essentially follow von Neumann's method in describing the foundations of computing science [4]: "we will base our considerations on a hypothetical element, which functions essentially like a neuron". We need different abstractions and approximations (furthermore, 'non-ordinary' laws of science) for describing biological processes.

However, an abstraction is usable in practice only when paired with a generalization: the more abstract the assumption, the more general and widely applicable the concept or conclusion. *By abstractions, we can reduce the unbelievably detailed world into manageable pieces, and we can learn anything general.* We must show which approximations are oversimplifications and which phenomena are misunderstood; measured, or interpreted in a wrong approach.

In our science-based non-disciplinary model (the 'abstract physical neuron', as we call it), a neuron (highly simplified) is an elastic sphere with two lipid layers (membrane) in an electrolyte, a similarly elastic insulating output tube (axon), and an incompressible, partially ionized electrolyte fluid with different compositions and concentrations inside and outside. In addition to ions, the electrolyte segments contain simple chemical molecules, and complex biological formations. The electolytes inside and outside may have gradients and the ions move at continuously changing speeds that can vary by several orders of magnitude during operation. The lipid layer is partially permeable (controlled or uncontrolled) for the components above (channels, mainly ion channels).

In our description, we consider the neuron as a closed surface (ideally: a sphere), see Fig. 2.1, made of an insulator (lipid bilayer) (the membrane), see Fig. 2.2, to which an identical outlet tube (the axon) is connected. Both are filled with saline and are surrounded by a saline solution of a different composition. In the processes related to its functioning, a small piece of it is considered a flat surface (see 2.2 figure), which has a specific structure, but for us only the protein insert openings (the ion channels) connecting the two saline solutions and the axon terminals of other neurons leading to the given neuron (the synapses) are important. The neuron is therefore considered an independent biological object, which sets and maintains its state with its own regulatory system, and its immediate external environment affects its state through the ion channels. It communicates with other neurons via axons: it sends and receives neural impulses.

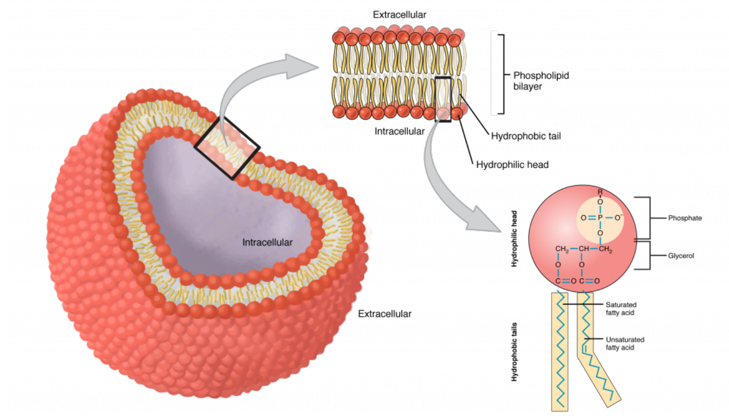

Figure 2.1: A simulated picture of the neuron membrane. We can consider the membrane as a closed surface (in our idealist view, a sphere), being slightly opened at an attached tube, the axon. For certain processes, we can also consider just a tiny section as a planar piece, see also Figure 2.2

In the case of mathematical modeling, we describe the measured data recorded about the phenomena on a "looks like" basis with a mathematical function (HH published *quantified measurement results* in their own opinion), and possibly some ad-hoc assumed physical mechanism is put behind it. In the case of physical modeling, we reduce the unknown operation in question to something that we already know, whose behavior we see as similar to that of the object under investigation in some respect, and whose mathematical description we can handle. Of course, similarity has limits, especially if we try to model a living object with an inanimate object described by the disciplinary methods of science. Based on the model, we can not only find limitations, but also how we can make our model more accurate; even if this requires combining the disciplines of inanimate natural science.

In the case of a living neuron, we can find many similarities with the functioning of an inanimate object: "whose thoughts come to mind" obviously means that the previous or even the most recent knowledge and available tools of the investigating scientist can decisively determine the scope of the modeling phenomenon. HH's work is hardly independent of the radar development work they carried out during the war years. Similarly, when developing the model they proposed, they took into account that the computational capacity at their disposal was a mechanical ("winding") computer. Unfortunately, their descriptions, which contained strongly simplifying assumptions, were encoded in the form of coefficients; thus, our modern calculations on advanced computers are based on them.

The theory describing its operation must answer questions such as

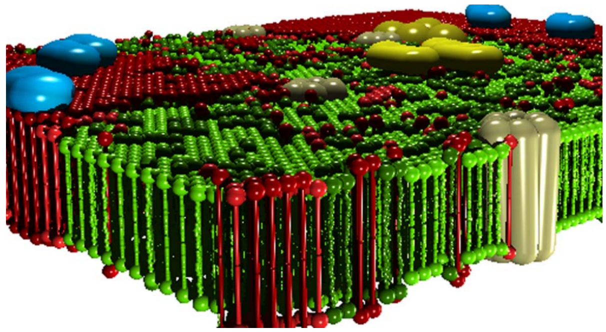

Figure 2.2: A closer look at the simulated picture of the neuron membrane shown in Fig. 2.1. Even the abstract model cannot stick to the flat surface and the uniform membrane thickness. In addition, lipids spontaneously form double layers; moreover, smaller biological objects can build up on the membrane. Lipids are amphiphilic molecules that like water on one side and repel water on the other side due to the hydrocarbon chains. In contact to water lipids spontaneously form double layers that are the basic building blocks of biomembranes.

- why a resting potential exists
- why the experienced ionic currents exist even in the so-called stable resting state
- why this state changes to an unstable transient state
- why the neuron can return to the same stable resting state
- why it can turn to a transient state again before reaching the resting state
- why there is a dead time before it can return to a transient state
- why the parameters of the transient state are constant
- why exactly such parameters describe the wave emitted in the transient state
- how the emitted wave can propagate practically without attenuation

2.2.1 Neuron's states

Under a small external influence, a neuron remains in its resting state and regulates itself through non-controlled ion channels in its membrane. If the external influence exceeds a threshold, the controlled channels in the wall suddenly release a large amount of ions into the internal space, putting the neuron into a transient state. The membrane potential changes suddenly (within a few dozen nanoseconds), and, due to the mutual repulsion of ions, the pressure acting on the membrane wall and the ion concentration in the layer near the membrane also increase significantly and suddenly. At this point, the disciplines of electricity and thermodynamics can be combined: the pressure and the electric field are proportional to the number of ions in the volume. The forces exerted on ions, calculated in a disciplinary way, are falsified in the sense that the other discipline also contributes inseparably; in a strict sense, neither of the two disciplines alone can describe ions' motion. The detailed analysis shows that the elastic energy during the action potential is about two orders of magnitude higher than the electrical energy, but the amount of ions is about three oders of magnitude lower than that of the non-dissociated molecules. It that situation, the two effects are of the same order of magnitude, so that they can modulate each other. The electric charge on the membrane's wall would produce a simple discharge current. However, its charge carriers are delivered by the damped oscillation of the electrolyte (a soliton) generated by the membrane. The amplitude of the soliton changes the number of the charge carriers delivered, so physiology measures an oscillating discharge current (that generates a potential wave known as AP). No significant mass transfer takes place.

Fig. 2.3 illustrates our abstract view of a neuron, in this case as a "stage machine". Notice that the double circles are *stages* (*states* with event-defined periods) connected by bent arrows representing *instant stage transitions*, while at some other abstraction level we consider them as *processes* having a temporal course with their own *event* handling. Fundamentally, the system is circulating along the blue pathway, and maintains its state by using the black loops, but sometimes it takes the less common red pathways. It reveives its inputs cooperatively (controls the accepted amount of its external inputs from the upstream neurons by gating them by regulating a stage variable), furthermore it actively communicates *the time of its state change* (that is: *not its state* as assumed in the so called neural information theory) toward the downstream neurons in a process parallel with its mentioned activity.

2.2.2 Stages of operation

Initially, a neuron is in stage "Relaxing" which is the ground state of its operation. (We also introduce a "Sleeping" or "Standby" helper stage, which can be imagined as a low-power mode in electronical or state maintenance mode of biological computing; or "creating the neuron" in biology; a "no payload activity" stage.) The stage transition from "Sleeping" also *resets the internal stage variable* membrane potential (to the value of the resting potential). In biology, a "leakage" background activity takes place: it changes (among others) the stage variable towards the system's "no activity" value.

An event (in form of a pulse of slow ions) arriving from the environment acts as a "look at me" signal and the system passes to "Computing" stage: an excitation begins. The external signal acts as triggering a stage change and, simultaneously, contributes to the value of the internal stage variable (membrane voltage). During normal operation, when the stage variable reaches the critial value (the threshold potential), the system generates an event: passes to stage "Delivering" and "flushes" the collected charge. In that stage, it starts to deliver a signal toward the environment (to the other neurons connected to its axon) and after a fixed period, passes to stage "Relaxing", without resetting the stage variable. From this event on, it is again in stage "Relaxing", where the "leakage" and the input pulses from the upstream neurons contribute to its stage variable. Process "Delivering" operates an independent subsystem ("Firing"): happens

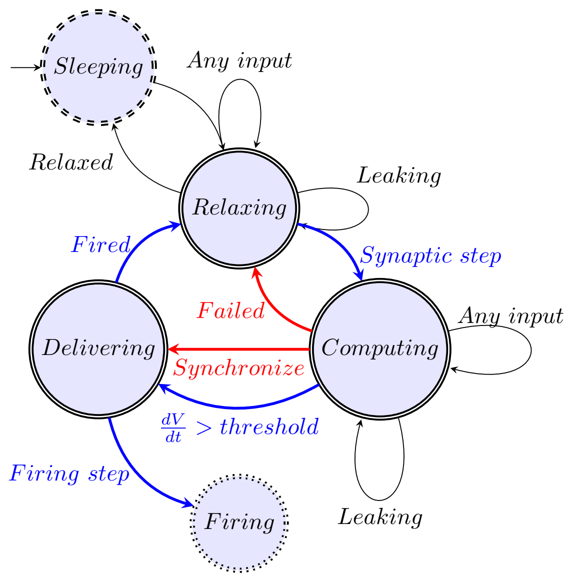

Figure 2.3: The model of neuron as an abstract state machine

simultaneously with process of "Relaxing" which, after some time, usually passes to the next "Computing".

The stages "Computing" and "Delivering" mutually block each other and the I/O operations happen in parallel with them. They have temporal lengths, and they must follow in the well-defined order (a "proper sequencing" [4, 22]) "Computing"⇒ "Delivering"⇒"Relaxing". Theoretically, *a three-state system is needed to define the direction of time* [23]; a fundamental issue for quantum-based computing is the lack of a third state. In electronical computing we can introduce this as "up" edge and "down" edge, with a "hold" stage between. A charge-up process must always happen before discharging. Stage "Delivering" has a fixed period, stage "Computing" has a variable period (depends mainly on the upstream neurons), and the total length of the computing period equals their sum. The (physiologically defined) length of the "Delivering" period limits neuron's firing rate; the length of "Computing" is usually much shorter.

In any stage, a "leak current" (through ion channels, but mainly the AIS instead of the membrane) changing the stage variable is present; ***the continuous change (the presence of a voltage gradient) has a fundamental importance for a biological computing system*** This current is proportional to the stage variable (membrane current); it is ***not*** identical with the fixed-size "leak current" assumed in the Hodgkin-Huxley model [14]. The event which is issued when stage "Computing" ends and "Delivering" begins, separates two physically different operating modes: inputting payload signals for computing and inputting "servo" ions for transmitting (signal transmission to fellow neurons begins and happens in parallel with further computation(s)). Given that stages "Computing" and "Delivering" take time, and the next "Computing" can only start through the stage "Relaxing", furthermore "Delivering" can only begin after "Computing" finished, *these stages mutually block each other* (and the length of the "Delivering" period limits neuron's firing rate).

There are two more possible stage transitions from the stage "Computing". First, the stage variable (due to "leaking") may approach its initial value (the resting potential) without firing and the system passes to stage "Relaxing"; in this case we consider that the excitation "Failed". This happens when leaking is more intense than the input current pulses (the input firing rate is too low or a random firing event started the computing). Second, an external pulse may have the effect to force the system (independently from the value of the stage variable) to pass instantly to stage "Delivering", and after that, to "Relaxing". (When several neurons share that input signal, they will go to "Relaxing" at the same time: they get synchronized; a simple way of synchronizing low-precision asynchronous oscillators.)

Anyhow: a neuron operates in cooperation with in its environment (the fellow neurons, with outputs distributed in space and time). It receives multiple inputs at different times (at different offset times from the different upstream neurons) and in different stages. In stage "Computing", the synaptic inputs are open, while in stage "Delivering", the synaptic inputs are closed (the input is ineffective). It produces multiple outputs (in the sense that the signal may branch along its path), in form of a signal with temporal course.

Stage 'Computing'

The neuron receives its inputs as 'Axonal inputs'. For the first input in stage 'Relaxing', the neuron enters stage 'Computing'. The time of this event is the base time used for calculating the neuron's "local time" (in describing the operation and in the simulator). Notice that producing the result is a cooperation between the neuron and its upstream neurons (the neuron gates its input currents). One of the upstream neurons opens computing, and the receiving neuron terminates it.

The physical implementation is a steep current gradient which evokes a steep voltage gradient dV/dt on the condenser 'membrane' in its intracellular segment, see section **??**. The membrane is connected to the axon through a resistor 'AIS' (these components, switched in serial, constitute the neuronal oscillator). Given that the current creates a potential gradient on the membrane, the increased potential starts a current ("Leaking") proportional to the voltage difference between the membrane and the axon (across the AIS). This current decreases the membrane's potential (discharges the condenser). In lack of further excitation, the membrane's potential decreases back to its resting value and the neuron returns to stage 'Relaxing'.

However, for repeated excitation, when the next 'Axonal input' arrive(s) before the neuron returns to stage 'Relaxing', the voltage increases further, and it might reach a threshold voltage. In such a case, the neuron enters stage 'Delivering' (the red section of the broken diagram line). The time at which point this happens depends on the arrival of spikes and the discharge of the membrane (when the difference of the received charge and the loss due to discharge evokes a sufficiently large voltage on the condenser's capacity). At that point, the computation is finished. *The result of the computation is the time* passed between receiving the first 'Axonal input' and the time when the neuron closes its input sources (and simultaneously, opens the ion channels in its wall, see below, to enter stage "Delivering"). No more input shall be received, so the neuron disables its synaptic inputs, and prepares for delivering the result of its 'computation' (nothing shall be stored: the result is the time of sending the message, but the delivery period takes time; is a fixed-length process, will follow immediately). In this stage, the role of the slow speed is not evident. The current arrives through an axon, passes the terminal and arrives at the AIS; the time components cannot be separated.

Stage 'Delivering'

In this stage, the result is ready: the time between the arrival of the first synaptic input and reaching the membrane's threshold voltage is measured. No more input is desirable, so the neuron's input synapses remain closed. Simultanously, the neuron starts it "servo" mechanism: it opens its ion channels and an intense ion current starts to charge the membrane. It is an 'instant' current. The voltage on the membrane quickly rises, but it takes a short time until its peak voltage reached. Given that the charge-up current is instant and the increased membrane voltage drives an outward current, the membrane voltage gradually decays. When the voltage drops below the threshold voltage, the neuron reopens its synaptic inputs and passes to stage "Relaxing": it is ready for the next operation. The signal transmission to downstream neurons happens in parallel with the recent "Delivering" stage and the next "Relaxing" and maybe "Computing" stages.

Delivering the result needs huge power because of the noisy environment and the huge distances, so at the beginning of the 'Delivery' phase, the neuron switches in a 'servo' mechanism. Exceeding the threshold voltage (either as a consequence of the 'spatiotemporal' summing several spikes or a single spike with sufficiently large voltage gradient [24]) opens the voltage-gated ion channels in the membrane's wall. Due to the vast voltage gradient across the two segments of the membrane, an enormous amount of ions rush-in into the intracellular space and the intense current increases the membrane's potential; for the details see section **??**. The amount of the rushed-in ions is several times more than those received during the stage 'Computing'.

The ion channels over the surface of the neuron open at the same time, and they are open only for a very short period (actually, they implement a sudden "step current"). However, this current on the surface is created at different distances from the AIS and (due to the final speed of the ions) and the time until the ions arrive at the AIS depends on the distance to AIS. Due to this effect, the current quickly increases; until it reaches a maximum. The time when the AP reaches maximum value (due to the event that the current from the most distant places of the membrane could reach AIS), and from this point it starts to decrease. (there is no other special threshold voltage value: the charge from the rush-in current distributes on the surface; the current injection can charge the fixed capacity membrane to that voltage). At this point the neuronal condenser is maximally depolarized.

The neuronal condenser is loaded to its maximum, and the resistor (the AIS) enables a current to flow out, so the condenser (the membrane) discharges (repolarizes). Here enters into the picture that the neuron is resemblant to a serial oscillator. The condenser stores the charge for some period, and releases it to the AIS at a later time. As a result, the direction of the current on the neuron's membrane reverses, see the shape of the output voltage in table **??**. The essential difference between the serial and parallel *RC* oscillators is that the differentiator circuit can produce opposite voltage on its output without assuming any additional mechanism, such as an outward current, given that *a current pulse has rising and falling edges which can natively produce positive and negative voltage derivatives.* As the result of the process, the potential reaching the AIS goes to negative; a phenomenon called hyperpolarization.

Notice that the current in the red section (plus also in the blue section) of the diagram line still originates from the rushed-in charge. The observer sees the sum of the resistive and capacitive currents; maybe new axonal inputs superimposed. The ion current's finite speed plays again. When the axonal arbors re-open, the ion flows into the membrane at some distant point, so its effect will be observed some time later; apparently when the neuron membrane is about its hyperpolarized state.

Stage 'Relaxing'

In this stage, the neuron re-opens its synaptic gates. Recall that the ion channels used for generating an intense membrane current are already closed. The neuron passes to stage 'Relaxing' and is ready for a new computation: the previous result is under delivering (parallelly, independently), the axonal inputs are open again. However, the membrane's potential at this point may differ from the resting potential. A new computation begins (the neuron passes to the stage "Computing") when a new axonal input arrives. Given that the computation is analog, a current flows through the AIS, and the result is the length of the period to reach the threshold value, the membrane voltage plays the role of an accumulator (with a time-dependent content): a non-zero initial value acts as a short-time memory in the next computing cycle. The local time is reset when a new computing cycle begins, but not when eventually the resting potential reached.

As the membrane's voltage drops below the threshold voltage, the neuron re-opens its synaptic gates (the ion channels in the membrane's wall are already closed). The neuron passes to stage 'Relaxing' and is ready for a new computation: the previous result is under delivering (parallelly, independently), the axonal inputs are open again. However, the membrane's potential at this point may differ from the resting potential. Given that the computation is analog, the membrane voltage plays the role of an accumulator (with a time-dependent content, given that the current flows though the AIS). A new computation begins (the neuron passes to the stage "Computing") when a new axonal input arrives. Given that the result is the time to reach the threshold value, the non-zero initial value acts as a short-time memory.

The ion current's finite speed plays again. When the axonal arbors re-open, the ion flows into the membrane at some distant point, so its effect will be observed some time later; apparently when the neuron membrane is about its hyperpolarized state.

Classic stages

why defining the length of the Action Potential is problematic. The effect of slow current affects the apparent boundary between our "Delivering" stage and "Relaxing" stages. Classical physiology sees the difference and distinguishes 'absolute' and 'relative' refractory periods with a smeared boundary between. Furthermore, it defines the length of the spike with some characteristic point, such as reaching the resting value for the first or second time, or reaching the maximum polarization/hyperpolarization. Our derivation of the stages (see Fig. 2.4) defines clear-cut breakpoints between them.

We can define the length of the spike as the sum of the variable-size length of period "Computing" and fixed-size period "Delivering". The "absolute refractory" period is defined as the period while the neuron membrane's voltage keeps the gates of the synaptic inputs closed (the value of membrane voltage is above their threshold). That period is apparently extended (and interpreted as a "relative refractory" period, which sensitively depends on the conduction velocity [25]) by the period when although the gating is re-enabled, but the slow current did not yet arrive to the AIS where it contributes to the measured AP, see Fig. 2.4. *Only one refractory period exists, plus the effect of the slow current.*

2.2.3 Resting Potential

2.2.4 Action Potential

2.2.5 Neuron's control

Biology uses a simple Proportional-Integral-Derivative (PID) controller with proportional, integral, and derivative components. In simple words, it means that the output of the system (the output voltage measurable on the AIS) depends on the input voltage (the present: input current if the resistance is constant), the integrated input current(s) (the past), and their derivatives (the future). Notice that calculating the derivatives of the interdependent parameters needs care, as discussed in [27]. The sum of those summands defines the resulting output voltage. The biological case is more complicated than the technical one because biology works with slow ions, and the components have their temporal dependence (applying the laws created for fast currents requires emulation, as

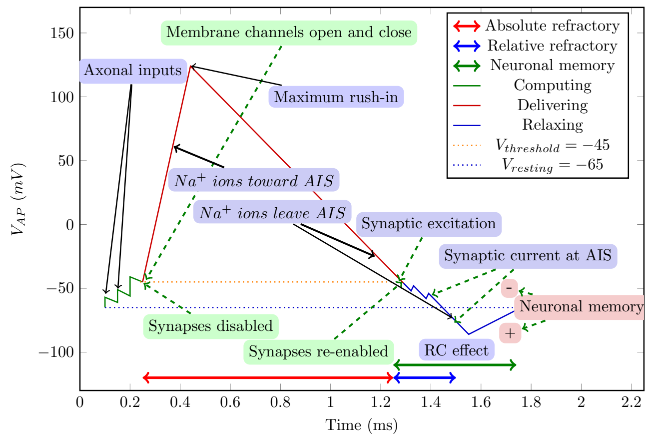

Figure 2.4: The conceptual graph of the action potential

discussed in section **??**). Furthermore, the summands have several constituents, and the neurons handle those current-related constituents autonomously. That is, unlike what is assumed in the classic physiology, the output voltage is not necessarily a simple function of the mentioned input variables. Furthermore, it is not sufficient to consider the "present" of the system. The classic models all omit the derivatives' contributions (a direct consequence of using 'clamping': in that "frozen" state, the "future" cannot be interpreted) except that of the capacitive current, which is the passive consequence of the input currents. Furthermore, the 'Integrate and Fire'-type models consider the integrated currents only in the 'charge-up' ('Computing') period of operation, and they entirely omit that different physical processes are going on in different stages of operation. That is, *even the best classic models can not describe neuronal operation completely and correctly*.

The gradients are used to adjust the process variable through their positive and negative contributions (corresponding to rising and falling edges), and the different speeds of the thermodynamic and electrical interactions minimize the delay (i.e., provide the maximum operational speed, vital for survival). The steady-state error is minimized by setting the process variable to the reference point using long-term stable parameters (geometry and overall concentration). The low-intensity current through the always-open resting ion channels provides dynamic stability in the steady state.

We considered that the neuron has a stable base state. On the one side, this resting state must be dynamically stabilized for little perturbations using as little energy as possible (and to provide a mechanism when the cell grows, divides, or ages). On the other side, it must be able to restore the state after rough perturbations as quickly as possible, causing short-time transients (when

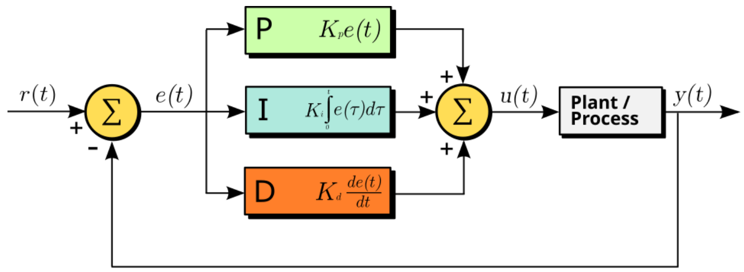

Figure 2.5: A block diagram of a PID controller in a feedback loop. r(t) is the desired process variable (PV) or setpoint (SP), and y(t) is the measured PV. (Wikipedia)

restoring the membrane's potential after issuing a spike). In both the resting and transient states, the system attempts to return to its balanced state, though the mechanisms required differ.

In its excited state, the system aims to provide an intense output that informs the downstream neurons. This overshoot is initiated by a large number of voltage-gated ion channels (distinct from the non-gated ion channels used to maintain the resting potential) distributed in the neuron's membrane. The overshoot current flows through those persistently open ion channels, which are concentrated in AIS (which has about two orders of magnitude higher channel density than the wall). The intense slow current produces a condenser-like behavior (capacitive current); the phenomena called "polarization" and "hyperpolarization" (not polarization; instead, a movement of completely separated charge in the surface layer of the electrolyte) of the membrane provide the necessary positive and negative error signals for the controller in the transient state by moving the actual potential value of the membrane above and below the resting potential value. The potentials and the electrical field's magnitudes depend on the concentration and the geometry (the finite size of the membrane). The ions' chemical nature comes into play only if the polarizability can differ for different molecules.

2.2.6 Neuronal components

Ion channel

Layer of life

Embedded in the wall of the membrane, a smaller amount of (controlled and uncontrolled) ion channels are scattered. Between the membrane and the axon, there is an array of a large amount of uncontrolled ion channels (the Axon Initial Segment AIS). The elastic wall of the membrane is covered with ion layers, and therefore, a potential difference is created there (the resting potential). The membrane represents *a confined space in which ions behave differently from in an infinite space*; furthermore, it significantly alters the properties of the few-nanometer-thick electrolyte layer near the membrane. In this layer, the concentration and potential change significantly with the distance from the membrane due to interactions between the membrane's ion layers and the electrolyte. Ion packages (spikes) sent by other neurons can enter the internal volume through controlled inputs (synapse).

2.3 Electrical model

The electrical phenomena associated with living matter ("animal electricity") have been known for centuries, so the first experiments in search of an explanation began with electrical measurements, although thermodynamic and mechanical studies were not far behind. Theoretical interpretation first became possible on the basis of electrical phenomena. Partly because it was not a suitable competitor to the theory, partly because it could really explain many phenomena, its use has now become almost exclusive. Neuroscience has tried to compensate for the contradictions experienced, on the one hand by ignoring inexplicable facts, and on the other hand by adding unscientific ad-hoc hypotheses.

Below, we attempt to establish a disciplinary model based on three different similarities, and then a unified model that seeks to find a balance between the most faithful description of phenomena observable in living organisms and the description provided by the disciplines of inanimate science.

Despite that we attach mechanical and other phenomena to the electrical ones, our motto is similar to the frequently cited saying 'Cherchez la femme': *Look for the charge*, meaning "*no matter what the problem, some charge is often the root cause*" (however, we consider also effects of the thermodynamics on charge, given that the charges are ions instead of electrons. Furthermore, we shortly consider the mechanical, optical, etc. consequences.). The expression comes from the novel *The Mohicans of Paris* by Alexandre Dumas and is frequently used in detective novels in the sense that there are always complications and unusual situations, but "no matter what the problem, a woman is often the root cause". Although, in most cases, the cause is seemingly different.

Similarly, the charge is always the primary quantity when speaking about neural operations. We follow this line and explain how processes, that traditionally belong to different classical science disciplines, redistribute charges and create electrical currents and voltages, then discuss how the cell handles the electrical charge. We note, however, that the different charge carrier (ions instead of electrons) needs care: the "slow current" shows unusual behavior and the needs using known laws in more or less different form. Also, we make hints that in most cases, the classical disciplines, alone, are not qualified to describe phenomena of living matter. The details are described in Chapter B, where the fundamental differences caused by the charge carrier are discussed.

We show that, alhough the thermodynamic effects are inseparable from the electrical ones, an appropriate combination of the disciplines satisfactorily describe the observations. We consider that neurotransmitters, receptors and specialized membrane proteins *only implement a kind of (time and energy-consuming) chemical/enzymatic decoupling of the signal transmission mecha-*

nism. The idea resembles opto-coupling in electronics: makes *the signal transmission independent from the local potential value.* Suppose neurons use galvanic coupling when the resting potential of one of the neurons equals the extracellular space. In that case, all connected neurons' resting potentials are equal to that of the extracellular space. Without this decoupling, the death of one neuron would immediately lead to the death of the entire neural network. Furthermore, the neurons could not make independent signal processing.

Because of its significance, the electrical model deserved separate chapter4.

2.4 Elastic and electric model

The mechanical (elastic) model takes us one step closer to the correct model. It assumes that the action potential is basically generated by the shock wave generated by the elastic membrane, which is significantly distorted by the electrostatic repulsion of the ions in the electrolyte.

The rush-in of Na^+ ions results in electric, chemical, and elastic gradients, see [16]. The final result (shown in Fig. 5.1) is that the membrane vibrates the electrolyte (with the cca. 10^{-3} part ions inside). Moreover, the electrostatic repulsion and the concentration gradient of ions provide an offset voltage that triggers an electrical discharge. The two processes are superimposed.

Furthermore, the elastic model explains the experienced constant intensity of AP (without the ad-hoc hypothesis of cooperating ion channels in the axon's wall). The vibrating electrolyte behaves as a 'soliton' [15], that is, transfers the pressure wave with a minimum material transport and intensity loss, and the dissociated ions inside the vibrated electrolyte generate the potential field observed as AP.

Pros and cons: Elastic and electric neuron

Pros:

- Connects mechanical and electrical phenomena
- Simple function form to calculate
-

Cons:

- Uses empirical coupling factor between disciplines
- Not a faithful form
-

Discussed in chapter 5

2.5 Thermodynamic model

The experience that not only electrical but also thermodynamic (temperature) changes occur in the neuron during the action potential has practically the same age as the observation of electrical phenomena. Later, with the development of measurement techniques, other phenomena (force effects, pressure waves, optical and density changes, etc.) were also observed. These phenomena cannot be explained by the single-discipline V1.0 theory, although they are perfectly consistent with the electrical effect and their time course is also the same; see figure 6.1.

As Hodgkin wrote: "Hill and his colleagues found [28] that an initial phase of heat liberation [of the AP] was followed by one of heat absorption. [...] a net cooling on open-circuit was totally unexpected and has so far received no satisfactory explanation." [29] "All authors came to similar conclusions: during the AP, *no significant net heat is produced.* Transient heat releases are mostly reabsorbed in the second phase of the AP. . . . The finding of a reversible heat release during the action potential of nerves is a striking and very fundamental fact. It is inconsistent with the HH model. The physics underlying the nervous impulse must rather be based on reversible processes". [30]

Discussed in chapter 6.

2.6 Unified model

The neuron is introduced as a lipid globule with a high elastic modulus filled with an aqueous solution and a long lipid extension. In our model, a force impact on the wall (generated by the rush of Na+ ions) in a container with an elastic wall initiates a rapidly damped vibration, which of course also vibrates the aqueous solution contained within. This model in itself belongs purely to the discipline of mechanics. In the case of the neuron, however, the aqueous solution also contains ions, and the small amount of ions in the vibrating liquid also performs an oscillatory motion, i.e. generates a voltage wave, although due to the ions moving back and forth, it results in only minimal mass transport.

The mass of the ions means a finite speed, and their charge – due to mutual repulsion – leads to the concentration of most of the free charge carriers of the electrolyte on the inner surface of the lipid membrane. This active layer represents a fundamental change compared to previous theories. It explains the physiological fact that significant changes occur in nanometer-sized layers close to the membrane. Slow-moving ions in the layer generate an electric potential wave, which, due to repulsion, also generates a local mechanical wave. We have shown that the action potential is essentially generated by elastic vibrations of the membrane (the elastic energy is two orders of magnitude greater than the electrical energy). The compression/expansion associated with the vibration naturally explains the phenomenon of heat generation/absorption, which has been mysterious for decades.

We have successfully linked the mechanical and electrical characteristics, thus providing a unified model of neuronal function. Based on the model, we have shown that the assumed K+ current is actually a capacitive current of the oscillatory circuit; the K+ current does not exist during the AP, but is merely an artifact caused by the clamping voltage; furthermore, the AP is not transmitted by the cooperation of synchronized ion channels, but by ions moving in an elastic wave (soliton) that create the illusion of the potential wave.

We have shown that the resting potential of the membrane is essentially created by the insulating lipid membrane of finite thickness placed in saline solution, and is maintained at a value determined by physiological parameters by a PID control circuit using ion channels embedded in the membrane (i.e., the GHK equation based on mobility and Nernst potentials is incorrect). We have shown that the action potential is very accurately described by the series oscillatory circuit known from electrical science (i.e., the parallel oscillatory circuit can only be used when describing the resting potential), and that the Axon Initial Segment, which is of fundamental importance in the operation, can be included in the theoretical description. We have also succeeded in providing a thermodynamic explanation for the heat engine-like operation of the neuron by measuring the electrical parameters; we have managed to determine the energetic efficiency of the neuronal cycle. The unified model, established by taking into account all relevant parameters, resolves the contradictions left unexplained by previous models and makes the introduction of ad hoc conditions unnecessary; its results agree with the known experimental results within their error limits. Based on our results, we need to revise the statements regarding neural functioning; also textbooks. Dissemination in the form of lectures, communications, tutorials and a textbook will be one of the main tasks of the project.

We have established that, contrary to the accepted belief, the information processing and transmission mechanism of the neuron is not directly related to the number of spikes (some kind of 'rate'), it is only a by-product. Information is distributed in space and time, the temporal branch of which we have largely explored. The information is carried by the time of spike emission. The task of the project will be to explore the principles of spatial distribution (related to the functioning of neural ensembles), communication and neural computation.

Discussed in Chapter 7.

Chapter 3

The Hodkin-Huxley model (V1.0)

Contents

Quotation:

This was how — Richard P. Feynman approached all knowledge: What can I know for sure, and how can I come to know it? It resulted in his famous quote, "*You must not fool yourself, and you are the easiest person to fool.*"

Pros and cons: Hodgkin-Huxley V1.0

Pros: Goods

- 1
- 2

Cons: bads

3.1 Equivalent electrical circuit

"These ionic concentration gradients across the cell membrane constitute the driving forces (or chemical potentials) for ionic currents flowing through open

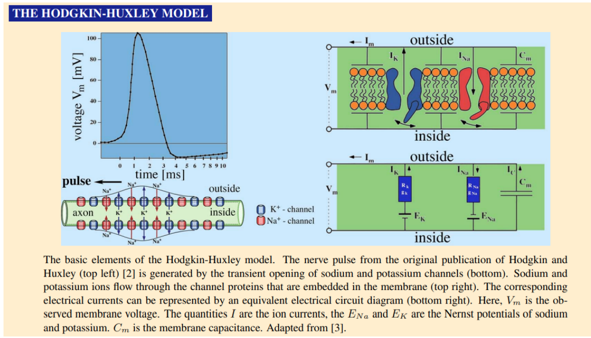

Figure 3.1: The time course of the electric gradient perfectly matches that of the mechanical and thermal changes; however, classic theory cannot explain the connection between them [4]. See Fig. 6.1 how the phase diagram of the voltage change in the function of its gradient matches the experimentally measured values.

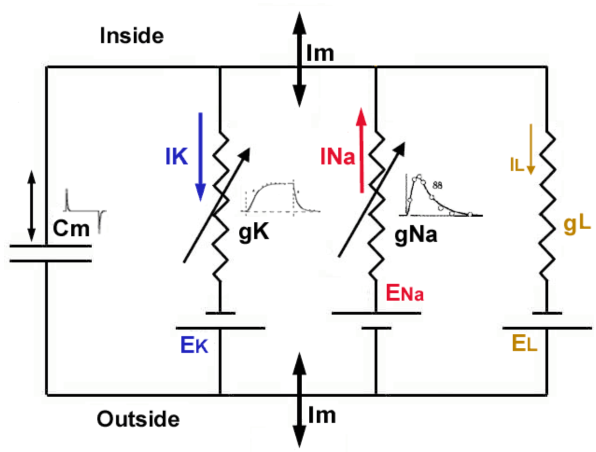

Figure 3.2: The equivalent circuit for the Hodgkin-Huxley model

channels in the membrane. In other words, the ionic concentration gradients act like DC batteries for cross-membrane currents." [7]

Fig. **??** shows the concept of the AP

The biological 'equivalent circuit' models assume that the circuits comprise point-like ideal *discrete elements* such as condensers and resistors, and some hidden power changes their parameters according to some mathematical formulas, furthermore ideal batteries with voltage that may again be changed by that power. All they are connected by conducting (metallic) ideal wires and their interaction speed is infinitely high (the Newtonian 'instant interaction'). Using that abstract model enables them to use the well-known classic equations, named after Ohm, Kirchoff, Coulomb, Maxwell and others. However, those abstractions have severe limitations. Biology applies Ohm's Law to its objects while claims that its objects are non-ohmic; understands that neural currents comprise ions while claims that the ions do not feel the Coulomb repulsion; measures ions' propagation speed but in its equations it claims that their effects are instant; hypotesizes non-existing physical mechanisms to make their nature. In general, biophysics abstracts from the world of the physics-inspired mathematical formulas a fictitious nature where biology lives and hypothesises complementary mechanisms to achieve some resemblance with the true biological word. The examples include that ions in the ions channels do not repulse each other; that charge, potential and current are independent of each other; that ion currents do not repulse each other neither when they travel from the presynaptic terminal to the AIS; that some hidden power opens the ion channels to let ions in into the intracellular space, that ions with the same charge move in opposite directions when ions rush-in into the intracellular segment of the neuron; that the ions pass ion channels without being accelerated by the potential difference accross the membrane; the telegraph equations are applied to the case of axons although neither external potential nor current loss exists; and so on.

Even that unfortunate idea of equivalent circuits leads to analyzing "electrotonic (electronic circuit equivalent) modeling of realistic neurons and the interaction of dendritic morphology and voltage-dependent membrane properties on the processing of neuronal synaptic input" [31]; that is, to study a simulated neuron built from discrete electronic components. However, the idea needs to put together many compone "raises the possibility that the neuron is itself a network". On the one side, such an idea misguides the neurophysical research (since actually a fake neural system is scrutinized, the validity of the approach is questionalized [32]), and on the other, since electroengineers understand the neuronal operation from the wrong model, it also misguides building neuromorphic architectures.

The equivalent circuits are a source of misinterpretations. As [7] formulates, "In other words, the ionic concentration gradients act like DC batteries for cross-membrane currents." We call the attention again, that *ions* represent the current, that is *that current changes changes the concentration, that changes the voltage of the DC battery, unlike in the case of the equivalent circuits.* This difference is significant in understand how the basic neuronal circuit works. Introducing equivalent circuits prevents explaining the fundamental electric phenomena.

One wrong consequence of forgetting that the charge transfer mechanism is entirely different, the charge carriers are large and heavy (and as a consequence: slow) ions instead of electron (cloud), furthermore they are not necessary present in the volume under test and the 'construction' of biological matter enables the tested medium to produce charge carriers, using 'equivalent circuits' for neuronal operation. This fallacy entirely falsifies the conclusions from the measurements.

Another wrong consequence of using 'equivalent circuits' to describe the electrical operation of neurons is believing that the currents in the biological circuit do not change the concentration, and through the concentration, also the potential; see also section B.3. The 'equivalent circuits', of course, use a constant potential (they follow the abstraction used in the theory of electricity, although the 'ideal batteries' also may produce their voltage using chemical processes), and unreasonably, some mystic process changes the resistance/conductance of components. This wrong abstraction results in numerous misunderstandings, among others, introducing ideas such as parallel oscillator equivalent of neuron, input resistance, delayed rectifier current, resting current, and time- or voltage-dependent conductance. Furthermore, we cannot interpret, among others, *how neuronal electricity works in lack of external potential*; *how slow currents operate neuron's infrastructure*, *how and why action potential is generated*. Deriving the time course of the Nernst-Planck potential opens the way to a quantitative understanding of neurophysical electrical processes, including their time course.

Again another wrong consequence is that the two secondary abstractions 'potential', and 'current', became independent from the primary abstraction 'charge' and each other (significantly contributing to the fallacy that 'physics cannot describe life'). Our equations and the underlying discussion point to the fact that *the potential and the current cannot be separated from the charge*. No 'delayed rectifying current' and 'voltage- (or time-) dependent conductance' exist. Those notions originate from the wrong interpretation of measured data derived from mismatching measured electrical data pairs and the misconception that biological structures and materials must behave like metals.

As we discussed in section **??**, the unbalanced charge in neurons and the charge injected suddenly into the intracellular segment during a rush-in action are in the same order of magnitude. We also discussed that the charge injection (the appearance and flow of charge in the proximal layer on the membrane's surface) causes well measureable mechanical, optical, etc. changes [33] on the membrane. Those changes are well observable on the low-concentration side, in the form of slow currents, and large-scale concentration and potential changes.

Those changes also mean that a large amount of charge passes from the high-concentration side to the low concentration side of the membrane, and causes a sudden drop on the high-concentration side of the membrane. As discussed, that change means simultaneously a change in the corresponding potentials across the membrane that solves the mystery why the AP AP stops. The potential across the membrane is defined by the difference of the potential of the two thin layers (see section **??**) on the membrane's surface. As we discussed, a huge electric field exists across the plates of the neuronal condenser, while a moderate one in the proximal layers. That means that the ions can quickly leave the high concentration when the rush-in injection begins. They produce an 'empty' layer (and a potential gradient) whithin that layer and the ions in the neighboring layer start to move using the 'downhill' potential. However, they do have a much slower speed, so they 'stop' at some distance.

Chapter 4

Electrical model

Contents

4.1 Calculations in a cell

In this section, we only use our knowledge of the neuron to the extent that we perform calculations for the ions of the 'layer of life' as if they were point charges located in a lifeless medium. In the calculations, we use the disciplinary formulas of technical electricity, although we keep in mind the different properties of discrete electric charge carriers, especially their slowness.

4.2 Method of electrical description

An essential element of our electrical model is that inside the insulating sphere shown in Figure 2.1, in direct contact with it, there is a nanometer-thick electrolyte layer, see section 2.2.6. We consider the layer ('layer of life') to be

Physics panel 4.1.1:
Estimating the force exerted on ions in electric field

By using that in an electric field $\vec{F} = q\vec{E}$, one can calculate the force exerted on a charge, that enables one to calculate also the particle's impulse, pressure, and energy. For example, in the case of Na^+ rush-in, the electric force field is 10^7 [V/m]. Since($[N/C] == [V/m]$), the value of the force exerted on a single ion, accelerating it across the ion channel, is

$$F_{Na^+} = 10^7[N/C] \times 1.602 * 10^{-19}[C] = 1.6 \times 10^{-12}\,\mathrm{N} \tag{4.1}$$

(Here we used data from Johnston&Wu constants from **Numerics panel 2.0.1**)

Physics panel 4.1.2:
Estimating the acceleration of ions in electric field

The force in the ion channel was calculated in **Physics Panel 4.1.1** and the mass of a Na^+ ion is $m_{Na} = 3.82 \times 10^{-26}\ kg$. It causes an acceleration

$$a = \frac{F_{Na^+}}{m_{Na^+}} = \frac{1.6 * 10^{-12}\ [N]}{3.82 \times 10^{-26}\ [kg]} = 0.42 \times 10^{14}\ [\frac{m}{s^2}] \tag{4.2}$$

The distance d to travel (without collision) is the thickness of the neuronal membrane $5 * 10^{-9}\ nm$ and we assume a continuosly accelerating ion. Using $d = \frac{1}{2} \times a \times t^2$

$$t_{traverse} = \sqrt{\frac{2 \times 5 * 10^{-9}[m]}{0.42 \times 10^{14}[\frac{m}{s^2}]}} = 1.54 \times 10^{-11}\ [s] \tag{4.3}$$

The acceleration of an ion is unbelievably large. It is sufficiently large to keep the potential at the same value over an extended object, provided that the ions must follow a small and slow change, such as due to the "leaking" current through the AIS. Similarly, the ions can 'instantly' follow quick changes such as a square wave gradient.

homogeneous, and within this the interior of the neuron ('bulk') is also homogeneous. This layer is responsible for the electrical phenomena. The ions that create the resting potential are located in the layer shown in Figure 2.2 on the surface of the neuron; these are held firmly in place in the resting state by the opposite charges on the other side of the membrane. The Na^+ ions that flow in at the beginning of the action potential "stop" in this layer and leave from there through the AIS. In other words, this layer essentially forms the capacitor

Physics panel 4.1.3:
Ion density and force on membrane's surface

Using data from **Numerics panel 2.0.1**), and assuming that the electrical repulsion keeps the surface density of ions uniform, one can calculate the area an ion occupies, and the distance between ions as the square root of that area

$$D_{ions} = \sqrt{\frac{8 \times 10^{-9}}{10^7}} = 2.8 \times 10^{-8}\ m$$

That is, the average distance between ions on the surface is about 150 times larger than the ion's diameter. Applying the formula

$$F_{ions} = \frac{1}{4\pi\epsilon_\circ}\frac{q^2}{r^2}$$

using the elementay charge and the distance calculated, results in

$$F_{ions} = 2.9 \times 10^{-13}\,\mathrm{N}$$

By comparing that value to the force acting across the membran as given in **Numerics panel 4.1.1**), one can see that the force perpendicular to the membrane is an order of magnitude larger than the force among the ions on the surface.

in the electrical description. Note that ions form a 'slow' current, which can be ignored for some phenomena.

Our electrical model describes the neuron as a circuit in which charges are present, voltages can arise and currents can flow, and we use concepts known from technical electricity. In the resting state, the intracellular and extracellular potentials of the neuron (in other words, the electrolyte potential on the inner and outer sides of the neuron membrane) are the same, so there is no potential difference that would drive current through the membrane. However, the voltage of the neuron is composed of one electrostatic and two thermodynamic (Nernst) voltages. Figure 4.1 graphically summarizes the formation of the potential, 4.2.1 physics panel describes the mathematics of the voltage relations, and Table 4.1 compares the theoretical results to the measured values.

Electric charges accumulate on both sides of the membrane, which create a potential difference. The orange arrow in the middle indicates the capacitor voltage. So, contrary to the assumption of HH, the potential difference is not created by a constantly flowing current, the "leakage current", but by charges attached to the membrane. Essentially, according to Kirchoff's voltage rule: by adding the sum of the Nernst voltages of the ions on both sides of the membrane to the voltage on the capacitor between them, we get zero voltage. The names of the ions that create the Nernst voltages are only shown on one side. The

Physics panel 4.1.4:
Estimating the acceleration of ions in electric field

The force in the ion channel was calculated in **Physics Panel 4.1.1** and the mass of a Na^+ ion is $m_{Na} = 3.82 \times 10^{-26}\ kg$. It causes an acceleration

$$a = \frac{F_{Na^+}}{m_{Na^+}} = \frac{1.6 * 10^{-12}\ [N]}{3.82 \times 10^{-26}\ [kg]} = 0.42 \times 10^{14}\ [\frac{m}{s^2}] \tag{4.4}$$

The distance d to travel (without collision) is the thickness of the neuronal membrane $5 * 10^{-9}\ nm$ and we assume a continuosly accelerating ion. Using $d = \frac{1}{2} \times a \times t^2$

$$t_{traverse} = \sqrt{\frac{2 \times 5 * 10^{-9}[m]}{0.42 \times 10^{14}[\frac{m}{s^2}]}} = 1.54 \times 10^{-11}\ [s] \tag{4.5}$$

The acceleration of an ion is unbelievably large. It is sufficiently large to keep the potential at the same value over an extended object, provided that the ions must follow a small and slow change, such as due to the "leaking" current through the AIS. Similarly, the ions can 'instantly' follow quick changes such as a square wave gradient.

curved arrows on both sides of the figure indicate the Nernst voltage of the two ions: the difference in the voltage coordinates of the end points corresponds to the Nernst voltage. The zero point is marked at the zero voltage value on both the outer and inner sides. Following the arrows between the two zero points, we obtain a circuit where no current flows.

The electric voltage contribution is unidirectional, the two-component thermodynamic contributions, depending on the concentrations, can be positive or negative. If choosing the zero potential at the middle of the electric potential, we can interpret that the concentration combinations maintain their ionic concentrations (and so also thermodynamic voltage contributions) independently on both sides starting from a zero potential level. They generate potentials autonomously in the segments proximal to the opposite surfaces of the membrane. The electric charge on the membrane creates another voltage independently. However, that voltage must fit between the two thermodynamic voltages. Any deviation in the potentials starts a current directing the concentrations and voltages toward a balanced state. The process is complex: the sum of surface concentration of ions must be the same on both sides for the condenser and the ratios on the intracellular and extracellular sides must be adjusted to satisfy the conditions the conditions shown in 4.2.1 physics panel.

Table 4.1 shows our results for three different famous published cases. It

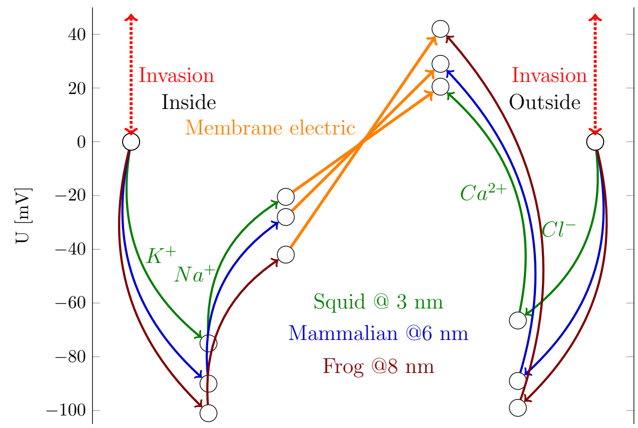

Figure 4.1: The mechanism for setting neuronal voltages is the same for all species shown in Table 4.1. It appears that the thicker membrane produces a higher membrane electric potential and requires a higher thermal potential to compensate for it. However, it can solve the task using a lower salt concentration. Given that the electric membrane potential accelerates the ions across the ion channels in the membrane, and the operation is faster. The electric field has a similar value, as shown in Fig. D.1. Furthermore, the total concentration decreases as the membrane thickness and potential increase.

Physics panel 4.2.1:
Voltage relations in a neuron

The local electric and concentration gradients control the local potentials:

$$\begin{aligned}
U^{K^+,Na^+}_{internal} &= -U_{electric} \quad +U^{K^+}_{internal} \quad +U^{Na^+}_{internal} \quad (+U^{invasion}_{internal})\\
U^{Cl^-,Ca^{2+}}_{external} &= U_{electric} \quad -U^{Cl^-}_{external} \quad -U^{Ca^{2+}}_{external} \quad (+U^{invasion}_{external})
\end{aligned}$$

In a balanced state, the left sides are zero: at the contact point between the electrolyte and the membrane there are no potential differences, while in perturbed state they are non-zero (the ions are slow) and so they represent the driving forces for restoring the balance on the respective side. From the derivation, follows that

$$\begin{aligned}
U_{electric} &= -2*(U^{K^+}_{internal} \quad +U^{Na^+}_{internal})\\
U_{electric} &= -2*(U^{Cl^-}_{external} \quad +U^{Ca^{2+}}_{external})\\
U_{electric} &= U^{K^+}_{internal} \quad +U^{Na^+}_{internal} \quad = -(U^{Cl^-}_{external} \quad +U^{Ca^{2+}}_{external})
\end{aligned}$$

(these relations are used in Table 4.1)

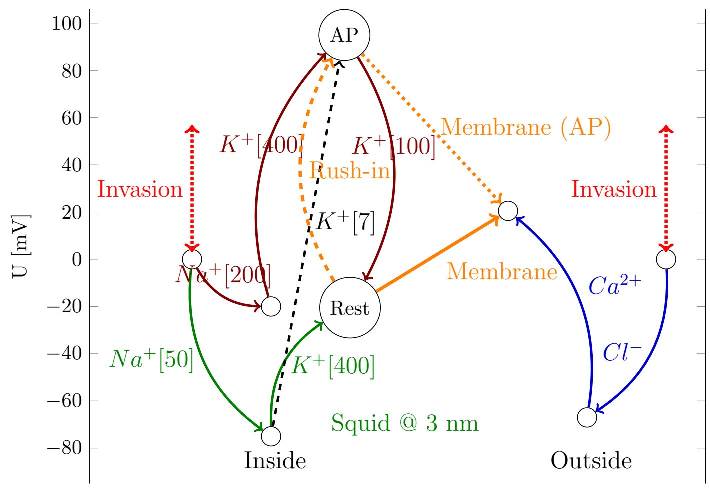

Figure 4.2: The figure is supplemented with the unbent black line showing that the clamping condition forces the neuron to find a new balanced state by compensating the clamping voltage by setting a concentration [Na50,K7]. For comparison, the case of natural AP is also shown. The mechanism of producing AP (using numbers referring to the case of squid shown in Table 1 in [16]). In the resting state, the inside thermal potentials follow the green paths. When the rush-in of Na^+ ions suddenly increases the internal concentration to [200], the potential on the internal side of the membrane increases to $+90\ mV$. The neuron must decrease its concentrations to restore the resting potential, mainly by releasing an action potential through the AIS. The clamping fixes the neuron's membrane voltage at AP, so the neuron can operate only with the concentrations.

looks like the calibration of the concentration of negative and positive ions needs scaling and the measurement of Ca^{2+} comprises issues.

4.3 Describing invasion: clamping

4.4 The outward K+ component during the Action Potential

Emitting Action Potentials is a fundamental aspect of neuronal operation. HH hypothesized that neurons operate with electrical currents and suggested the idea [14] that the neuronal operation is more or less reminiscent of an electric oscillator. Given that the parallel RC circuit they hypothesized cannot produce an output voltage with opposite sign, they had to introduce the concept of a delayed current in the opposite direction. Even, they demonstrated that K^+ current flows in the second half of the AP. The net electrical model led immedi-

Table 4.1: The summary data table for equilibrium (Data from Table 2.1 in [7]. For the graphic representation of potential balance see Figure 4.1)

Ion	[In] [mM]	[Out] [mM]	$U_{therm}\frac{RT}{zF}log(\frac{[C]_o}{[C]_i})$ [mV]	U^{theor}_{electr} [mV]	U^{exper}_{electr} [mV]	U_{membr} [mV]
	Squid [14] T=293 K^o Width= **3** [nm] Field= **14.0** [MV/m]					
K^+	400	20	$-75.5 = 58log(\frac{20}{400})$	**41.9**	**41.4**	-62.0
Na^+	50	440	$+54.8 = 58log(\frac{440}{50})$			
Cl^-	40	560	$-66.5 = 58log(\frac{560}{40})$	**41.9**	**42.0**	63.0
Ca^{2+}	0.27*	10	$+45.5 = 29log(\frac{10}{0.27})$			
	* used for the published value 0.4					
	Frog muscle (Conway) T=293 K^o W = **8** [nm] E= **9.6** [MV/m]					
K^+	124	2.25	$-101 = 58log(\frac{2.25}{124})$	**88.9**	**83.6**	-125.4
Na^+	10.4	109	$+59.2 = 58log(\frac{109}{10.4})$			
Cl^-	1.5	77.5	$-99.4 = -58log(\frac{77.5}{1.5})$	**88.9**	**82.7**	124.1
Ca^{2+}	0.021*	2.1	$+58.0 = 29log(\frac{2.5}{0.25})$			
	* used for the published value 4.9					
	Mammalian cell T=310 K^o W= **6** [nm] E= **11.1** [MV/m]					
K^+	140	5	$-89.7 = 62log(\frac{5}{140})$	**57.7**	**57.3**	-85.8
Na^+	15	145	$+61.1 = 62log(\frac{145}{15})$			
Cl^-	4	110	$-89.2 = -62log(\frac{110}{4})$	**57.7**	**56.9**	85.9
Ca^{2+}	0.04*	5	$+60.8 = 31log(\frac{5}{0.04})$			
	* used for the published value 0.0001					
	Na^+ ions only [34] T=310 K^o W= **6** [nm] E = **10.0** [MV/m]					
Na^+	12	145	$+66.6 = 62log(\frac{145}{12})$	60.7	66.6	

ately to contradiction with the empirical evidence that, as Hodgkin wrote [29], "Hill and his colleagues found [28] that an initial phase [of the action potential] was followed by one of heat absorption. [...] a net cooling on open-circuit was totally unexpected and has so far received no satisfactory explanation." Not to mention the mechanical, optical, etc. effects [33]; accompanied by AP. Neglecting the AIS and sticking to AIS-less model led to clue-less ad-hoc assumptions and desperate claims such as "The sodium influx associated with depolarization is exothermic [...] the capacitive energy stored in the membrane is released as heat. However, the efflux of K^+ ions is endothermic [...] *converting heat in the environment into capacitive energy at the membrane.*" [33] (That anti-dissipation effect would be something new for physics and could convert the harmful global warming to an inexhaustible energy source for humankind.) The correct physical explanation can be derived using the unified model of neuronal operation [35, 16]: under the mechanical pressure caused by the mutual repulsion of the ions, the electrolyte is compressed and expanded.

One of the big mysteries in the creation of the V1.0 model was why AP became negative when only Na^+ influx was seen when AP was formed. Given that they found only positive charges nearby, they supposed that an outflux of positive charges, namely of K^+ ions, was the reason. Since the process is too fast, HH used a voltage clamping method for the experimental demonstration [36] of the hypothesized K^+ current. The essence of this is that they imitated the crossing of the membrane voltage threshold (reaching the "AP" state, see Fig. 4.3): "(after completion of the quick pulse through the membrane capacity)", then raised the intracellular domain of the membrane to a potential higher than the resting potential with a negative feedback circuit: "the potential difference across the membrane is suddenly changed from its resting value, and held at the new level by a feed-back circuit ('voltage clamp' procedure)", and measured the time dependence of the current in this state.

4.4.1 The formal description of the process

Figure 4.2 shows the changes that occur in a neuron that is displaced from its equilibrium state. These processes are naturally slow, their role in the transient process is discussed in the **??** section, here we only illustrate our statements regarding the K^+ current. Our statement is essentially based on Kirchhoff's Voltage Law, according to which the sum of the voltages in a closed circuit is zero. In our case, the circuit extends from the inside of the neuron to the external environment. If for any reason the external and internal potential levels are not the same, then a current flows until they are equalized (current carries charge and causes a potential change; it is not an ideal voltage generator). Of course, the equalization is not instantaneous, the current has a time dependence.

On both the left and right sides of the figure, the resting potential level is at the zero value of the vertical scale. In the resting state (without invasion), in the intracellular segment the [Na50,K400] concentrations are present inside the cell. Using these values, we reach the "Rest" state along the green arrows (based on the Nernst voltages, see 4.1 Table), which is located at a height of $-21\,\mathrm{mV}$. On the right side, in the extracellular segment, from the point also at a height of 0, we arrive at a point at a height of $21\,\mathrm{mV}$ along the blue arrows in the same way. The potential difference between the two mentioned points is $42.5\,\mathrm{mV}$, which HH measured. It is a correct value, but it is the result of an electrostatic charging and not of a voltage drop due to the "leakage current" they assumed. The voltage across the membrane (represented by the solid orange arrow) points upwards, preventing positive ions in the intracellular space from moving towards the extracellular space and preventing negative ions in the extracellular space from moving towards the intracellular space. The Ca^{2+} ions in the extracellular segment, however, are permeable, requiring continuous pumping. There is no potential difference between the outside and inside, so no (significant) current flows.

If, for any reason, the membrane voltage suddenly rises above the threshold value, due to the extremely fast Na^+ influx, the membrane potential increases by about 100 mV (a [Na200,K400] state occurs, see the brown arrows), i.e. the neuron jumps into the AP bubble. The orange arrow that was pointing upwards now points downwards as a broken line: the outflow of positive ions (both K^+ and Na^+) is given a free path, but the sloping path for Ca^{2+} ions is eliminated. However, the free ions in the surface layer of the membrane are the vast majority of the Na^+ ions that have just arrived, so the ion current through the AIS mainly consists of such ions. A significant increase in the membrane voltage leads to a significant increase in the potential of the inner space of the neuron. A current is initiated that tries to restore the initial voltage and concentration relationships. The extracellular environment of the membrane is extremely stable (the concentrations do not change), but inside the cell the voltage increases to the same extent, in the direction of "Invasion". The membrane potential has changed significantly due to the change in the Na^+ concentration, so the cell then moves them towards equilibrium.

In the case of natural AP, there is no obstacle for the neuron to restore its previous state by changing any parameter, so the system concentrations start towards the state [Na50,K400]. This can be easily achieved by trying to make Na[50] from Na[200], while K^+ is still/already at the appropriate concentration. That is, it releases (mainly) Na^+ ions through the AIS until the potential difference drops to zero. Reducing the Na^+ concentration also reduces the membrane voltage, i.e. after the release of the excess ions (admitted during the Na^+ rush-in) the neuron returns to its initial state; a cycle takes place.

"Clamping" fixes the membrane voltage, meaning that the neuron cannot restore its initial resting state. Instead, it has to find a new balance by changing the concentrations. After the artificial AP, clamping (using a negative feedback) keeps the membrane potential at a fixed value, so the neuron can only produce a new balance by changing the ion concentrations: a current flows until the Nernst voltages of the Na^+ and K^+ concentrations can balance the clamping voltage. From this [Na50,K400] state, the neuron has to adjust the potential so that the sum of the Nernst voltages plus the membrane's voltage plus the clamping voltage is zero (so another balance has to be set!), while the voltage is fixed. To do this, it must reduce the K^+ concentration, while the Na^+ concentration is already at a good value (the opposite of the previous case). And not by a small amount: [K7] only causes a voltage change equal to the clamping 100 mV imposed on the cell. This voltage is given by the green curved $Na^+[50]$ and the black dashed straight $K^+[7]$ path. The neuron can achieve this [Na50,K7] concentration by starting an intensive K^+ release.

Another important difference compared to the natural case is that there is no electrostatic acceleration across the membrane (which is why the Na^+ influx stopped), only a thermodynamic suction effect outwards due to clamping. Indeed, K^+ ions attempt flow out through the membrane, but they cannot exit until the electric potential drops below the equivalent thermodynamic driving force of the concentration. Moreover, the nearby K^+ ions get there sooner, but they leave a gap (reduced concentration) behind them, where the nearby neigh-

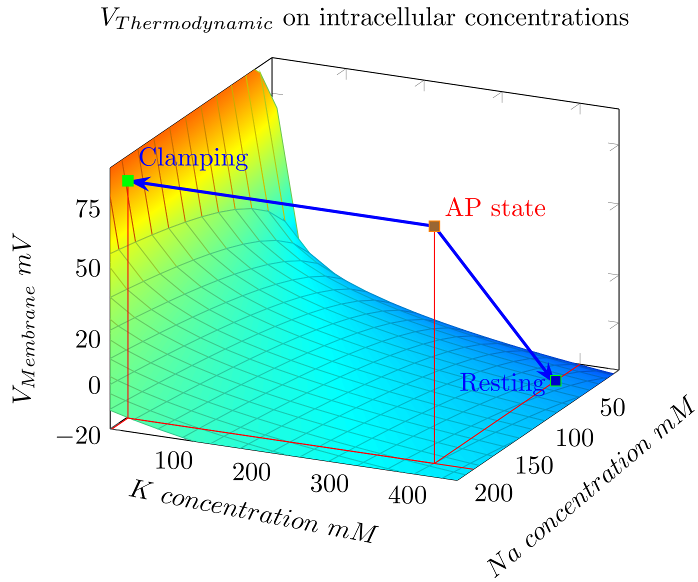

Figure 4.3: Artificial and natural degradation of the AP state. In the case of natural degradation, the neuron has to get rid of the (unnecessary) Na^+ ions; by releasing them, it returns to the initial state; cycle takes place. Clamping keeps the membrane voltage fixed, so the neuron can only establish a new equilibrium situation by changing the intracellular concentrations which, due to the very reduced K^+ concentration, is only possible with an intense K^+ release.

bor K^+ electrodiffuses. This is why the physiologist sees a "delayed current". It is a strange game of nature that the resultant of the thermodynamic and the electric (plus the elastic) gradients moves, simultaneously, the $Na+$ ions parallel to the membrane surface, toward the AIS, and the K^+ ions perpendicular to the surface toward the membrane, approximately until the peak of the action potential voltage is reached. The Na^+ starts to leave through the AIS, while the "excessive" K^+ ions leave through the membrane. *The two currents do not flow in the same path, so they cannot interfere as currents in opposite directions and cannot cause hyperpolarization.* Hyperpolarization is caused by the reversal of the Na^+ current, which can be interpreted in the electrical sense as the capacitive current of the capacitor in the serial RC circuit, and in the mechanical sense as the negative period of the elastic membrane oscillation.

That is, the K^+ current is an artifact caused by the clamping, which is caused by the 'clamping' measurement conditions, and does not occur in the neuron under natural conditions.

So, this is where the urban legend "K^+ flows out through the membrane in the second phase of AP, with a significant delay compared to the Na^+ influx" comes from. It is really a real measurement, it is really K^+ ion, but they did not measure what they thought. HH and their late descendants did too. So there is no question of specific Na^+ and K^+ (synchronized with each other) channels: in both cases, the concentration and potential relationships decide what ions flow and in which direction. HH was aware that theirs was not good, but they didn't know any better. Unfortunately, they spread the misconception.

In the classic experiment of HH, clamping means that the left-hand "Invasion" suddenly raises the membrane voltage to the peak value measured at AP, without the concentrations changing, so the voltage of the left-hand segment will also be higher. The current only stops when the difference in the Nernst voltages of the inner segments equalizes the clamping voltage. Due to the slowness of charge carriers, the process is not instantaneous.

Fig. 4.3 shows a pictorial summary of the process. The points on the surface give the Nernst voltage acting on the ions, for different combinations. The points on the surface are in equilibrium. The onset of the AP moves the point that was in equilibrium until then well outside the surface. The neuron can find the new equilibrium position without the forcing effect of clamping by restoring the parameters of the previous state. In the case of clamping, it must set new concentrations at which the sum of the Nernst voltages can counteract the clamping voltage. In both cases, a relaxation process takes place, which in the case of the native process is measured by physiology as AP.

4.4.2 The physics of the process

The essential functioning of the cell largely takes place in layers of a few nanometers close to the cell membrane, where differences in concentration and voltage gradients (and consequently material flow rates) of up to several orders of magnitude can occur for a few dozen microseconds. Furthermore, the slow material flow creates potential and concentration changes that change from moment to moment, and whose effects can be significantly different in each layer due to the chemical quality of the ions. For this reason, the description of the events is unique in each case, for each layer and for each time period; it is not uncommon for scientific disciplines to be changed when describing individual stages, or for them to be combined in a different way.

The two ends of the ion channels embedded in the membrane have different potentials and concentrations, so that the ions passing through acquire significant energy and speed. However, this gradient disappears when they exit the channel, so that the ions "stop" in frustration. The passage continues until the passing ions increase the electrochemical potential of the other side so much that the gradient disappears. Inside the electrolyte, the other, orders of magnitude smaller gradients continue to operate, of course, but can only create very slow material transport. The changes in the "bulk" part of the electrolyte tend towards slow equalization (different rules remain in effect near the boundary membrane). For this reason, a slow diffusion effect is also associated with all processes.

When Na^+ ions penetrate the intracellular segment, they do so under the influence of a significant electrochemical gradient. In this way, immediately after the passage (lasting a few nanoseconds), the ions create a Na^+-rich layer, from which the charge of the ions (partly by direct collision) knocks out the ions there, mainly K^+. Basically, a Na^+ layer is formed directly under the intracellular part of the membrane, which increases the voltage on the capacitor plates, and since the AIS at the end of the membrane acts as a sink for the ion current, an immediate Na^+ current is initiated, as discussed in the electrical model. Na^+ ions move along the membrane surface due to the high electrical gradient through the high-conductivity AIS, but small amounts of Na^+ and K^+ ions apparently diffuse into the adjacent layer. The ion concentration and potential of the layer decrease with the outflow current. Due to the slowness of the ion current, the ions only reach the AIS in tens of microseconds. The membrane is initially closed to positive ions (after the Na^+ ions have passed through), and the Na^+ layer represents an additional potential barrier. However, over time, the decreasing Na^+ concentration due to the current in the layer that causes the AP increasingly allows K^+ ions to move towards the membrane and the experienced K^+ current to start, which is only possible after a few dozen microseconds.

"Delayed K^+ channels" therefore mean ion channels that structurally do not distinguish between ions, and the delay is caused by transient processes taking place in the ion layers. These channels indeed release K^+ ions, which only start when the clamping voltage is switched on to the neuron (i.e., they create a non-equilibrium state of the neuron segment), and only K^+ ions because the thermodynamic gradient is only favorable to them.

HH found that the AP can be decomposed into two currents: "the ionic current during a depolarization consists of two more or less independent components in parallel, an early transient phase of current carried by sodium ions, and a delayed long-lasting phase of current carried by potassium ions". Their conclusion is more than surprising, that although the current component arising from Na ions appears quickly, the component consisting of K^+ ions remains delayed and persistent. According to that "explanation", the neurons are disposable devices: after emitting a single AP, the neuron emits a short-lasting Na^+ and (according to their figures 5 and 6) an infinitely long-lasting K^+ current, contrary to experience. In other words, for $100\,\mathrm{pA}$ (that is $1 \times 10^{-10}\,\mathrm{A}$) saturation current and 1×10^{11} neuron it would mean 10 amper permanent current in the brain. Furthermore, it would keep the resting potential above the value of the resting potential. The difference would be the extra voltage of the clamping; that is, it would disappear at zero claping voltage. In other words, the extra voltage and the K^+ current disappears in the absence of clamping, as observed.

In our reading the current lasts as long as the voltage invasion caused by clamping persists or the K^+ content of the intracellular segment of the neuron is depleted.

Chapter 5

Elastic model

Quotation:

"Brain cells communicate with mechanical pulses, not electrical signals" [37]

We introduced the neuron as a lipid ball filled with an aqueous solution, which performs a damped oscillatory motion in a transient state and of course also vibrates the aqueous solution inside. In our model, in a container with a flexible base, a vibration is initiated by a force impact on the base, which quickly damps down. This model itself belongs purely to the discipline of mechanics. In the case of the neuron, the aqueous solution contains ions, and the small amount of ions in the vibrating liquid also performs an oscillatory motion, i.e. generates a voltage wave, although due to the ions moving back and forth, it only involves minimal material transport.

The mechanical (elastic) model takes us one step closer to the correct model: it is a forerunner of the unified model. It assumes that the action potential is basically generated by the shock wave generated by the elastic membrane, which is significantly distorted by the electrostatic repulsion of the ions in the electrolyte.

The rush-in of Na^+ ions into the cells results in electric, chemical, and elastic gradients, see [16]. The final result (shown in Fig. 5.1) is that the membrane vibrates the electrolyte (with the cca. 10^{-3} part ions inside). Moreover, the electrostatic repulsion and the concentration gradient of ions provide an offset voltage that triggers an electrical discharge. The two processes are superimposed.

The sudden rush-in of Na^+ ions has (at least) a double effect. One effect is that they press the surface of the membrane that elastically increases its diam-

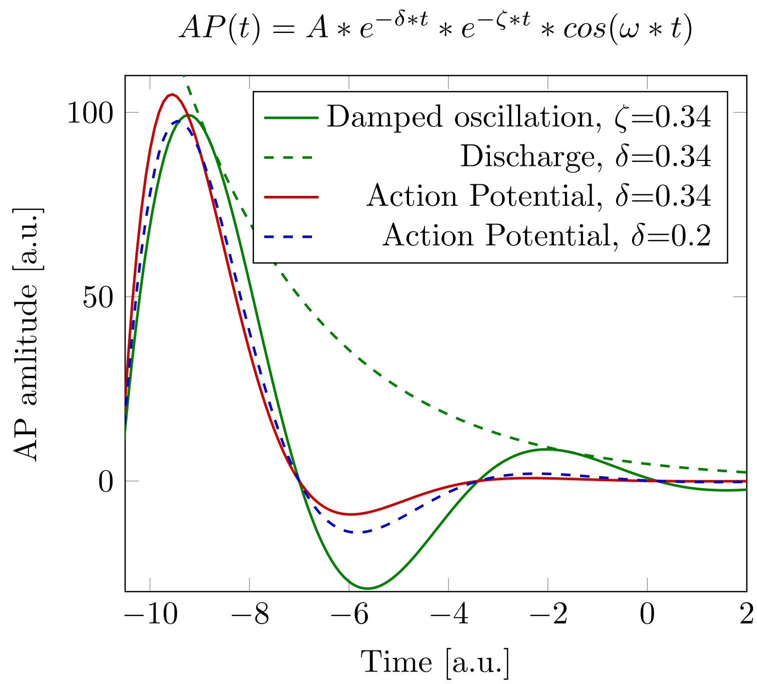

Figure 5.1: Describing the generation of an AP as the superposition of an elastic vibration of the membrane and an electrical discharge process.

eter [38]. Due to their mutual repulsion, the ions experience a strong electrical force toward the membrane. It represents an impulse $J = F\,\Delta t$ [N s], that enables estimating the time and energy of the change the rush-in causes and explains that the energy needed to generate an AP is suddenly produced electrically, and stored temporarily as elastic energy (the neuron produces the required energy in the background, when Adenozine Triphosphate (ATP) produces ions by hydrolysis under the effect of the electrical field near to membrane); see [16]. The case is practically identical to the deformation of an elastic plate induced by the hydrostatic pressure of a water column. Initially, the aluminum plate is suddenly subjected to hydrostatic pressure, and it reaches equilibrium after initial oscillations. The diagram line of the process is shown in the figure [39], and the method of simulation is discussed in [40]. Compare the diagram line to the gradient in Fig. **??**, middle inset.

Physics panel 5.0.1:
The elastic force in neurons

The elastic force is described by the

$$F(t) = F_{max}^{elastic} \times e^{-\zeta * t} \times cos(\omega \times t)$$

formula. The classical model entirely neglects this elastic gradient. However, it is almost four orders of magnitude higher than the electrical and thermodynamic gradients (see Appendix **??**), explaining why the damped elastic oscillation almost perfectly describes the time course of the AP. However, the elastic force oscillates and decays exponentially, so the dominance of the elastic term changes with time. The electrical model can perfectly describe AP since the pressure and the electric field are proportional to the number of ions inside a volume.
The other effect is that a large amount of positive charges appears in the neuron, significantly increasing the membrane's potential. The axon remains on the resting potential, so the potential gradient drives a current. A discharge process takes place where the amount of charge inside the neuron exponentially decreases

$$F_{electric}(t) = F_{electric}^{max} \times e^{-\delta * t}$$

The two effects act at the same time, that is

$$F_{resulting}(t) = F_{resulting}^{max} \times e^{-\delta * t} \times e^{-\zeta * t} \times cos(\omega \times t)$$

(the two exponential factors have physically different reasons, but they can be contracted mathematically), resulting an oscillation shown in Fig. 5.1. This model combines the effects of two distinct disciplines, but may not be perfect, because of the dual role of ions: the electric effect changes the oscillating mass and the oscillation changes the amount of charge. However, it suggests a way to synthesize the two effects, and explain a till now unexplained thermodynamical observation.

Furthermore, the elastic model explains the experienced constant intensity of AP (without the ad-hoc hypothesis of cooperating ion channels in the axon's wall). The vibrating electrolyte behaves as a 'soliton' [15], that is, transfers the pressure wave with a minimum material transport and intensity loss, and the dissociated ions inside the vibrated electrolyte generate the potential field observed as AP.

Pros and cons: Elastic/electric neuron model

Pros:

- Connects mechanical and electrical phenomena
- Simple function form to calculate

Cons:

- Uses empirical coupling factor between disciplines
- Not a faithful form

Chapter 6

Thermodynamic model

Contents

The experience that not only electrical but also thermodynamic (temperature) changes occur in the neuron during the action potential is practically the same age as the observation of electrical phenomena. Later, with the development of measurement techniques, other phenomena (force effects, pressure waves, optical and density changes, etc.) were also observed. These phenomena cannot be explained by the single-discipline V1.0 theory, although they are perfectly consistent with the electrical effect and their time course is also the same; see figure 6.1.

As Hodgkin wrote: "Hill and his colleagues found [28] that an initial phase of heat liberation [of the AP] was followed by one of heat absorption. [...] a net cooling on open-circuit was totally unexpected and has so far received no satisfactory explanation." [29] "All authors came to similar conclusions: during the AP, *no significant net heat is produced*. Transient heat releases are mostly reabsorbed in the second phase of the AP. . . . The finding of a reversible heat release during the action potential of nerves is a striking and very fundamental fact. It is inconsistent with the HH model. The physics underlying the nervous impulse must rather be based on reversible processes". [30]

6.1 The neuron as Carnot-engine

In simple words, a neuron combines different disciplines; it works like a two-tact biological internal combustion engine, and can be described theoretically as a special Carnot-type thermodynamic engine, see section **??**. The synaptic input currents operate the ignition. The influx of Na^+ ions acts like an explosion,

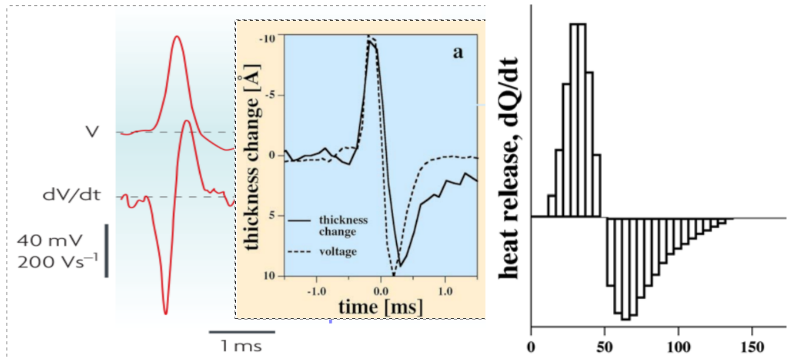

Figure 6.1: The time course of the electric gradient perfectly matches that of the mechanical and thermal changes; however, classic theory cannot explain the connection between them [41]. See Fig. ?? how the phase diagram of the voltage change in the function of its gradient matches the experimentally measured values.

producing a large pressure impulse due to electrical repulsion between ions; see section **??**. The neuron's volume remains practically unchanged (representing an iso-volume process), and the pressure wave vibrates the particles (including dissociated ions) as a longitudinal wave, and the resulting offset potential moves the ions out of the neuron through the only available output channel: the axon. The pressure wave is similar to a sound wave, with negligible material transport. The force analysis shows that the pressure wave causes a longitudinal vibration, while the changed potential adds a potential-dependent longitudinal force component for moving the ions.

The two-tact engine, in the first stroke, the pressure impulse invests energy into compressing the elastic membrane and starting a compression wave (a soliton). In the first tact, the elastic membrane pumps out some electrolyte, and in the second tact it pumps in part of the electrolyte (the driving force is a combined thermoelectrical/mechanical force rather than some magic battery). The pumping goes through the ion channels in the AIS, which "resists" it, thereby generating a potential wave known as AP; well described by electricity. The force analysis shows that the pressure wave causes a longitudinal vibration, while the changed potential adds a potential-dependent longitudinal force component for moving the ions. Compressing the electrolyte produces heat, as evidenced by a rise in temperature. The expansion consumes heat, as evidenced by a decrease in temperature. These are well-understood processes in thermodynamics. However, they are still unexplained in biophysics, although they were observed seven decades ago [28]. The elastic membrane functions as a damped oscillator, which, in its negative amplitude phase, reverses the direction of electrolyte flow (and, consequently, the direction of the current carried by the ions). In the discipline of electricity (when using the serial RC model), it can be interpreted as a capacitive current, which in physiology is observed as hyperpolarization that inverts the polarity of the output voltage. The ions represent both mass and charge simultaneously, and, correspondingly, one can observe, in addition to the electrical potential wave, mechanical, optical, and other features changing. In different disciplines, pressure and voltage represent the same phenomenon, using concepts from various disciplines. Both waves propagate with the speed of the carrier.

6.2 Heat emission/absorption

The early experimental observation [28], that positive and negative heat production is associated with a nerve impulse, suggested a thermodynamic model. In the framework of the classical theory, Ohmic currents flow through resistors that dissipate heat due to friction, no matter in which direction ($W = I^2 \times R$) the ion currents flow. The existence of "cooling", alone, undermines the credibility of the classic theory [42]. The issue here is that the dissipation was attributed to the "leakage current." As Hodgkin wrote: "Hill and his colleagues found [28] that an initial phase of heat liberation [of the action potential] was followed by one of heat absorption. [...] a net cooling on open-circuit was totally unexpected and has so far received no satisfactory explanation." [29] "All authors came to similar conclusions: during the action potential, no significant net heat is produced. Transient heat releases are mostly reabsorbed in the second phase of the action potential. ... The finding of a reversible heat release during the action potential of nerves is a striking and very fundamental fact. It is inconsistent with the HH model. The physics underlying the nervous impulse must rather be based on reversible processes". [30] At that time it was commonly known that compressing a medium produces heating and extending it causes cooling. Although modeling neuron as a thermodynamic engine that is heated and cooled, and produced no significant net heat, would be trivial, no such model came to light. The probable reason is that no reasonable idea shined up that could attach the observed heat absorbtion with the observed evident signs of the electrical operations.

Heat liberation and absorption are direct experimental evidence of the "slow current" and the mechanisms described above. The lack of leakage current not only validates our model, but also solves the long-standing mystery of reversible heat release during the AP. Pressure changes accompany the electrical process and naturally account for the observed temperature changes. The energy consumption of the nervous impulse underpins that the complex physical process comprises mostly reversible disciplinary processes: most of the energy of AP is stored as reversible elastic potential energy. Hence, its dissipation is also negligible: "no significant net heat is produced" [30].

The shocking conclusion in [37], that "brain cells communicate with mechanical pulses, not electrical signals", more precisely sounds that the elastic membrane generates a pressure wave in the volume of the membrane, and all components having mass (the electrolyte) vibrate "in place". The charged components (the ions, a 10^{-3} part of the electrolyte) repel each other (as long as there is excess charge on the membrane) toward the AIS and along the membrane. Actually, the disciplines of thermodynamics and electricity cooperate in both producing and transmitting AP. Neither of them can do that alone. No compromise is needed. The nervous pulse transmission over the axons also must be revisited. The classical mechanism based on coordinated action of ion channels is undoubtedly wrong. Among others, the physical conditions of applicability of the telegrapher's equations are not fulfilled, plus (see Appendix **??**) the electrical forces are orders of magnitude weaker than the elastic ones. The enormous pressure, the incompressible electrical fluid, and the repulsion of the delivered particles suggest an alternative and more reasonable mechanism. The resultant of the electrical and thermodynamic, and first of all, the mechanical force generated by the elastic membrane, does the task. No protein mechanisms and cooperating ion channels are needed.

Chapter 7

Unified model

Quotation:

”Breaking down topics into digestible chunks and explaining them helps you identify gaps in your knowledge. This creates new neural pathways in your brain, which makes connecting ideas and concepts easier.”
R.P.Feynman

We follow Feynman’s hints, and break processes into chunks, and we connect them by scientific assumptions if one must change science disciplines between chunks.

Chapter 8

Abstract neuronal components

Contents

8.1 Ion channels

Embedded in the wall of the membrane, a smaller amount of (controlled and uncontrolled) ion channels are scattered. Between the membrane and the axon, there is an array of a large amount of uncontrolled ion channels (the Axon Initial Segment AIS). The elastic wall of the membrane is covered with ion layers, and therefore, a potential difference is created there (the resting potential). The membrane represents *a confined space in which ions behave differently from in an infinite space*; furthermore, it significantly alters the properties of the few-nanometer-thick electrolyte layer near the membrane. In this layer, the concentration and potential change significantly with the distance from the membrane due to interactions between the membrane's ion layers and the electrolyte. Ion packages (spikes) sent by other neurons can enter the internal volume through controlled inputs (synapse).

Appendices

Appendix A

Modeling for neurons

Contents

Quotation:

"What I cannot create, I do not understand" R. P. Feynman

Quotation:

"Biological laws are much more difficult to find than physical ones" [6] @1995

A.1 Modeling methods

"There is a widely accepted distinction between merely modeling a mechanism's behavior and explaining it. . . . Some models are data summaries. Some models sketch explanations but leave crucial details unspecified or hidden behind filler terms. Some models are used to conjecture a how-possibly explanation without regard to whether it is a how-actually explanation."[43].

A.2 Disciplinary modeling

As Feynman R. P [44] wrote, "the separation of fields [of science] ... is merely a human convenience, and an unnatural thing. Nature is not interested in our separations, and *many of the interesting phenomena bridge the gaps between fields*". "When we go to investigate *we should not pre-decide what it is we are trying to do* except to find out more about it. ... The first principle is that you must not fool yourself, and you are the easiest person to fool." Physiology, despite HH's and Feynman's warning, pre-decided that only the science discipline 'electricity' is authorized to describe neurons' operation. However, electricity has no place for the mass and speed of its charge carriers. After seven decades, it should be realized, admitted, and fixed.

Appendix B

Physics for neurons

Contents

B.1 Introduction

Knowledge of the fundamental (physical) functioning of the biological neuron is essential for understanding its electrical processes and the excitation process, information processing based on signal transmission between neurons, and the low-level functioning of the brain. Although the fact that neurons show electrically measurable changes (i.e., belonging to the subject of electrical science) has been known since the beginning of instrumental studies, it was soon demonstrated that changes falling within the subject of thermodynamics and mechanics (i.e., another, separate discipline of physics) can also be measured. In the traditional view of physics, each discipline interprets only a certain range of quantities and can only derive relationships and laws between them. Nature, of course, does not care about the disciplinary nature of science, so science is essentially puzzled by the "miracle" that during the operation of a neuron interpreted as a purely electrical model, quantities falling within the scope of other disciplines (temperature, pressure) also change. The logic of things follows that then one must try to describe the experiences with the tools and laws of each discipline. In this way, it is possible to perform a "complete measurement", i.e. to experimentally measure all quantities that are related to the operation of the neuron, but no connection is established between the disciplinary laws and quantities.

Although it is obvious that all phenomena are related to the same physical process, and the quantities are clearly correlated, a qualitative step forward is needed that does not simply put together the disciplinary laws developed for inanimate science, but rather adapts them to the conditions of living matter and creates the necessary connections between them. The infinitely distant, infinitely small change-causing concepts of physics, and disciplinary separation, are certainly not applicable. The amount of electric charges changing their location during operation is of the same order of magnitude as the amount of free charges on the membrane surface, so we can only use Lotka-Volterra type differential equations to describe electrical changes (the usual use of "equivalent electric circuits" is incorrect and misleading), and furthermore, between the closed biological boundary walls of neurons, the interaction of ions on each other creates a counterforce on the wall and the change in the counterforce creates a pressure change in the electrolyte filling the cell. The elastic lipid layer forming the neuron wall undergoes a size change due to the pressure change (although it has a constant modulus of elasticity: during operation, the neuron's size of about 25 micrometers changes by only about 1 nanometer).

Although it is known from the beginning of scientific thinking that the material world around us comprises discrete objects (particles) that cannot be divided without limitations at a given abstraction level, most of the scientific laws were devised for a continuous material, such as the laws about pressure and current. Our laws about mass and charge (and so: on energy, momentum, and so on), are about continuous concepts. In our everyday practice, the discrete nature of the matter that is considered to be continuous, causes no trouble; the laws abstracted in the continuous view perfectly describe our phenomena. Thermodynamics was the first discipline which connected two of those separated attributes, the discrete and continuous views. Similarly, a new discipline was born when Planck had to hypothesize that energy also has discrete constituents. We must follow a similar way of thinking when discussing phenomena of a very small amount of discrete objects in limited volumes of biological matter.

Furthermore, science has created abstract concepts such as space and time, mass and charge, continuous and discrete views. *Classical science considered only one such concept at a time.* Classical disciplines can discuss only one abstracted attribute of ions: classical mechanics can discuss its mass and classical electricity its charge. Neither of the mentioned disciplines, alone, can describe systems containing ions. Essentially, the electricity, thermodynamics and mechanics of ions in closer resemblance with the mentioned non-classical isciplines than with a classical one. Because of this, one can take over laws from the latter ones with great care; especially what wanting to apply them to the special structures of living matter.

The roles and interpretations of time, space and matter are subjects of endless debates in science. Considering finite interaction speeds is against using a "nice and classical physics" with its nice mathematical formulas, but omitting the different speeds misled and may mislead research in several fields. Biology produces situations where the complexity of phenomena and the needed carefulness meets the ones needed in cosmology. The difference is that, in biology, the consequences of phenomena are immediate and they can be studied experimentally.

To describe the related phenomena, we must scrutinize, case by case, which interactions are significant and which interaction(s) can be omitted; instead of setting up ad-hoc models for living matter that contradict each other and fundamental scientific principles when used outside their narrow range of validity. To provide their correct physics-based description, we must understand the corresponding behavior of living material, including that it works with slow ion currents, electrically active, non-isotropic, structured materials, and consequently, its temporal behavior (the speeds of interactions) matters. We must consider macroscopic and microscopic phenomena (continuous and discrete features) at the same time, different science fields, and their interplay (or mixing). "Living complex systems in particular create high-ordered functionalities by pairing up low-ordered complementary processes, e.g., one process to build and the other to correct". [21] We must check the validity of our abstractions.

First of all, we introduce (and, in this aspect, correct the current common understanding) that neurons' "output product" [?] *is created and transmitted by thermoelectric processes instead of net electric phenomena or net thermodynamic*. Feynman was right, also in this point: "*nature is not interested in our separations, and many of the interesting phenomena bridge the gaps between fields.*" [44] We derive the correct mathematical and physical handling processes, not considering the disciplinary boundaries, pointing to where the disciplinary omissions and approximations are not valid, deriving the required correct handling of the physical processes together with with the mathematics required to handle them. We use the theory developed in section **??**, where (among others) the time derivatives of the processes are introduced, in this way enabling the quantitative mathematical description of the neurons' abstract operation, that serves as a basis for discussing quantitatively the neural computing processes (as we discuss in section **??**, these derivatives are the laws of motion of ions) and their information handling. However, these corrections must be carried out in disciplinarily separated chapters.

As Schrödinger and Feynman implicitly suggested, we revisit the approximations that led to classical physics (where we derived the well-known 'ordinary' laws). We scrutinize the "construction" and "working" of living matter in a cross-disciplinary way, using non-ordinary approximations and abstractions. Then, laws based on the same first principles in a different approximation can describe living matter. We show that nature and its observable phenomena are complex; it is pointless to dispute which of the scientific disciplines describes neuronal behavior. The non-disciplinary approach shows that the firm and fast electric interaction sufficiently well describes neuronal operation after introducing the equivalent "thermodynamic electrical potential".

B.2 Schrödinger on life

In this chapter we reinterpret some concepts of physics to show that science can describe living matter. However, we must derive the appropriate approximations from the first principles of science and construct the correct form of laws for the conditions of living matter, instead of taking over the laws constructed by using different approximations from the first principles for the inanimate matter.

Quotation:

"Is life based on the laws of physics?"..."*New laws to be expected in the organism*"...E. Schrödinger: What is life?[1] @1944

Physics panel B.2.1:
Schrödinger on 'non-ordinary laws'

"From all we have learnt about the structure of living matter, we must be prepared to find it working in a manner that cannot be reduced to the ***ordinary*** **laws of physics**. And *that not on the ground that there is any 'new force' or what not*, directing the behavior of the single atoms within a living organism, *but because the construction is different from anything we have yet tested in the physical laboratory.*"..."How can the *events* in *space and time* which take place *within the spatial boundary* of a *living organism* be *accounted for by physics* and chemistry?" ..."living matter, while not eluding the 'laws of physics' as established up to date, is likely to involve 'other laws of physics' hitherto unknown, which, however, once they have been revealed, will form just as integral a part of this science as the former." (for the interpretation of those points, see **Physics panel B.2.2**) E. Schrödinger: What is life?[1] @1944

Physics panel B.2.2:
Schrödinger's points on living matter

In his very accurately formulated question (see **Physics panel B.2.1**, Schrödinger focused on (at least) these significant points

- '*events*' Unlike non-living matter, *living matter is dynamic, changing autonomously by its internal laws*; we must think differently about it, including making hypotheses and testing them in the labs. Those laws are non-ordinary 'because the construction is different', but its principles must not differ from the ones we already know.

- '*space and time*' Those characteristic points are significant changes resulting from processes that have material carriers, which change their positions with finite speed, so (unlike in classical mechanics) the events also have the characteristics 'time' in addition to their 'position'. In biology, the spatiotemporal behavior is implemented by *slow ion currents.* In other words, instead of 'moments' we must consider 'periods' and define the 'events' that define the beginning and the end of those periods.

- '*within the spatial boundary*' Laws of physics are usually derived for stand-alone systems, in the sense that the considered system is infinitely far from the rest of the world; also, in the sense that the changes we observe do not significantly change the external world, so its idealized disturbing effect will not change it. The 'construction' also results in constraint forces. Furthermore, we must consider the issue of finite resources.

- '*accounted for by physics*'[by extraordinary laws] We are accustomed to abstracting and testing a static attribute, and we derive the 'ordinary' laws of motion for the 'net' interactions. In the case of physiology, nature prevents us from testing 'net' interactions. We must understand that some interactions are non-separable, and we must derive 'extraordinary' laws. The forces are not unknown, but the known 'ordinary' laws of motion of physics are about single-speed interactions.

- '*living matter*' To describe its dynamic behavior, we must introduce a dynamic description. Processes happen inside it, and we can observe some characteristic points. The internal processes also manifest in changing physical quantities which are inseparable from the ones the experimenter intended to change.

- '*yet tested in the physical laboratory*'[including physiological ones] We test those 'constructions' in laboratories, in their actual environment, and in 'working state'. The measurement represents an intervention into their internal processes that falsifies the measurement results. As we did with inanimate matter, we must develop and gradually refine the testing methods as well as the hypotheses. Not only in measuring them (including measurement design and evaluation) but also in handling them computationally, we need more or less different thinking and algorithms.

B.3 Science concepts

Science has created abstract concepts such as space and time, mass and charge. It based its laws on abstractions/idealizations that were concluded by approximations and contained *only one* of these concepts. Inventing non-independence of space and time led to creating the theory of relativity. Under certain circumstances one must treat particles as a kind of continuous wave, and continuous waves as a particle. All those inventions had far-reaching consequences. These micro-objects interact with their environment, including our measuring instruments. With a measurement method suitable for particle detection, we see them as particles, and with an instrument suitable for wave measurement, we see them as waves. (How one micro-object sees the other remained cryptic.) This kind of micro-level behavior is inherent in nature; our methods of description and measurement must be adapted to nature and not the other way round. We understood that the Newtonian approach has its limitations; finite-velocity interactions can only be described by using a different kind of mathematics: by introducing the concept of space-time. In another approach, mass is also related to space and time; taking this into account, we can describe nature with curved space-time. We can also say that according to classical physics, we can describe phenomena with sufficient accuracy using only the aforementioned concepts (the zeroth derivatives). According to modern physics, however, for a more accurate description, we must take into account the first derivative with respect to time, and even the second derivative with respect to time for a general description.

Science's first principles could serve as a firm base for all its disciplines. As we discuss, *its disciplines use abstractions based on limited-validity approximations* based on the same first principles. However, *the approximations are different even for science's different disciplines, and even more for biology and physics.* In physics, most processes we observe are fast enough so that we can use the abstraction that they are essentially jumps between states. In most cases, the approach can be –more or less– successful. Notice, however, that in the case of a jump we neglect the need to explain why and how the transition happens. For the slower, well-observable processes, we have the laws of motion that describe how the processes happen under the effect of some driving force. We also experienced that nature is not necessarily linear (in the sense that it depends only on space coordinates but not on their derivatives), which we can describe by "nice" mathematical formulas. A century ago, A. Einstein invented that the approximations I. Newton introduced two centuries earlier are not sufficiently accurate for describing the movement of bodies at high speeds. In other words, a new paradigm, the constancy of the speed of light (today: the maximum speed of interactions), must have been introduced that caused a revolution in physics and led to the birth of "modern physics".

Classical vs modern science concepts

Science, unfortunately, is separated into classical and modern science based on whether the theoretical description assumes infinitely fast interaction (the Newtonian model) and continuous quantities or acknowledges the finite interaction speed (the Einsteinian model) or acknowledges the discrete nature of some quantities or connects the discrete and continuous views. Experience shows that the generated forces, regardless of their origin, can be summed, and one can apply Newton's laws of motion. *The finite interaction speed is erroneously associated with the speed of light and frames of reference moving relative to each other with speeds approaching the speed of light.* Assuming that the interaction speed is finite is sufficient to build up the special theory of relativity [**?**] (using the speed of light as the value for its external parameter). Still, the Minkowski-mathematics [**?**] behind the special theory of relativity works with any speed parameter c. The same mathematics describes technical [22] and biological [**?**] computing systems, where there are no moving reference frames, but the finite interaction speed has noticeable effects on the operation of the system. Furthermore, connecting the discrete and continuous views needs care in the case of biological objects and processes.

The so-called modern physics was born when experiments began to contradict the idea that all phenomena of nature could be described in such a simple way. Neuroscience has similarly accumulated a number of experiences that contradict theory, although mainstream neuroscience has either ignored them or explained them with ad-hoc flawed and unfounded hypotheses. It happens despite Hodgkin's opinion about the unexplained observations. In this field (and in the physical foundation of biology in general), a similar modern approach is needed, which first requires reviewing the fundamental assumptions of the underlying physics.

Science disciplinarity

Science disciplines attempt to discover the infinitely complex nature in a disciplinary approach. To describe a *well defined* range of phenomena, we use approximations and omissions, and we create abstractions which can then be described by known laws using the universal language of mathematics. Newton's laws deal with masses only. Coulomb's laws deal with charges only. We use the abstraction "charge" and "charge carrier" for electrons, protons, ions, etc., and we can describe the electricity-related abstract features of the carriers. The charge has no mass, but the charge carrier does. We must not forget, however, that *those laws have been derived based on approximations and omissions*, and so they also *have their range of validity.* To apply laws from different fields of science, we must scrutinize whether all laws we use are applied within their range of validity. "Biological laws are much more difficult to find than physical ones" [6]. We add that mainly because those laws cannot be uniquely mapped to the disciplines of physics. The charge and mass can be separated conceptually, but not physically.

Classical physics classified phenomena into disciplines that largely described nature well. Modern physics has revealed new relationships, but they have also organically integrated into the fabric of science as a whole. To understand the living organism, in the spirit of R.P. Feynman, we must make "excursions" between disciplines. E. Schrödinger expressed his conviction that no new force or unknown interaction is emerging; only a previously unknown regularity concept must be found, then *the non-ordinary laws organically integrate into the fabric of science, together with the already known laws.* In this chapter, we make an attempt to find such a concept in the hope that scrutinizing the features of ions leads to a solution.

Regarding ions, separately, we can handle their charge and mass well, we know the relevant laws. Their properties are analogous to the particle-wave duality: depending on the interaction (i.e. the measurement method), we see an ion as either a charge or a mass. *We see the same object, but we measure it based on the concepts of different disciplines.* Therefore, we must connect charge and mass, just like space and time in the case of relativity. They do not exist independently of each other and their combined behavior differs from what we expect based on the classical disciplinary approach. Connecting two abstract notions generally result in counter-intuitive experiences. Interestingly, similarly to relativity, their velocity establishes the connection between the two quantities here.

Furthermore, we must introduce a connection for the currents represented by ions, such as the one introduced by statistical mechanics when connecting the discrete and the continuum approaches. However, we base the correspondence on geometrical considerations and, due to the unseparable connection between the two properties, we cannot apply statistical laws based on one property in an unchanged form. Moreover, in most biological cases, the number of particles involved are so small that their statistical interpretation is severely limited. *The fundamental approximation of physics that the force field acts on the particle, but the particle's effect on the field is negligible, is not met.* In some cases, closed spaces give rise to counterforces that fundamentally change the processes that take place. *Ignoring these properties of living systems has led to the erroneous statement that science cannot describe living matter and its processes.* Describing living matter is undoubtedly more complicated than describing inanimate nature and requires a revision of several approximations. However, it is possible. *Physics is much more than a collection of formulas concluded for inanimate matter and valid within the frames of a discipline that can be applied without thinking about whether they can be applied to the conditions of living matter.*

Locality and finite resources

Quotation:

> "How can *the events in space and time* which take place ***within the spatial boundary*** of a living organism be accounted for by physics and chemistry?" [1]

The closed volume of the biological objects and the finite speed of the charge carriers is also a problem of finite resources. In biology, we cannot use the fundamental assumption of physics that, although a field acts on the ion in question, the ion does not affect the field (the other ions generate). The material transport represents simultaneously mass and charge, so the transport itself gradually changes the gradients (the field itself). The transferred ions decrease the field in the volume they departed from and increase it in the volume where they arrived (a problem of finite resources); furthermore, the concept of 'field' is slightly different from the classical one, see below. We must consider that the resources are finite. A main source of confusion is that the phenomena happen in a limited and closed region of space (*within the spatial boundary*), and we study processes (in a period instead of a moment) where the environment is not "infinitely far" from the studied object (*the studied process interacts with its environment*), and the forces exerted on it belong to different disciplines. The boundaries of a closed volume may exert constraint forces and fundamentally affect the processes inside the volume.

Due to the field-dependent speed within the electrolyte, we must consider the autonomous change between the microvolumes where the ion traverses. This process keeps the entire volume of the electrolyte in a (more or less intense) continuous change, which makes life possible. The more distant parts of the biological cell will see any change in the local values of the state variables with a delay. Furthermore, biological objects inside the cell can absorb ions and charge up. With their potential, they alter local gradients, accelerating or decelerating ions. Not to mention that biological objects can be active in the sense that (depending on the environmental conditions) they can let ions from one separated volume part into the other, thus changing the number of charge carriers in the volume under study. Co-existing processes may affect each other by changing the available resources: changing one quantity changes the other. Mathematics descibes such processes by linked differential equations, such as the Lotka-Volterra (or "predator-prey") equations. For details, see section **??**.

These processes are what E. Schrödinger coined as "*the construction is different from anything we have yet tested in the physical laboratory*" [1]. Consequently, measurements must be designed and carried out with care; the routine methods used for measuring objects from inanimate nature, cannot surely be applied in unchanged form to living objects.

Fields in science

The concept of force fields is fundamental for science since I. Newton introduced that remote objects can interact with each other. Traditionally, a field is defined by the force a test charge (or mass) would experience at a specific point in space, even if no charge/mass is currently there (assuming an infinitely small charge or mass, despite that we know their discrete nature). R. P. Feynman questioned whether fields were fundamental physical entities or simply convenient mathematical descriptions of the "condition of space". He cautioned against "mistaking the map for the territory," suggesting that while (electric and magnetic) fields are useful, they might not be the ultimate reality. Anyhow, the fields are created by nearby ions which can change their positions during the interaction, furthermore, the ions under test interact with them and this way change their environment. Although the fundamental interactions forces and laws are the same that we are used to when testing objects from the inanimate nature, the "construction" is entirely different. This difference needs much attention because the laws concluded for objects of inanimate nature are not valid for objects in living nature in unchanged form.

Mass and charge may have the corresponding fields given that they have well-known remote interactions. However, thermodynamics does not consider either gravitational or electrical fields: the interaction are implemented by short-range interaction (direct collisions). Although there are theories for both fields where (typically virtual) particles represent the action of the field, those particles are not identical with the sources of that field. To extend the concept of field to thermodynamics, one must extend the idea of deterministic forces with stochastic forces; so the resulting 'field' is of stochastic nature. In the case of ions in electrolytes (particularly in structured biological matter), they move under the effect of electric and thermodynamic fields. In their laws of motion, one has to add electrical and thermodynamic fields, i.e., a deterministic and a stochastic field. The result is stochastic, anyhow. The case is further complicated by the fact that the propagation speeds are different for the two cases. One can tell only stochastically valid statements and must derive special laws of motion (say, for the selectivity of ion channels; see section D.4.3).

B.3.1 Dichotomies in modeling

For centuries, science developed its methods for deriving abstract concepts by reducing the features of a real object to an abstract (idealized) one that cannot be reduced further, such as mass or charge. Ions are an exception when using the concepts of classical disciplinary physics: *the ions are charge and mass simultaneously, without a further possibility of reduction.* The consequences of this item are listed in Table 1 of paper [9] as the "Electrical vs Mechanical" dichotomy. Science has derived laws for the forces acting on those abstract objects, such as Newton's universal law of gravitation and Coulomb's law of electricity. Then one could apply Newton's laws of motion. Experience shows that the generated forces, regardless of their origin, can be summed, and one can apply the laws of motion using the resultant force (see below for a discussion of the resultant force and its application in biology). However, notice that those experiences are based on the Newtonian 'instant interaction', i.e., that the interactions' speeds are equal.

Paper [9], which, in the context of this section, tends to be the precursor of our ideas, sheds light on some remarkable dichotomies, scrutinizing which can result in "a sound basis for unification of the physics of nerve impulses". By deeply agreeing with that they have a "potential impact on our understanding of (the physical nature of) neuronal signaling", we list and uncover more dichotomies; furthermore, we uncover their fundamental, partly philosophical reasons.

B.3.2 Complete measurement

In physics, measurement means quantifying something relevant to the process under study. A 'complete' measurement measures all relevant quantities. *The different disciplines of physics restrict the measured quantities to the ones which are, in general, that the discipline studies. The remaining quantities remain outside the scope of the discipline.* The fundamental physical quantities of mechanics are length, mass, and time, which form the basis for defining all other quantities like velocity, force, and energy in that field. The fundamental physical quantities in thermodynamics are Temperature, Energy (Internal Energy, Heat, Work), and Entropy, which characterize systems at equilibrium and describe energy transformations; Pressure and Volume are also key variables. Those of electricity include electrical charge (Coulomb), the basis of all electrical phenomena; electrical current (Ampere), the rate of charge flow; voltage (Volt), the potential difference driving current; resistance (Ohm), opposition to current; power (Watt), the rate of energy transfer; and energy (Joule), the capacity to do work. As seen, there is little overlap of the studied quantities between the disciplines.

When making a measurement, one must keep in mind that measuring devices are designed for the purposes of determining a quantity interpreted in one of the disciplines of science. It is the task of the experimenter to figure out which physical quantities (that maybe interpreted in different disciplines) play a role in the process and to define the set of quantities that describe the process, instead of pre-deciding which discipline's set of inanimate science will be used for biology which is not one of the disciplines of inanimate science. In biology, that set does not necessarily matches any of the sets used in one of the disciplines. *A disciplinary study (such as the electrical and thermodynamic ones in physiology) provides an incomplete set of measured quantities.* None of the disciplines alone can describe electrolytes, since they consider different and incomplete sets of physical quantities: thermodynamics misses charge, electricity misses mass. As discussed in connection with the Onsager's relations, a disciplinary (incomplete) measurement does not discover that an unexpected change (an unexplainable "miracle") in the value of one of the quantities belonging to another discipline is accompanied by a change in the value of the other quantity. This change is clearly the case with the measurements performed in the spirit of HH: the values the electrical instruments provide are accompanied by the values of mechanical/thermodynamic quantities which were not measured, partly due to the obvious measuring difficulties and because they are outside the scope of the discipline electricity. The theory, however, should not remain disciplinary. If one attempts to reduce ion-related phenomena to a single abstraction (see the mentioned theoretical descriptions), one experiences that some quantity (the charge or the mass) changes in an uncontrolled way. The fundamental reason of the incompatibility on the disciplinary theories is that they consider an incomplete set of physical quantities. The disciplinary separation in classical science does not apply to the study of living matter. *We must make a complete measurement and interpret them in a cross-disciplinary way.*

Thermodynamics provides a framework for handling ions and determining their thermodynamic properties. However, as good thermodynamic textbooks (including the one on the thermodynamics of the membrane [30]) emphasize, *thermodynamics derives its concepts for non-interacting particles*, so one cannot expect its validity for ionic solutions [17] (Boltzmann assumed that, in the absence of long-range interaction between the particles, the sizes of cells in the phase space do not change). In addition, he required the presence of a vast number of particles in a homogeneous, isotropic, infinite volume, which is typically not the case in biology.

Electrical vs Mechanical

In the electrical abstraction, no mass is present, so one can use the equations assuming 'instant interaction' (aka an infinitely large interaction speed, and, consequently, an infinitely large acceleration), which leads to non-physical explanations of the physiological observations (such as 'delayed current'). In the mechanical abstraction, mass is to be moved, making the finite-speed interaction evident. The same physical phenomenon, the interaction (or movement) of ions is described using an 'infinite' electrical speed and a million times less mass propagation speed, respectively and simultaneously, which leads to an unresolvable discrepancy, given that physics is not prepared to handle different speeds in the same interaction event [17]. This item is listed in Table 1 in [9] as the "Electrical vs Mechanical" dichotomy.

To deliver a current, one needs moving charged particles that need exerting of some external (electrical, magnetic, or chemical) force or a mixture of forces. We have speeds of EM interaction, thermal motion of the charge carriers, macroscopic current, drift speed, and their mixing, simultaneously, in the same phenomenon. In the theoretical description of processes, instant interaction (i.e., the abstraction of non-physical, infinitely large interaction speed) is used in most cases, which is a good approximation only if the charge carriers are electrons. In the cases when absolutely needed, the generic notion of "speed" is used without specifying which one of the mentioned speeds it means.

The low speed of ions in electrolytes introduces further problems. Any change in the local value of the state variables will be seen by the farther parts of the cell with a delay, The material transport represents simultaneously mass and charge, so the transport itself changes the gradient. This process keeps the entire volume of the electrolyte in (more or less intensive) continuous change. Furthermore, the biological objects inside the cell can absorb ions and charge up. With their potentials, they change the local gradients, accelerating or decelerating the ions. Not to mention that biological objects can be active in the sense that (depending on the environmental conditions) they can let ions from one separated volume part into the other. The same physical phenomenon, the interaction (or movement) of ions is described using an 'infinite' electrical speed and a million times less mass propagation speed, respectively, which leads to an unresolvable discrepancy, given that physics is not prepared to handle different speeds in the same interaction event [17].

Atomicity

Although science is aware that the apparently continuous matter in nature cannot be divided infinitely, even though it knows that the abstracted discrete 'material points' (as A. Einstein coined) have well-defined discrete values, it occurs rarely that both views must be applied in describing a single phenomenon. The continuous and discrete approaches (also called macroscopic and microscopic views) seem to be independent of each other. Connecting those views was one of the tasks performed by thermodynamics for particles with no long-range interactions. Inventing discreteness of energy led to quantum mechanics. Connecting the features of the microscopic discretete constituent objects to the macroscopic continuous features led to thermodynamics. However, there is no similar discipline for electricity, although the behavior of charge carriers is similar to that of neutral discrete particles. It is not evident for electrons, but it is significant for ions. This item is listed in Table 1 of paper [9] as the "Macroscopic vs Microscopic" dichotomy.

Reversibility

The third dichotomy is "Reversible vs Irreversible". A reversible process can proceed in either direction in time. Cellular operation seems to be cyclic (i.e., mostly reversible) in the short run, while in the long run, spanning from birth to death, it is irreversible. Experience suggests that a mixture of reversible and irreversible processes describes neuronal operation. As reviewed in [9], in contrast with the electricity-centered disciplinary conception HH model [14], all other disciplinary models discussing phenomena suggest reversible operation.

Reversibility seems to be closely related to the dichotomy "Electrical vs Mechanical" and to the nature of the exclusively used electrical HH theory. In the framework of that theory, Ohmic currents flow through resistors that irreversibly dissipate heat due to friction, no matter in which direction ($W = I^2 \times R$) the ion currents flow. By introducing the concept of 'delayed current' and the hypothesis that some hidden power controls the operation of neurons by altering their conductance, physiology gave rise to the fallacy that science and life sciences are almost exclusive fields. Furthermore, their (unintended) model provokes questions (for a review see [12]) whether it is a model at all, and what controversies it delivers. As Hodgkin wrote [29]: "Hill and his colleagues found [28] that an initial phase [of the action potential] was followed by one of heat absorption. [...] a net cooling on open-circuit was totally unexpected and has so far received no satisfactory explanation." Since that discovery, experimental evidence from the missing "leakage current" [47] to the conversion between elastic and kinetic energies [33] to demonstrating the pressure-wave-like behavior of AP witnesses that at least part of the phenomenon is of thermodynamic origin; that is, it requires a cross-disciplinary explanation. For discovering the reciprocal relations in thermodynamics [**?**], also in electrolytes, Lars Onsager was awarded the 1968 Nobel Prize in Chemistry. Those relations, together with the theoretical understanding, manifesting in the cross-disciplinary Nernst-Planck relation, paved the way to a combined theory. However, no thermodynamic explanation has been given so far. Our discussion provides, in line with the experimental results, a complex unified electrical/thermodynamic description of neuronal operation.

B.3.3 Speed

The role of speed and time, particularly in the context of an object's changing location over time, has long held a mystique in the realm of scientific discovery (and recently returned to be mystic again in cosmology). This intrigue can be traced back to historical debates, such as Zeno's paradoxes. The acknowledgment that an object's movement speed can influence our observations is a topic that has sparked significant scientific discourse over the years. In this section, we aim to draw parallels between the historical debate on the finite speed of light and its contemporary implications in various scientific disciplines, such as the finite speed of ionic currents in biology.

It has been a long-standing mystery that interactions with different speeds play their role *simultaneously*. The issue forces researchers to give non-scientific explanations to everyday phenomena only because *they routinely assume that the interactions have the same speed, and they use the laws about strictly pairwise interactions*. They have no choice: there is no formalism to handle non-equal speeds.

We need different abstractions (finite-speed interaction in modern physics) for different phenomena that require different mathematical handling, which is not as simple and friendly. The *speeds of observation* and propagation of electric fields remain the same in biology, and it is easy to extrapolate, mistakenly, that *all* interactions have infinitely large interaction speeds. However, also *slow* interaction speeds exist, furthermore, *different interaction speeds can intermix in the same phenomenon. Neglecting that effect introduces the need to assume fake mechanisms and effects* for explaining some details; which are naturally explained by assuming finite interaction speeds and their combinations. More discussion follows in section **??**.

Instant Interaction

The above dichotomies are rooted in deeper layers of science. Notice that in the electrical abstraction, no mass is present, so one can use the equations assuming 'instant interaction', which in biology led to using non-physical explanations of the observations (such as 'delayed current'). In the mechanical/thermodynamic abstraction, mass is to be moved, making the finite-speed interaction evident. Classical physics is based on the Newtonian idea that space and time are absolute, so everything happens simultaneously. Consequently, when nature's objects interact, it must be instantaneous; in other words, their interaction speed is infinitely large. Furthermore, electromagnetic waves with the same high (logically, infinitely high) speed inform the observer. This self-consistent abstraction enables us to provide a "nice" mathematical description of nature in various phenomena: the classical science.

We learned that the idea resulted in "nice" reciprocal square dependencies, Kepler's and Coulomb's Laws. We discussed that the macroscopic phenomenon "current" is implemented at the microscopic level by transferring (in different forms) discrete charges; furthermore, that solids show a macroscopic behavior "resistance" against forwarding microscopic charges. We also learned that *without charge (and, without atomic charge carriers), neither potential nor current exists.* We did not learn, however, that *also thermodynamic forces can move the ions*, given that they have inseparable mass.

The low speed of ions in electrolytes introduces problems. The same physical phenomenon, the interaction (or movement) of ions is described using an 'infinite' electrical speed and a million times less mass propagation speed, respectively, which leads to an unresolvable discrepancy, given that physics is not prepared to handle different speeds in the same interaction event [17]. (Historically, Onsager's work in 1931 referred to thermoelectricity and transport phenomena in electrolytes, thus connecting experimentally two separate disciplines.)

Physics notoriously suffers from the lack of handling different simultaneous interactions; facing such a case leads to misunderstandings, debates and causality problems. Such a famous case is the interaction speed of entanglement. In that time, E. Schrödinger introduced his famous Law Of Motion into quantum-mechanics entirely analogously as I. Newton introduced his Laws of Motion. Similarly to the Newtonian 'absolute time', the quantum mechanical interaction is supposed to be 'instant' (this is the price for having 'nice' equations in classical and quantum mechanics), i.e., its speed is supposed to be infinitely high. However, at that time was already known that the electrical interaction (propagation of electromagnetic waves) is finite, so if an object has quantum mechanical interaction (aka entanglement) and electrical interaction, the corresponding forces start at the same time, but arrive at the other object at different times. The entanglement arrives instantly, the electromagnetic effect arrives at the time we can calculate from the interaction speed and the spatial distance of the objects. This leads to causality problems: the effect of the two interactions of photons entangled earlier in an exploded supernova should be measured at two different times; meaning a "spooky remote interaction" as A. Einstein coined, and leads to contradictions such as the Einstein–Podolsky–Rosen paradox. Actually, the issue roots in the improper handling of mixing interaction speeds: the Schrödinger-equation introduces the infinitely large interaction speed, while the EM interaction has finite speed.

The confusion and question marks in connection with describing the life by science mostly arise from the interpretation of notion 'speed' in physics. When discussing the underlying physical laws, we go back to the very basic physical notions instead of taking over the approximations and abstractions used in the *classical physics for non-biological matter* and less complex interactions. As we emphasized many times, we construct laws and conclusions based on somewhat simplified abstractions about nature, in all fields of science. The notions and laws depend on the circle of phenomena we know and want to describe. The Newtonian and Einsteinian worlds are basically distinguished by considering their *speed dependence* that actually means *explicite time dependence*. Interesting consequences are that in the Einsteinian world, the mass is not constant, the time and space are not absolute, and so on. We can be prepared to some similar counter-intuitive experiences in physiology: "we must be prepared to find it working in a manner that cannot be reduced to the **ordinary** laws of physics"[1]. Here we scrutinize the basic notions and discover some differences between physics and biology as consequences of the required different abstractions and approximations.

During our college studies, we mentioned that light is an electromagnetic wave with a vast but finite propagation speed. Still, we forgot to highlight that, at the same time, it is the propagation speed of the electric (, and magnetic, and gravitational) interaction force fields as well. The effect of "Retarded-Time Potential" is also known in physics and communication engineering. Algorithms "marching-on-in time" and "Analytical Retarded-Time Potential Expressions" are derived to handle the problem; for a discussion, see [**?**]. The Telegrapher's equations (unfortunately, also used to describe biological signal transfer) explicitly assume a finite propagation speed millions of times slower than the (implicitly assumed) EM interaction's. The issue is not confined to large distances: designers of micro-electronic devices also must consider the effect: they introduced clock time domains and clock distribution trees; see, for example [**?**, 22].

Science uses 'instant' in the sense that one interaction is much faster than the process under study; we consider the faster interaction as instant. The approach of classical science is based on the oversimplified approximation that the interaction speed is *always* much higher than the speed of changes it causes and that the processes can *always* be described by a single stage. In our approach, for biology, we put together a *series of stages* to describe the observed complex phenomena, where the stages provide input and output for each other, involve more than one interaction speed, and use per-stage-valid approximations. We simplify the approximations by omitting the less significant interactions and introduce ideas for accounting for the different interaction speeds. This way, we reduce the problem to a case that science can describe mathematically. *This* *procedure is fundamentally different from applying some mathematical equations derived for an abstracted case of science to a complex biological phenomenon without validating that we use the appropriate formalism.*

Speed of light

In 1676, the Danish astronomer Ole Rømer was making meticulous observations of Jupiter's moon Io and concluded not only that the speed of light is finite, but he measured its value with a sufficient accuracy. Rømer never published a formal description of his method, possibly because of the opposition of his bosses, Cassini and Picard, to his ideas. Cassini knew Rømer's idea and the measurement data. However, instead of accepting the finite value of the speed of observation, he made periodic corrections to the tables of eclipses of Io to take account of its irregular orbital motion: *periodically resetting the clock* . The speed of light must remain infinitely large.

However, the theory of finite speed quickly gained support among other natural philosophers of the period, such as Christiaan Huygens and Isaac Newton [**?**]. Although Newton surely knew that the observation speed was finite, in his "Philosophiae Naturalis Principiai Mathematica" [**?**], published in 1687, he decided to refer to observations that they happen "at the same time" despite knowing that what we observe at the same times, happen at different times. Using instant interaction results in "nice'" mathematical laws and enables us to describe most of nature's experiences with sufficient accuracy. Einstein, in 1905, discovered [**?**] that the speed of observation (in moving reference frames) may play a decisive role in interpreting scientific phenomena. The results he derived using Minkowski-coordinates [**?**] were counter-intuitive, with many unexpected consequences. Instead of introducing improvement(s) or correction(s) to the existing classic principles and methods, he introduced a new principle: the finite (limiting) interaction speed. The *disciplinary analysis of the reception of Minkowski's Cologne lecture reveals an overwhelmingly positive response on the part of mathematicians, and a decidedly mixed reaction on the part of physicists* [**?**] has turned to the exact opposite. Today, physics generally accepts the description, that is, the existence of finite interaction speed (resulting in the birth of a series of modern science disciplines). However, other science disciplines, including biology and computing science, refute (or at least do not use) it; despite that its effects are evident.

Speed in neuroscience

Helmholtz, in 1850, sent a short report off to the Academy [**?**] "I have found that a measurable time passes when the stimulus exerted by a momentary electrical current on the hip plexus (Hüftgeflecht) of a frog propagates itself to the nerves of the thigh and enters the calf muscle." His teacher "had thought that the speed of nervous conduction might be in excess of the speed of light and could probably never be measured. Helmholtz's father, on hearing of the experiment and the surprisingly slow measured speed, wrote to his son that he would as soon believe this result as that one can see the light of a star that burned out a million years ago" [?].

With the development of measurement technology, it became evident that finite speed is a general feature of the "nervous connection". (Somehow, "the speed of nervous conduction" has been renamed to "conduction velocity", neglecting the clear distinction that physics makes between the two wording.) With the dawn of instrumental electronics and computing, the McCulloch-Pitts model [?] introduced the picture that the brain can be modeled by a network of simple perceptron nodes connected by wires; that is, it comprises a two-state equipotential membrane connected with perfect wires. The experimental research also quickly (re-)discovered that those wires forward signals in a particular way; the speed of the potential wave is finite. Furthermore, *the axons are not equipotential during transmission.* Although its structure is practically identical with axons, *biology assumes that, unlike an axon, the membrane remains equipotential during its operation, although the evidence shows the opposite*: 'the action potential spreads as a traveling wave from the initial site of depolarization to involve the entire plasma membrane' [?].

When seeing that assuming an equipotential membrane was wrong and a single equipotential surface (in other words, classical physics' instant interaction) cannot describe neurons adequately, multi-compartment models (typically comprising equipotential cylinders with different potentials) have been introduced [?]. (Notice that it is a consequence of the wrong oscillator model hypothesized by Hodgkin an Huxley: the membrane is modeled as a series of resistors and capacitors.) Then (forgetting that Ohm's Law is valid only for the classical physics's 'instant interaction', furthermore, that no external potential is connected to either of the compartments, and no charge is present at the beginning, except at the input of the first compartment), the individually equipotential compartment pieces were connected by individual resistors. This model shows that the more compartments, the better the agreement (accuracy) with experiments. It happens because the shorter is the size of the compartment (approaches a differential equation), the less noticeable is the deviation from the true non-equipotential surface. This conclusion means that charging the capacitance attached to the compartment takes time, resulting in a delayed distributed current. Using infinitely many compartments, we would arrive at the differential equations describing a delayed distributed current on the surface of the non-equidistant membrane. However, biology did entirely not give up its position. It admitted that membrane current exists, but only between compartments, and its speed must be infinite (or, at least, the speed of EM interaction). However, at least the compartment pieces must be equipotential. Instead of fixing the wrong hypothesis, biology is "periodically resetting the clock". Instead of accepting that the charged ions represent a "slow current" (compared to the "fast current" represented by electrons and their charge transmission method), biology introduced changing conductance, delayed current, rectifying current, and so on; effectively blocking the understanding of the neural operation.

Finite-speed interactions

When speaking about speed, especially about the speed of charged objects inside biological objects, one needs to consider microscopic and macroscopic levels of understanding. On the boundaries of the two levels, we need to make distinctions between different kinds of speeds, among others (in units of m/s), the propagation speed 10^8 of the electric field (aka potential gradient), the speed 10^5 of thermal motion and potential-accelerated motion, the apparent speed of current (potential-assisted speed of a macroscopic stream, both in metals and electrolytes; mainly due to the repulsion of nearby ions in the stream) 10^1, for ion current inside a neuron (see Fig. 1 in [**?**]) 10^{-2}; diffusion speed of electrons in a wire 10^{-4}, drift speed of the individual carriers in aqueous solutions 10^{-7}; and of ions moving in a narrow tube filled with viscous liquid 10^{-8}. Fortunately, in most *but not all* cases, different mechanisms (such as the Grotthuss mechanism or free electron model; for a review, see [**?**]) at the level of microscopic structure help to create the illusion of a high macroscopic propagation speed (million times higher than the speed of its microscopic carriers). *The same carrier can have macroscopic speeds differing by orders of magnitude, depending on the context*; see a biological example at ion channels. When more than one of those speeds plays a role in the phenomenon we study, we must carefully consider its context and prepare for handling *fast* and *slow* effects, furthermore, their mixing.

When an object can interact with another in a way abstracted by science as more than one interaction type, we need to find the relation (the 'extraordinary' law) between them. Such a famous case is electricity and magnetism. Their interrelation is defined by the Maxwell equations: how an electrical field creates a magnetic one and vice versa (notice that the law is about their *space derivatives, aka space gradients* instead of the entities themselves). While we understand that the speeds of electromagnetic and gravitational interactions are finite, we can use the 'instant interaction' approximation in classical physics because one effect of the first particle reaches the second particle simultaneously with the other effect, leading to the absence of a time-dependent term in the mathematical formulation.

An apparently similar case is found in electrodiffusion, where ions can be abstracted as mass and charge, one belonging to thermodynamics and the other to electricity. There is, however, an essential difference between those cases: the interaction speeds are the same in the first case (moreover, in the spirit of classical physics, the interactions are instant) and differ by several orders of magnitude in the second one. Of course, the Maxwell equations can be nicely solved and modeled also for biology if one introduces [**?**] that the axial currents have the same speed (BTW: which was measured as 20 m/s) as the electrical and magnetic waves, furthermore the longitudinal current is (?)defined(?) to have no attenuation. Furthermore, it is likely also defined that current needs no driving force and this is why the positive and negative ions flow in the same direction. It is really a novel paradigm leading to "(mis)understanding cell interactions", but definitely describes some alternative nature.

Processes vs states

The finite speed introduces something unusual into science: processes, rather than jumps between states must be considered. The processes connect states and we need the corresponding laws of motion to connect them.

Speed in laws of science

Actually, the famous Coulomb's Law is expressed as

$$\frac{F(t)}{Q_1} = k\frac{Q_2}{\overrightarrow{r}^2} \tag{B.1}$$

$\overrightarrow{r}$ is a space-time distance. That is, in the Newtonian approximation, time is identical at all places, so we used to omit it. However, physics knows also the notion of *retarded time*. Considering the finite field propagation speed requires revisiting the fundamental physical laws. (in a Lorentz-transformed form) should be written as

$$\frac{F(t)}{Q_1} = k\frac{Q_2}{r^2}(t-\frac{r}{c}) \tag{B.2}$$

It is an implicit equation usually solved numerically or using perturbation methods for moving sources.

The electrostatic field that charge Q_1 experiences due to the finite propagation speed c of the electric field (or interaction) corresponds to that Q_2 at a distance r *generated* $\frac{r}{c}$ *time ago* (k is the constant describing the electrical interaction). This term has no role if the two charges do not change their position; similarly to that in the special theory of relativity, only the relative movement leads to complications. If the distance changes, its effect is so tiny that the term can usually be omitted. So, our college knowledge can serve as a good first approximation.

This speed term pops another law from classical physics into our minds: Kirchhoff's junction rule. The law is perfect in the approximation 'instant interaction' that classical physics uses, but not for biology. First, because it expresses charge conservation, *it is invalid when charges are "created" inside biological objects* (ions diffuse into the junction; see the role of ion channels in the wall of membranes). Second, it is not valid for input currents arriving with finite speeds into finite-size space regions, but it is valid for a single point in space-time (in other words, in differential equation form). As we discuss in section **??**, using a wrong definition of current means assuming 'instant interaction', that is, that neural signals propagate with the speed of light. The currents (and the voltages), measured at two different points in space-time, are different. Consequently, for extended objects (such as a line-like finite-size neuron), it is valid only with a time delay

$$I_{out}(t) = -I_{in}(t - \Delta t) \tag{B.3}$$

The time delay in biology is in the 1 *msec* range. *We must not describe the axon or the membrane with the non-differential form of the Kirchoff equation:*

the input and the output currents flow at different times (the charge carriers need time to reach from input to output); only the differential equation form expresses charge conservation (furthermore, in the case of "producing" ions, even by the differential form is invalid). For its exact interpretation, see sections on axonal charge delivery and on the true membrane current, Fig. **??**, and the text around it.

Studying electrical phenomena on structured media, such as biological cells, needs much care. We must not apply laws derived from entirely different conditions (mainly metals). The "ghost image" can be the "zeroth order" approach to understanding the AP: we assume that a"slow current" causing to appear as Post-Synaptic Potential (PSP), flows out through AIS some Δt time later than in flowed in "at the other end" of the neuron. The time delay is needed for the slow current to reach from the entry point to the exit. In fact, the figure displays Kirchhoff's junction rule for the slow currents in biology: the rush-in current triggered by exceeding the threshold leaves the junction a few tenth of millisecond later. The difference between the numerical solution and the simple difference is mainly due to the distributed nature of the input current. As the conduction speed increases, the difference current decreases, and at the speed of "instant interaction", results in Kirchhoff's junction rule as known in the technical electricity.

B.3.4 Spatiotemporal

Classical physics is based on the Newtonian idea that space and time are absolute so everything happens simultaneously. Moreover, their observationis also instantaneous. Consequently, when their objects interact, it must be instantaneous; in other words, all interactions have the same (and, logically: infinitely high) speed. Furthermore, electromagnetic waves with the same high (logically, infinitely high) speed inform the observer. This self-consistent abstraction enables us to provide a "nice" mathematical description of nature in various phenomena: the classic science. In the first year of college, we learned that the idea resulted in "nice" reciprocal square dependencies, Kepler's and Coulomb's laws. We discussed that the macroscopic phenomenon "current" is implemented at the microscopic level by transferring (in different forms) discrete charge, and that movement of charges has no effect on the environment. Furthermore, that *without charge (and, without atomic charge carriers), neither potential nor current exists.* In the next semester, we learned that the speed of light is finite and that solids show a macroscopic behavior "resistance" against forwarding microscopic charges. One semester later, we learned that nature behaves differently at high speeds, at low sizes and energies; furthermore, that the transition between the continuous and discrete views needs new concepts.

In biology, it was evident that the transfer (conduction) time must be considered together with the computing (synaptic) time (in this sense, presynaptic to postsynaptic transmission time). The name "spatiotemporal" and a (separated) time dependence is commonly used [18], in the sense that Precise Firing Sequence (PFS) "*tended to be correlated with the animal's behavior*"; further-

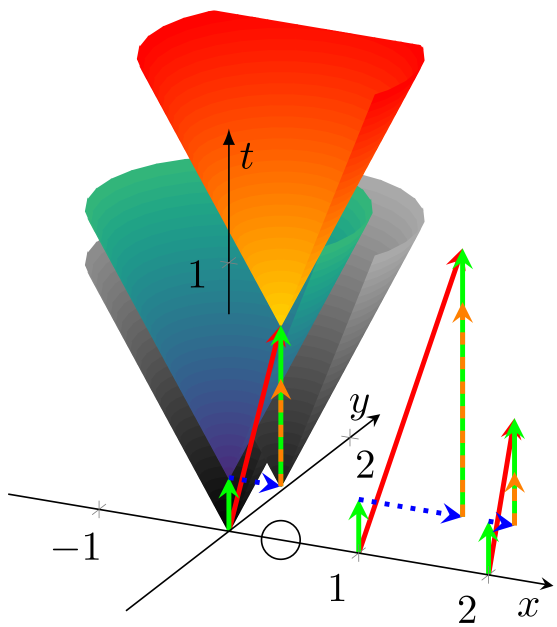

Figure B.1: The temporal diagram, i.e., the way of calculation to combine the spatial distance (transmission time, blue arrows) and computing time (green arrows) illustrated in the time-space coordinate system. The orange-green vertical arrow shows that the second computing unit must idly wait until the transmitted result reaches its position, because of the finite transmission time. The axes x and y refer to space coordinates (transformed to time using the conduction velocity), the axis t refers to the time itself. The arrows starting from points 0, 1 and 2 on the x axis illustrate timing for three different propagation speeds. The red vector points from the beginning to the end of the process. Its length may serve as a statistical entity to describe temporal distance of the units.

more, that "*the results suggest that relevant information is carried by the fine temporal structure of cortical activity*" [**?**]. The "*neural dynamics*" was studied and "*spatiotemporal spreading of population activity was mapped*" [**?**] by methods used to describe the *static computing methods*: interspike intervals histograms, auto-correlation and cross-correlation. Because of the peculiarities of this information handling, *there are severe doubts whether the notions of the classic neural information theory are valid for biological computing systems* [**?**]. The correct method of describing biological computation is still missing, given that the significant item of the computing is missed: *the time and position are connected through the information transfer speed* (called conduction velocity).

Will be based on [**?**, **?**, **?**]

In Fig. B.1, for visibility, two spatial and one temporal coordinate are shown. In the following figures, some illustration enable to omit one more spatial dimension, i.e., effectively to draw the events as a two-dimensional diagram.

B.3.5 Mathematics

Galilei said, "Mathematics is the language in which God has written the universe". However, it is not sure that when we attempt to read a piece of the universe written in that language, we use the right piece and dialect of the language, and even that humans already invented the needed piece. Not mentioning whether one understands the given piece correctly. For example, mathematical calculus (integral and differential) was invented mainly for the practical needs of analyzing spatial motion of celestial bodies. Similarly, Minkowski's mathematics theory proliferated widely [**?**] only after inventing the special theory of relativity. Although the mathematical description was developed earlier, there was no practical need to apply it. The classical laws of motion were valid only until more meticulous observations required to consider speed and acceleration (time derivatives of position) dependence in addition to position dependence. Newton's *static* laws remained valid, but for the *dynamic* description we must revisit the second law of motion.

Also, we must not forget that "mathematics is not just a language. Mathematics is a language plus reasoning. It's like a language plus logic. Mathematics is a tool for reasoning." (Richard P. Feynman) Mathematical formulas work with numbers but math theorems and statements begin with "If ... then". They have their range of validity, even when they describe nature. Using mathematics to describe the classical equations of motion, to calculate forces and times that speed up bodies above the speed of light is possible, but in that case mathematics is applied to an inappropriate approximation to nature. When approaching the speed of light, different physical approximations (that calls for a different mathematical handling) are to be used. *A mathematical formula, without naming which interactions it describes and naming under which conditions and approximations the formula can be applied, are just numbers without meaning. It surely describes something but only eventually describes what we studied.* Galilei made measurements with objects having friction, but his careful analysis extrapolated his results to the abstraction that no friction was present. *We know his name because of making meticulous abstractions and omissions (and mainly: recognizing the need to do so!)* instead of publishing a vast amount of half-understood measured data.

The discrete-continuous dichotomy raises the interesting question: how long quantities describing matter remain differentiable? At macroscopic quantities, for example, in technical electricity, surely they are: for a large number of charge carriers flowing through a cross section, the statistical fluctuation of the number of charge carriers obviously is sufficiently small to operate with the approximation that the fluctuation is zero, so the definition that, at a cross section, $I = \frac{dQ}{dt}$ defines the current has sense. Even one can approximate the current by saying that dQ cannot be smaller than the elementary charge (recall what a revolution happened due to assuming the the energy of a radiation cannot be arbitrarily small). Mathematically, one can say that the time course of a single charge carrier is a δ-function, which is surely not differentiable. Then, is the rush-in current with its 10^7 charge carriers differentiable? If it happens in fractions of a nanosecond, over a surface of the membrane? The same question for the single ion channels? Or, when that amount of rush-in current is distributed over a several millisecond range? In a microsecond, in average, about thousand electrons represents the current; at the beginning and the end of an AP, only a few dozens. These are reasonable questions when measuring voltage gradient that such a low current current generates on a resistor represented by some ion channel.

The interdependent behavior of charge and mass has an interesting, entirely mathematical consequence as well. The definition of a partial derivative of a function of several variables is its derivative with respect to one of those variables, *with the others held constant*. In the case of ions, changing the function with respect to mass or charge means simultaneously changing also the other; that is, the partial derivatives depend on each other. In other words, one cannot calculate the partial derivatives; only the total derivative. As a consequence, equations that use the partial derivatives of concentration and electrical potential are either incorrect or only more or less good approximations. Classical mathematics cannot be applied. Consequently, in the case of ions and electrolytes, one must not use the well-established concepts (enthalpy, entropy, etc) of thermodynamics in unchanged form. (BTW: the discrete nature of charged particles leaves the question open, under which conditions remains the ion current differentiable.)

Experience shows that thermodynamics describes correctly the macroscopic behavior of physical media when an enormous number of microscopic objects are present in a macroscopic volume. Are the biological objects large enough to neglect the statistical fluctations of the objects they contain? The Nernst-Planck equation (and the corresponding transport equations) relates concentration and voltage gradients. Are those laws valid at the size of ion channels, where it is hard to interpret those derivatives? Do those statistical laws remain valid for quasi-continuous infinite volumes, for non-infinite closed volumes, or for individual ions? Recall that thermodynamics follows a hybrid path that calculates the *probable number* in a volume in the phase space and works that as if it were the *actual number* of particles. Mathematics and physics do not have methods behind the pairwise, instant interactions.

Science has a well-established method for deriving its laws. It groups resemblant phenomena to major disciplines and they branch along even more resemblance to sub-disciplines. The hierarchy is not strict, given that "nature is not interested in our separations, and many of the interesting phenomena bridge the gaps between fields." [44]. Mathematics is a discipline on its own right. It abstracted its notions from nature's concrete objects and phenomena, its abstract methods serve as a solid base for the fellow sciences and are frequently used to describe new discoveries. It is commonly accepted that physics is the very basic tool to study nature at some elementary level. Chemistry, at another level of abstracting nature, is based upon its laws, and constructs its laws for the phenomena it researches. The laws of physics are borrowed and respected, and the laws of chemistry cannot conflict with the laws of chemistry. Biology is the discipline studying living matter, and is considered to be an even higher level abstraction, with its own laws. These laws, as expected, cannot conflict with the laws of chemisty and physics. Howevever, it looks like some laws, mainly the ones belonging to thermodynamics, are are not perfect for describing biological phenomena. Especially, the life itself seems to contradict the second law of thermodynamics. E. Schrödinger suspected that some non-ordinary laws are to be derived to describe life.

Classical physics is simplifying the things until it can attribute a single substance (such as mass or charge) to the tested object and in its laboratories derives accurately the laws and determines the attibutes of the substances. As E. Schrödinger has drawn attention to, "the construction [of living matter] is different from anything we have yet tested in the physical laboratory". Science experienced many times that the used approximations do not describe some phenomena accurately. At that times, scrutinizing the fundamental assumptions (such as whether the Galillei's relativity principle remain valid when approaching the speed of light, or the matter remains continuous when decreasing extremely the tested sample s size) has led to establishing new scientific fields, such as theories of relativity or theory of quantum mechanics. When describing living matter, one needs some more approximations.

B.3.6 Biology

The above items are what E. Schrödinger coined as "*the construction is different from anything we have yet tested in the physical laboratory*" [1]. Consequently, measurements must be designed and carried out with care; the routine methods used for measuring objects from inanimate nature cannot surely be applied to living objects without changing them. We discuss some of the relevant terms and notions of physics, differentiating which approximation is appropriate only for physics (mainly electricity and thermodynamics) and, instead, which approximation should be used for biology. *Biophysics simply translated the corresponding major terminus technicus words from the theory and practice of physics' major disciplines, mainly from electricity, which were worked out for homogeneous, isotropic, structureless metals, and for strictly pair-wise interactions with a single (actually, 'instant') interaction speed; to the structured, non-homogeneous, non-isotropic, material mixtures and for multiple interaction speeds.* Those notions do not always have unchanged meaning, and how much they do, depends on the actual conditions. The precise meaning needs a case-by-case analysis.

A common fallacy in biology is that physics cannot underpin the operation of living matter, citing E. Schrödinger. However, the claim falsifies his opinion by omitting the most essential word 'ordinary'. Schrödinger wanted to emphasize the opposite: there is no new force (no unknown new interaction), only studying living matter needs different testing methods in the physical laboratory. He suggested to answer the question "Is life based on the laws of physics?" affirmatively, but expected to discover the appropriate forms of physical laws describing the 'non-ordinary' (in our reading: non-disciplinary) behavior of living matter. No doubt, the basic notions and terms must be interpreted precisely for living matter, much beyond the level we used to at college level. However, after that reinterpretation, we can interpret features of living matter, although we need a more careful, many-disciplinary analysis to do so. We must use the appropriate abstractions and approximations for the phenomena, depending on the level needed in the given cooperation of objects and interactions.

The physical models consider infinitely large volumes, surfaces, distances; furthermore, and most importantly: instant interactions. Is the cell large enough to consider it infinitely large (at least on the scale using ion's size); that is to apply laws of science derived for the abstraction 'infinitely large'? When working with charge, we know that charge is quantized, while the macroscopic quantities voltage and current are continuous (derivable). Do cells contain a sufficient number of charge carriers to apply macroscopic notions? Do the thousands of times smaller ion channels transfer enough charge to speak about well-defined current? When a couple of ions are transferred through an ion channel, do they change significantly the potential that accelerates them?

Life, including the brain's operation, is dynamic. As Schrödinger formulated, the "construction of living matter" differs from the one science used to test in its labs. The scientific abstraction based on "states" (i.e., on instant changes) fails for the case of biology, where "processes" happen (i.e., the changes are obviously much slower). The commonly used measuring methods such as clamping, patching, and freezing, reduce the life to states (and correspondingly, the related theories describe states with perturbance [18]). On the one side, this technology fixes the cell at some well-defined static state and enables one to observe a static anatomic picture of the cell. On the other, it eliminates the dynamic processes, i.e., *hides for forever the essence of the life that the cell exists in a continuous change governed by laws of motion.* Those methods stop the processes for studying them, in this way killing their dynamicity to be measured. It was forgotten that using feedback for stabilizing an autonomously working electrical system means introducing foreign currents and this way falsifying its operation.

Furthermore, it is hazardous to introduce technically (and incorrectly) derived and misinterpreted macroscopic features and interpret them as fundamental electrical notions. In general, instead of understanding and developing the correct scientific base of the operation, saying that science cannot describe it. The idea of conductance has been introduced to neurophysiology almost a century ago. It was taken from physics, where the notion was derived for metals (conductors, instead of electrolytes). Since then, its original interpretation has been forgotten, and today (in contrast with physics, fundamentally due to the wrong concept of 'equivalent circuit') it has become a primary entity for describing electrical characteristics of biological cells. We explain how the right physics background enables us to discover wrong physical models and misinterpreted notions of physics in neurophysiology; furthermore, how the right interpretation opens the way to the correct interpretation of neuronal information. We set up an abstract electrical model of neuronal operation.

We derive the needed 'non-ordinary laws', which are derived by using the same first principles as the 'ordinary laws', but are abstracted for the approximations valid for living matter. As we discuss, those 'ordinary' laws were derived for strictly pair-wise interactions at very high speeds. In biology, we can observe interactions at million times smaller speed, in inhomogeneous, non-isotropic, structured material. *Biology does not have the conditions for which the ordinary laws of physics were derived.* We show that the ordinary laws are also the result of approximations (including omissions) and by using the appropriate approximations for the biological cases we can derive those 'non-ordinary' laws of physics. Which laws are more complex to derive and need joining different disciplines, furthermore, we must use several stages (with approximations changing from stage by stage) instead of one single stage in the case of the 'ordinary' laws. However, *all laws follow the same principles.*

Biology, and especially neuronal operation, produces examples where wrong omissions in complex processes results in absolutely wrong results. In those cases some initial resemblance between our theoretical predictions and our phenomena exist, but the success in simple cases provides no guarantee that the model was appropriate: "the success of the equations is no evidence in favour of the mechanism" [14]. Finally, all laws are approximations and the accuracy of verifying their predictions is limited. Several theories can describe the same phenomenon with the required accuracy. We also show in the section B.3.3 about the finite interaction speeds that the mostly known laws (from Newton, Coulomb, Kirchoff, etc.) are also approximations. They have their range of validity, although it is often forgotten.

One such neuralgic point of omissions and approximations is the vastly different interaction speeds; furthermore, that where the speed is considered at all, *the same speed is assumed for all interactions.* The laws are abstract also in the sense that, say, the objects in the laws of physics have either mass or electric charge, but not both. It is the researcher's task to decide which combination of laws can be applied to the given condition. For example, one can assume in most cases that the speeds sum up linearly, except at very high speeds. Biology provides excellent case studies where different interactions shape the phenomenon and special care must be exercised. We give a short review of history and kinds of interaction speeds.

Another point is that science started with the assumption that the non-living matter is continuous, although it was early discovered that there are smallest pieces of that matter. When we approached that size, we experienced that different subsets of science laws describe that matter and the atoms they contain; it is one of the hardest tasks to establish relations between those subsets. Again, we used abstractions that the continuous matter is infinitely large and that the isolated atoms are infinitely far from each other and from the external world. We also experienced the semi-infinite cases, and studied the behavior of surfaces and interfaces; which, again is different from both that of the atoms and their large masses. Given that biological objects are between those microscopic and macroscopic sizes, and they are surrounded by surfaces, we must be prepared that no simple rules describe their behavior.

Neuronal operation is at the boundary where sometimes, in the same phenomenon, one interaction can be interpreted at macroscopic level, some another must already be interpreted at microscopic level. Furthermore, a series of stages (instead of a single state) and processes (instead of stages) describe the subject under study. Given the vital role of charge and current in neuronal operation, we give their precise interpretation. Furthermore, we must consider that the processes happen in a finite volume, "within the spatial boundary". Special emphasis is given to the true interpretation of conductance, one of the central terms in biology.

B.4 Addendum

In this section, we add some comments, pinpointings and fixes to the chapters of book [7], needed mainly to make enhanced abstractions for biology, with the goal to derive the needed 'non-ordinary' laws of physics. The quotations are from that book, without citing the book per quotation, providing only page number. Our fixes and notations are typeset in blue, and confine themselves to the most needed text, so the best way or reading is to read the two texts simultaneously.

Correct neuroscience textbooks such as [**?**] call the attention to that the analogy (from the point of the used equations) between the equivalent electrical circuits and biological neurons is not complete .

B.4.1 Ion movement in excitable cells

Page 9 "In excitable cells, movement of ions across the plasma membrane results in changes of electrical potential across the membrane, and these potential changes are the primary signals that convey biological messages from one part of the cell to another part of the cell" Given that those messages are delived by ions, the conveyed biological messages also mean changes of concentration. After crossing the membrane, the ions (as slow currents) also physically move in the layer on the surface of the plasma membrane. To consider the process and the dynamically formed component is essential in understanding why and how APs form, as well as how those signals are processed and transmitted. Their effect is significant. As shown in Page 12, the total number of ions in the spherical cell is $2 * 10^{13}$. As we estimate in section **??**, the number of ions conveyed in a single signal is about 10^7, which seems to be negligible. In line with our estimate, in the order of 10^4 pulses in quick succession are needed to drastically change the number of ions (aka concentration or potential of the bulk) in the cell. However, the number of uncompensated ions is estimated, in line with our estimate for the order of number of ions per pulse, as $4.7 * 10^7$, so it is absolutely necessary to consider the finite resources, as we do it in section B.3.

B.4.2 Physical laws that dictate ion movement

Page 10 "The first two laws concern two processes: diffusion of particles caused by concentration differences and drift of ions caused by potential differences."

Actually, there are three processes: it remains outside the scope of the HH theory that the mutual repulsion of ions generates a shock wave in the electrolyte. The generated wave mechanically moves the ions that per definitonem means a current. As our force analyzis shows, that mechanical force can dominate the ion movement in the transient state. The third law concerns the relationship between the proportional coefficients of the first two processes, the diffusion coefficient D and the drift mobility μ." Unfortunately, here the reciprocal relations, which we discuss in section **??**, between the processes is missing, as well as the role of the finite resources, which we discuss in section B.3. This is a fundamentally wrong abstraction. The worst context of this abstraction is that it hides that the speeds of diffusion coefficients and speeds differ by several orders of magnitude, as it refers to the Nernst-Planck equation at zero flux, where, really, both speeds are zero. The equation, in addition, does not consider that by moving ions, we change the potential and concentration simultanously and inseparably; that is, handling the case with partial derivatives is not allowed.

Page 15"The negative sign indicates that I flows in the opposite direction as $\frac{\partial V}{\partial x}$ and in the opposite (same) direction as $\frac{\partial [C]}{\partial x}$ if z is positive (negative). This equation describes the passive behavior of ions in biological systems." Unfortunately, there are doubts here if partial derivation can be applied at all: when changing one of the abstract entities the other changes simultanously, given that the ion represents mass and charge simultanously.

B.4.3 The Nernst equation

Page 15"The NPE gives the explicit expression of ionic current in terms of concentration and electric potential gradients." As we discuss in section **??**, NPE is valid only in the 'mean-field' approximation, i.e., when we assume that the diffusion and electrical interactions have the same speed. Furthermore, although it was derived for equilibrium state, it implies that for the short time of forming an AP, the concentration also changes, given that the potential changes. According to their Table 2.1, for a short period, up to two orders of magnitude change in the concentration can take place. See also section **??**.

B.4.4 Ion distribution and gradient maintenance

Page 17 "These ionic concentration gradients across the cell membrane constitute the driving forces (or chemical potentials) for ionic currents flowing through open channels in the membrane. In other words, the ionic concentration gradients act like DC batteries for cross-membrane currents." First, the relation is mutual; it could be considered that there are chemical batteries. Second, the ionic currents flow not only through the membrane; they can flow also through the extracellular space. Furthermore, one must tell also when it flows. Third, the driving force of those DC batteries changes during operation.

B.4.5 Membrane permeability

Page 24 ”Within the membrane, if one assumes that $[C]$ drops linearly with respect to x”Since the ions are simultanously also charges, the potential also drops linearly, as explicitly said on page 26. Ions must accelerate across the membrane surfaces.

B.4.6 Space-charge neutrality

”In a given volume, the total charges of cations is approximately equal to the total charge of anions . . . The principle of space-charge neutrality therefore holds in any volume in the biological system except within the plasma membrane, where the electric field is nonzero.” The paper correctly states that the statement is valid only approximately, but misses the point that the volume must be choses appropriately and that the essence of life is the temporaly inbalance of charges; either the dynamic ”charge pumping” processes in the resting state, or the transition state leading to issuing an AP.

Appendix C

Thermodynamics for neurons

Contents

C.1 Deriving the "thermodynamic electrical field"

In a segmented electrolyte, electrical charges are typically globally balanced; however, they may become locally unbalanced due to physical factors. An important case for the biology is when a semipermeable membrane separates two segments of electrolytes with very different ionic concentrations. The differing concentrations generate voltage according to the well-known Nernst-Planck equation, as shown in **Physics panel C.1.1**. **Physics panel C.1.1** describes how the chemical and electrical spatial gradients depend on each other, **Physics panel D.1.1** shows a practical approximation for calculating it, with the calculated figures in **Numerics panel D.1.1**.

In simple words, the Nernst-Planck equation states that the changes in concentrations of ions create changes in the electrical field (and vice versa), and in a stationary state, they remain unchanged. Notice that in the case of an ion mixture, *a joint charge-generated electrical field exist, while an electrical field generates per ion different concentrations.* Note that inside a neuron the gradients change sharply, so there only the differential form can be used; using the statical macroscopical intagral form is misleading.

Contents

C.1 Thermodynamic field 93

Physics panel C.1.1:
The Nernst-Planck law

The Nernst-Planck electrodiffusion equation in a balanced state

$$\frac{d}{dz}V_m(z) = -\frac{RT}{q*F}\frac{1}{C_k(z)}\frac{d}{dz}C_k(z) \tag{C.1}$$

describes in one dimension the mutual dependence of the *spatial gradients* of the electrical and thermodynamic fields on each other. In good textbooks (see, for example, [26], Eq (11.28)), its derivation is exhaustively detailed. In the equation, z is the spatial variable across the direction of the changed invasion parameter, R is the gas constant, F is the Faraday's constant, T is the temperature, q the valence of the ion, $V_m(z)$ the potential, and $C_k(z)$ the concentration of the chemical ion. The two sides are the derivatives of the equation

$$V_m = \frac{RT}{q*F}\ln\left(\frac{C_k^{ext}}{C_k^{int}}\right) \tag{C.2}$$

known as *Nernst equation*. In other words, **Eqs.(C.1) and Eq.(C.2) are the differential and integral formulations of the same knowledge**. The limits of the integration are chosen arbitrarily (by choosing the reference concentration C_k^{int}).

Appendix D

Resting Potential

Concerning the origin of the membrane's resting potential, we disagree that "the resting membrane potential is the result of the [low intensity] passive flux of individual ion species through several classes of resting channels" [2]. This claim cannot be true: the flux is not passive (to move the ions, energy required), furthermore, no setpoint is defined, so some (low) current flows due to the interplay of the electrostatically set setpoint and the reversal potential of the ions inside and outside. The claim was already question-marked whether the membrane permeability is a primary property for the membrane potential generation, for a review, see [?]. That interpretation leads to causality issues [12].

It was already guessed that the membrane's resting potential is of electrostatic origin [?, 34, ?, ?, ?]. When ions are confined in a closed volume, they exist in a state of thermodynamic and electrical equilibrium. However, those equilibrium states cannot be discussed independently. We must discuss them in a non-disciplinary way because the participating ions belong simultaneously to the disciplines of electricity and thermodynamics; furthermore, we must cross the borderline between discrete and continuous points of view of nature [17]. Furthermore, the features of living matter may change when measured in physics laboratories as usual. When measuring them, we must consider that "the construction is different from anything we have yet tested in the physical laboratory" [1].

D.1 Deriving mebrane's "thermodynamic electrical field"

In section C.1, we introduced the idea of the "equivalent thermodynamic electrical field"; now we calculate it for the case of neuron membrane as shown in .

Physics panel D.1.1:
The membrane's eqivalent electrical themodynamic field

We assume that, in a balanced state, the concentrations on the two sides of the membrane are C_k^{ext} and C_k^{in}, respectively; furthermore we assume that

$$\frac{dC_k}{dz} \approx \frac{C_k^{ext} - C_k^{in}}{d}$$

To simplify the math writing, we neglect the change caused by considering the difference (maybe we should apply a near-unity factor); we use the larger concentration value instead. Using the constants and $T = 300$, we can derive the "equivalent thermodynamic electrical field" across the membrane for the case of a permeable ion channel in the membrane

$$E_{thermal}^{C_k} = -\frac{RT}{q*F}\frac{1}{C_k^{ext}}\frac{C_k^{ext} - C_k^{in}}{d} \approx -\frac{RT}{q*F}\frac{1}{d} \tag{D.1}$$

The equivalent thermodynamic electric field does not depend on concentration, but rather is linear in temperature and inversely proportional to membrane thickness (given in nm). We provide the numerical values in **Numerics panel D.1.1** for the case of ion channels in the membrane's wall.

Numerics panel D.1.1:
Neuron's equivalent thermodynamic electrical field

It is valid separately for all C_k

$$E_{thermal}^{C_k}(d) = -2.585 * 10^{-2}\frac{1}{d} \quad \left[\frac{V}{m}\right] \tag{D.2}$$

Figure D.1 depicts the field's dependence on the membrane's width and how it compares to the charge-generated electrical field. The figure illustrates the setpoint(see section **??**), the boundaries of the changes, and the interesting phenomena that occur within the shaded cross-section. The figure is an illustration, but its figures are not far from the true ones. For a 5 nm membrane thickness, it results in $5.17 * 10^6\ \frac{V}{m}$, which compares well to the value provided by Eq. (**??**). (Notice that the membrane's thickness may change during the action potential, see [41]: the rush-in extra charge compresses the membrane. Similarly, the concentration of the electrolyte in proximity to the membrane also changes.) Correspondingly, by using that $U = E * d$,

$$U_{thermal}^{C_k}(5) = -25.85 \quad \left[mV\right] \tag{D.3}$$

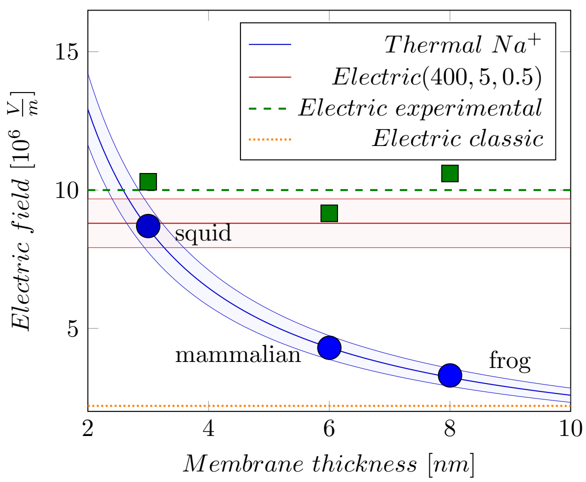

*Figure D.1: The thermodynamic "electrical field" depends on the membrane's thickness, see Eq.(D.2). The electrical gradient depends on the charged layer's concentration and thickness; see Eq.(**??**). The dots mark measured data shown in Table 4.1.*

In simple words, the Nernst-Planck equation states that the changes in concentrations of ions create changes in the electrical field (and vice versa), and in a stationary state, they remain unchanged. Notice that in the case of an ion mixture, a joint charge-generated electrical field exists, and they equal, per ion, the concentration-generated "electrical fields", see section **??**.

In the steady state, in contrast with the case of an ion in the infinite space, some other forces also contribute to the mentioned ones. To discuss how nature restores the steady state when a microscopic change occurs in a balanced state of a biological solution, we write the well-known Nernst-Planck equation (see Eq.(C.1)) in a slightly extended form:

$$\underbrace{\left(-F_{constr}(z)\right)}_{Constraint} = \underbrace{e_{el} * E_{Gap}^{Total}(c, \Delta z)}_{Electrical} + \underbrace{e_{el} * E_{thermal}^{C_k}(d)}_{Thermodynamic} \tag{D.4}$$

$$\underbrace{\left(+F_{Transp}(z)\right)}_{Transport} \underbrace{\left(+F_{Invasion}(z)\right)}_{External} \tag{D.5}$$

We multiplied the usual two terms by the elementary charge, so its terms are expressed as forces, plus we added an external force (its role is discussed below). Furthermore, changing one force triggers a corresponding counterforce and/or causes the ion to move in a viscous fluid. That is, the speed of the material transport gradually changes as the steady state approached; furthermore, it is by orders of magnitude smaller than the speed of the interactions. When neuroscience teaches that "pumps that maintain ion gradients ... transport ions against their electrical and chemical gradients" [2], page 101, one shall ask *what kind of force acts on the ion*, keeping in mind E. Schrödinger's opinion that not "any 'new force' or what not" [1] affects ions' motion.

An interesting option is when the counterforce combines two disciplines. When Na^+ ions rush-in at the beginning into the membrane, they produce simultaneously a huge electrical and thermodynamic gradient. The repulsion among the ions creates a huge mechanical pressure that presses the elastic membrane. The counterforce starts a mechanical shock wave (soliton) [15], that is simultaneously an electrical potential wave [16] that is measured as AP. Only using cross-disciplinary discussion (non-ordinary laws) enables understanding neuronal operation. That means when describing an ionic transfer process, *we must not separate the electrical current from the mass transfer*: they happen simultaneously and mutually trigger each other. Notice that the thermodynamic term is ion-specific while the electrical term is not. To be entirely balanced, the system must be balanced to all elements. In this way, changing one concentration implicitly changes all other concentrations and the electrical field.

It was a colossal mistake to introduce equivalent circuits with their fixed-value voltage generators. It forces one to assume that the conductances of the players (membrane, synapses, AIS) change without any reason and prevents understanding how the competition between thermodynamic and electrical processes governs neuronal operation. It leads, among others, to attributing conductance change to membranes, which are simple isolators with no charge carriers that can implement charge transfer, see section **??**; this way attributing the change of an electrical entity to the biological material. This assumption neglects the driving force the mentioned forces provide. Instead, it attributes the magic ability to the ion channels that they can change their transmission ability as the actual situation requires.

When discussing balanced states, no transport occurs, so the transport force cancels, and the mobility has no role. The counterforce adapts to the situation. That force may be a mechanical one: the ions sitting on the surface of the membrane press the surface due to the attractive force of ions, and the membrane mechanically provides the needed counterforce. If the boundary of the segments is not freely penetrable, the counterforce equals the difference between those two forces. In this way, no force acts on the ions; the two gradients persist. When the ions can move freely between the segments, they will move until they produce a concentration gradient (a thermodynamic force) for the given ionWhen neuroscience teaches that "pumps that maintain ion gradients ... transport ions against their electrical and chemical gradients" [2], page 101, one shall ask what kind of force acts on the ion, keeping in mind E. Schrödinger's opinion that not "any 'new force' or what not" [1] affects ions' motion. that counterbalances the electrical gradient (the electrical force) and then the transport stops. A transport force is needed to reach the Stokes-Einstein speed (see Eq.(**??**)) in a viscous fluid.

We must not forget that initially, the Nernst-Planck equation described a transfer process (i.e., a dynamic equation), assuming the same speed for mass and charge transfers. We used it only to describe the balanced state (i.e., a static equation at zero speed) and added the term describing external (such as mechanical constraint) force that may be involved; it affects the process, but it does not belong to either of the respective fields. Commonly known mechanical constraints appear in a non-conducting layer called the membrane, where that counterforce prevents ions from penetrating the membrane layers. A membrane is a perfect isolator, i.e., no charge carriers exist between its two surfaces, so it is senseless to introduce the notion of its conductance. (There may be ion channels that deliver ions built into the membrane, but it is a different subject.) The separation results from *charge separation*, a mechanism distinct from *polarization*, as typically described in most biology textbooks. In the form we derived, the equation correctly describes the ion transfer, a current conveying ions with the speed controlled by the actual gradients, instead of assuming an ion current traveling with the apparent speed of electrical current in solids.

We must not forget that originally the Nernst-Planck equation described a transfer process (i.e., a dynamic equation), assuming the same speed for mass and charge transfers. We used it only to describe the balanced state (i.e., a static equation, at zero speed), and added the term describing external (such as mechanical constraint) force may be involved; it affects the process, but it does not belong to either of the respective fields. Commonly known mechanical constraints are a non-conducting layer, called membrane, where that counterforce prevents ions from penetrating the membrane layers. A membrane is a perfect isolator, i.e., no charge carriers exist between its two surfaces. (There may be ion channels, that deliver ions, built into the membrane, as we discuss in section **??**, but it is a different subject.) The current clearly results from charge separation; a mechanism distinct from polarization most biology textbook uses. In the form we derived, the equation correctly describes the ion transfer, a current conveying ions with the speed controlled by the actual gradients, instead of assuming an ion current traveling with the apparent speed of electrical current in solids.

D.2 Charge layers in segment

Figure the ionic basis of a membrane potential shows and explains the case introduced by separating the cell into two segments by a finite-width membrane; although does not explain even qualitatively, *why* the ions behave apparently against the laws of thermodynamics: the left and right sides of the figure are apparently identical. On the left side (the case of infinitely thin membrane), at an exact balance of charges on each side, the potential across the membrane is zero. We explain that the finite thickness will result in a lack of balance (introduces inhomogeneity and creates a voltage and concentration gradient) proximal to the surfaces of the membrane, even if the concentrations on the two sides are the same. Changing the bulk concentration or potential in one of the segments creates a corresponding gradient across the separating membrane that increases the inhomogeneity proximal to the membrane. The ions will experience an extra force due to the gradient, but the mechanical counterforce of the membrane will keep them back in the segment. The *concentration and potential, inseparably and having the same time course*, will change across the two sides of the membrane just because of the gap's physical features the membrane represents.

We consider that the segment is composed of electrically conducting discs (the ions are free to move on the surface) and the charged discs' contribution to the electrical field at point P (see Fig. **??**) can be calculated as known from the theory of electricity. Due to the symmetry in direction of z, in a homogenous solution, the resulting electrical field in the plane perpendicular to direction z is zero: we have contributions of the same size with opposite signs.

Here we used the abstraction of an infinitely thin "charged sheet" and that there is a step-like gradient in the electrical field in the gap. The physical reality is that the charge is represented by ions, which simultaneously represent a mass; furthermore, the thickness of the surface layer they form is by orders or magnitude thicker than the layer that the electrons form. In an equilibrium state, the forces due to the voltage and concentration gradients (see Eq.(C.1)) must be counterbalanced by an external force. An interference between science disciplines can also manifest here. We know at the same time that at the boundaries of electrolytes, different interfaces, including electrically neutral electrical double layers, can be formed by only partly known processes [**?**]. The presence of those structures makes drawing quantitative conclusions hard.

D.2.1 Simple conducting sheet (a classical condenser)

When discussing the internal electrical operation of a neuron, first we consider a condenser that obeys laws of classical electricity and has the geometrical size of a biological neuron. We apply the physics terms to that pseudo-biological object and test whether the values we derive are consistent with physiological phenomena. A tiny (according to [7], page 12, about $2 * 10^{-3}$) portion of the positive ions leave their negative counterions behind and form a thin positively charged ion layer on the surface of the membrane (of course, the same happens on the opposite side with negative ions). Experience shows, that – also in biology – two very thin (but not infinitely thin) layers of ions are formed on the two sides of the membrane, see the caption of figure (Fig. 11.22 in [**?**]): "The ions that give rise to the membrane potential lie in a thin ($< 1\ nm$) surface layer close to the membrane". The two surfaces get covered by a sub-nanometer-thick conducting layer, on the top of a 5 nm nanometer thick insulator. So, we model the cell with two finite-width "conductive sheet" layers. These layers represent an electrical condenser (so we can calculate the internal electrical field between the plates) and they counterbalance each other's electrical field outside the condenser. We may assume that the solid surface represents the needed counterforce to keep the charges in rest.

One cannot expect geometrically plane surfaces: there are some biological objects of up to 1 nm size on the surface. So, we assume a physical 'uniformly charged sheet' with an average thickness $\Delta z = 0.5\ nm$ on the surface. For the sake of simplicity, we assume a step-like function for the electrical field. It is zero in the segment outside the "conducting sheets" (the 'bulk' portions) as well as inside the sheets. Between the sheets, it jumps to the value of the electrical field of a charged sheet with surface density σ_A (see Eq.(**??**)). (In this picture we see that there is a jump in the value of the electrical field (and so: in the force acting on a charge) in the two sides of the physical layer. Here we still assume that a mechanical counterforce due to surface's roughness keeps the charges in place inside the charged layer.)

We assume that all unbalanced (dissociated) ions are in the layer, and the rest contains no dissociated ions, furthermore, that the ions' concentration is unchanged in the segment. By assuming a 0.5 nm thick layer on the surfaces of the membrane, our model delivers an electrical field (see Eq.(**??**))

$$E^{Gap}_{Classic}(c, \Delta z) = E_z(c, \Delta z) = 4.45 * 10^3 * c * \Delta z \quad \left[\frac{V}{m}\frac{1}{nm * mM}\right] \tag{D.6}$$

In the case of a 400 mM solution and a 0.5 nm 'thin physical layer' it evaluates to an electrical field

$$E^{Gap}_{Classic}(400, 0.5) = 0.11 * 10^7 \left[\frac{V}{m}\right]$$

and at a 5 nm membrane thickness the voltage across the plates becomes

$$U^{Gap}_{Classic}(400, 0.5, 5) = 2 * E^{Gap}_{Classic}(400, 0.5) * 10^{-3} * d = 10.84\ \left[mV\right] \tag{D.7}$$

The number of uncompensated ions needed for the cell, using the method in [7], is $0.88 * 10^7$. Remarkably, the calculated values are about by a factor of 5 lower than the experimentally derived values. "An electrical potential difference about $50 - 100\ mV$ ... exists across a plasma membrane only about 5 nm thick, so that the resulting voltage gradient is about $100,000\ V/cm$" [**?**]. "The number of uncompensated ions needed for the cell is $4.7*10^7$" [7]. We are in the right order of magnitude but we arbitrarily assumed a layer thickness, a non-dielectrical medium. Furthermore, given that the charges are distributed uniformly in the layer, the electrical field changes linearly between the two sides of the conducting sheet. The deviation from the experienced values suggests that the dipoles' presence in the electrolyte bulk significantly changes the achievable electrical field and potential across the plates. Our simple model seems to be a strong oversimplification.

If the distance between the plates is finite, the resulting electrical field will differ from zero. With such a model, a usual parallel plate condenser can be derived, as shown in Fig. D.2. We have two charged disks (infinite conducting sheets), and an insulator layer between them. In the classical picture, the opposite charges'amount on the two plates must be the same. As shown, a constant electrical field is present inside the membrane (across the plates of the condenser) and zero electrical field inside the parallel conducting plates, as well

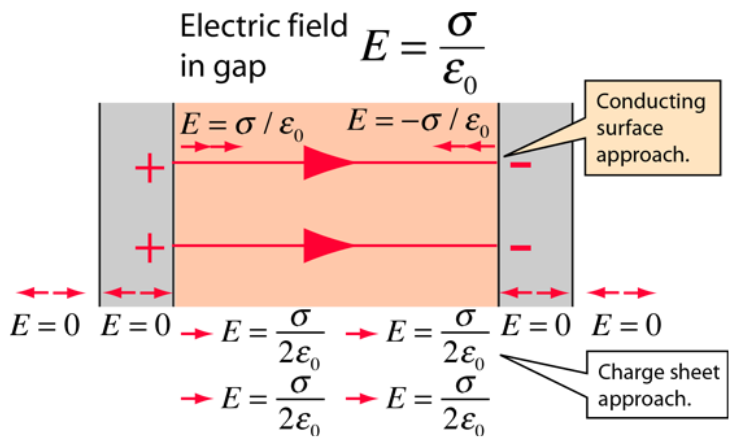

Figure D.2: The oppositely charged parallel surfaces of the membrane are treated like conducting plates (infinite planes, neglecting fringing), then Gauss' law can be used to calculate the electrical field between the plates. Presuming the plates to be at equilibrium with zero electrical field inside the conductors, then the result from a charged conducting surface can be used.

as outside the condenser. In this ideal picture, the charges are aligned on the border of two (infinitely thin) conducting layers and cannot move. Anyhow, the attractive force between the opposite charges on the plates keeps them fixed in direction of z. The repulsive force between the charges with the same sign keeps their surface density in the (x, y) plane uniform: the infinitely thin plates are equipotential. The strong electrostatic force can produce an enormous acceleration for the individual ions, but the thermodynamic gradient can only change with a several orders of magnitude lower speed, allowing measurable changes in the current intensity on the surface). This picture is valid for the equilibrium state of charges, infinitely small non-dissociating charge carriers and perfectly smooth surfaces.

Figure D.3 (the blue diagram lines) show the electrical field around the neuronal membrane. In this picture we see that there is a jump in the value of the electrical field (and so: in the force acting on a charge) in the two sides of the physical layer and we assume that the physical layer somehow emulates the well-defined boundary on the side opposite to that proximal to the membrane.

D.2.2 Electrical field in a two-segment electrolyte

Fig. D.3 displays how the function shapes of electrical field change in the function of the distance from the membrane. As discussed, the physical distance between two two segments with electrical charge causes a charge condensation on the surface of the separating membrane. In the case of classic condenser, this primary charge density is abstracted as an infinitely thin layer of charges (electrons) on an ideally plain layer, and the rest of the volume is abstracted that no charge carrier is included. This case is perfectly described by the classic electricity theory. In the case of electrolytes, the separating membrane's surface is not ideal, furthermore the electrostatic force can press some biological objects to the surface, so we must assume that the primary charge forms a uniformly charged layer with finite thickness on the surface. In the uniformly charged layer the electrical field is a linear function of the distance from the surface. The presence of dielectrical molecules in the rest of the volume creates secondary (virtual) dipole charges, that contribute to the electrical field, as shown. The two fields join to each other at the boundary of the charged layer. The electrical field on the surface of the membrane is the sum of those two fields. The opposite charges on the other plate generate another electrical field in the opposite direction.

As shown in the figure, the field steeply rises near to the membrane. At the membrane surface, it reaches its maximum value; on both sides of the membrane. Inside the membrane, there are no charges, so the field remains contant, for both types of condensers. Notice that the classic condenser can be considered as the abstraction of the real condenser to the case of zero dielectricity.

The electrical field acts on the ions, but there may be present a mechanical counterforce, plus a thermodynamic force aring from the difference of ions' concentrations. (We just notice here that ***the electrical driving force does not depend on the chemical type of the ions, but the thermodynamic force does***. The mechanical counterforce is adaptive, per ion, so it can keep different types of ions in rest.) If there are 'holes' (ion channels) in the membrane, there is no mechanical counterforce, so the ions could move under the effect of the difference between the electrical and thermodynamic forces. The latter depends on the chemical nature of the ions, so it natively provides selectivity: the resulting forces are different for ions of different types. The usual "downhill" method of explanation must be corrected: the *resulting potential* instead of the *electrical potential* defines how the ions move.

The figure shows two ions (green and red) at the axis of the ion channel in two positions (the empty circles along the axis of the ion channel), at the beginning and end of a movement, when the ion channel is open. On the diagram lines, filled circles show the electrical fields on the diagram lines at the position of the ions.

In the charged layer the ion *density* is constant, that is the thermodynamic force is zero while the electrical force decreases with the distance from the membrane's surface

If the channel opened, the mechanical counterforce cancels, the ions will move under the effect of the resulting force, that comprises not only the thermodynamic and electrical forces, but also the force to move the ion with the Stokes-Einstein speed. In the position at the entrance of the ion channel (see the red ions), if the channel is open, the large thermodynamic and electrical forces 'push' the ions into the channel. Here the concentration gradient quickly cancels, but the huge potential gradient will accelerate it toward the other segment. After passing the membrane with a high speed, the ion faces a quickly decreasing field, and its speed quickly decreases. The leave of ions changes both concentration and charge density in the vicinity

In the position farther from the membrane's surface (see the green ions) the ions must travel 'uphill'; that is, they cannot travel instantly toward the membrane. They must wait until the gradient change caused by the ions pushed into the gap reaches their position. As the figure shows, the ions quickly move out from the region with high charge density and very slowly refill that region with ions from the region farther from the membrane. This speed difference results in a quick change in the local electrical field. In the region of the other plate, the ions decrease the (opposite) local potential. As the result, there are less ions ready to travel on the one side, and the potential difference continuously decreases on the other, so the flow of ions stops: the ion channel shuts itself down (and, on both sides, starts to regenerate the potential as it was before the channel's cap was removed). That is, the ion channel is opened by a slow mechanical cap, but it is closed by a fast electrical shutter. While the potential self-closes the channel, the electrical fields on the two sides of the membrane have to relax: the ions and dipoles have time to change their position and even their polarization and ionization state. The local gradients will rearrange concentration and electrical field in the segment.

As depicted, in the two regions, there is a significant difference in the value of the effective force acting on the ions (they are using $E_{accelerate}$ and E_{assist}, so they produce different Stokes-Einstein speeds (see Eq. (**??**)). Based on those speeds, one can introduce "fast" and "slow" ions and correspondingly, speak about *'slow' and 'fast' currents* that the ions represent at a macroscopic level. The corresponding speeds also differ by orders of magnitudes. For this study, we assume the diffusion, potential-assisted and potential-accelerated speeds, in m/s to be 10^{-4}, 10^{-1} (also inside neurons [**?**]), 10^{+3}, respectively (used only to estimate the order of magnitude of some respective operating times). When staging, we assume the greater of the mixing speeds as 'infinitely large' and omit the time that the process needs, while discussing how the slower process proceeds.

We are in line with the estimation given in [**?**] that in the case of resting potential, the scale of the gradient that accelerates the ions across the ion channel is calibrated approximately as kV/cm. Recall that we are still speaking about the resting state and only about the extra gradient evoked by the finite-width membrane. We are at the boundaries of the macroscopic and microscopic worlds. We derived our integrand from the picture of discrete charges but integrated it into the picture of continuous charge distribution. We assume an atomic layer (a skin) on the surface. However, the layer itself can also be modeled as having just a few ions under their mutual repulsion on the surface or a few atomic layers on top of each other, depending on the concentration and voltage in the bulk on the two sides. (The diagram line is valid in the plane crossing the membrane and the ion channel.)

We assumed that the membrane's width is 5 nm. An ion channel is depicted in the middle of the figure with a diameter of about 1 nm. Furthermore, we assume that the ion's size and, correspondingly, the thickness of the atomic layer in the electrolyte on the surface of the membrane is about 0.1 nm. For

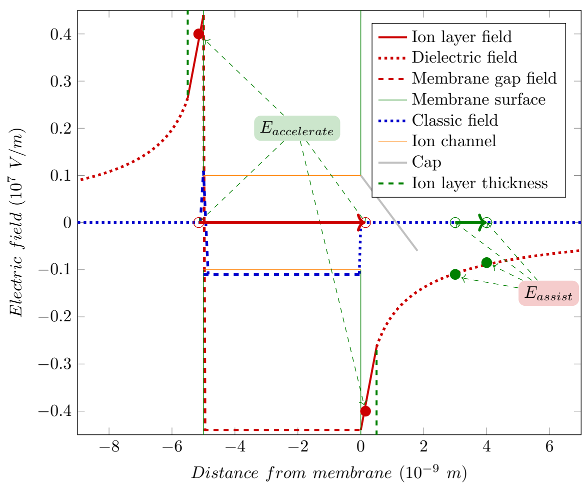

Figure D.3: The neuronal membrane's electrical field *in the function of the distance from the membrane's surface and the bulk potential. The thickness of the atomic layers proximal to membrane's surfaces are also shown.*

comparison, recall that the size of the tip of the clamp pipette is in the range of $1,000\ nm$ and the size of the soma in the range of $10,000\ nm$.

Our results align with the observation (see caption of 11.22 in [**?**]: "A small flow of ions carries sufficient charge to cause a large change in the membrane potential. The ions that give rise to the membrane potential lie in a thin ($<$ $1\ nm$) surface layer close to the membrane". See the dotted line in Fig. D.3; notice that the x-scale on the figure spans $1\ nm$ only. The amount of unbalanced ions is in the range of 10^7, and so is the amount of rush-in ions. In addition, those ions on the high-concentration side rush-in to the low-concentration side and cause the large change in the membrane potential. Their absolute amount is small compared to the total number of ions in the cell, but it is significant compared to the number of unbalanced ions.

D.2.3 Membranes with ions channels

The attraction between the ions in the two skin layers prevents the ions in the layers on the two sides from diffusing into/from the bulk without a current drain in the layer for an extended period. This steady state results from the interplay of the concentration and the potential described by Eq. (C.1). The gradients change gradually within the segments and drop suddenly across the membrane. No current can flow through the membrane; there is no leakage current.

Here, we assume that no ion channels are in the excellently isolating wall (ion channels would mean a current drain and, therefore, a voltage drop).

D.3 Goldman-Hodgkin-Katz potential

It was already stated, based on experimental evidence, that "the membrane permeability to the ions has nothing to do with the potential generation and the ions adsorption on the membrane surface generates the membrane potential", for a review see [**?**]. The idea itself is nonsense: in a balanced state, no ion transport happens, so even without permeability, the balanced state persists; the resting potential has nothing to do with either ions' mobility or permeability nor with ion absorption [**?**]. Furthermore, there is no idea in Goldman-Hodgkin-Katz (GHK) about whether the 'setpoint' (why that specific concentration or potential difference) is present and why the same potential is reset after rough perturbations such as issuing an AP or replicating a cell. The causality is reversed: the potential is a static concept which is set electrically, and the two gradients form the experienced concentrations and potentials from the available solvent molecules. For the time course of forming the gradients, mobility and permeability play a role, but not in determining the concentrations or the resting potential.

In our clear physical picture, the thermodynamic forces on the one side and on the other side of the membrane are summed. They counterbalance each other; furthermore, jointly the effect of the neuronal condenser, see Fig. 4.1. As we emphasized, the concentration gradients are ion-specific. Furthermore, as we discussed in connection with Eq.(C.2), the Nernst equation comprises a per-ion indefinite constant (a potential difference). To calculate a linear combination of terms comprising an arbitrary constant is nonsense, and so is adding absolute concentrations on the different sides of the membrane, or changing the base of the logarithm used in a calculation to match the experimental value. It is not more than number magic. The potential is described by coupled equations as discussed in connection with Eq. (**??**).

The Ca^{2+} ions do not fit into the GHK picture. One of the reasons why GHK cannot be good is that $[Ca^{2+}]$ is omitted. Biologically, it is hard to believe that Ca^{2+} does not participate in the game of life (especially since biology sees the need for Ca^{2+} pumps). Physically, one concentration alone on one side of the membrane cannot keep balance; as the different ion exchange processes, Table 4.1 and Fig. 4.1 show. When adding a new ion to the solution, the sum concentration increases, and so the electrical force increases, forcing the previous elements to find new concentrations on both sides. The appearance of a new chemical element indirectly changes the concentrations of the others (a good example is the role of the negligible amount of Ca^{2+}).

In the light of our findings, we must reconsider a few key conceptions. We derived (see Eq.(**??**)) that the gap voltage depends on the thickness of the membrane and the overall concentration. Furthermore, we demonstrated that the gap voltage is equivalent to the resting potential (correctly: potential difference) and is independent of ion mobility or ion composition in the solution. Derived values, such as the Goldman-Hodgkin-Katz potential, should be rethought. The idea is nonsense: in a balanced state, no ion transport happens, so even without permeability, the balanced state persists; as experiments proved, the resting potential has nothing to do with either ions' mobility or permeability nor with ion absorption [?]. The causality is reversed: the potential is a static notion and is set electrically, and the two gradients form the experienced concentrations from the available solvent molecules. For the time course of forming the gradients and maintaining the balance, mobility and permeability play a role but not in determining the concentrations or the resting potential.

The Ca^{2+} ions do not fit into the GHK picture. However (see our Table 4.1), it provides a thermodynamic contribution in the same order of magnitude as the charge separation by the membrane. If we add the potential due to Ca^{2+}, we arrive at the correct result that the total balance of the thermodynamic contributions equals zero, that is, the resting potential is zero. The electrical potential contribution is an additional term, resulting in the resting potential.

When adding a new ion to the solution, the sum concentration increases, and so the electrical force increases, forcing the previous elements to find new concentrations on both sides. The appearance of a new chemical element indirectly changes the concentrations of the others (a good example is the role of the negligible amount of Ca^{2+}). No linear summing or similar assumptions can be made. The potential is described by coupled equations as discussed in connection with Eq.(**??**). As we discussed in connection with Eq.(C.2), the integral form of the Nernst equation implies an undetermined additive constant. To calculate a linear combination of terms comprising an arbitrary constant is nonsense, and so is changing the base of the logarithm used in a calculation to match the experimental value.

D.4 Moving in the membrane's potential

In Fig. D.3, an ion channel is depicted in the middle of the figure with a diameter of about 3 nm for visibility. Furthermore, we assume that the ion's size and, correspondingly, the thickness of the atomic layer in the electrolyte on the surface of the membrane is about 0.1 nm (although with their electrical double layers, they can also reach the size of 1 nm). For comparison, recall that the size of the tip of the clamp pipette is in the range of $1,000$ nm and the size of the soma in the range of $10,000$ nm.

D.4.1 Passing through ion channels

Figure D.3 also hints at how ions can move in the proximity of the membrane. On the left side, the electrical field increases toward the membrane, so the ions cannot move in that direction without an initial force being applied. The case is the same on the right side, because the charge is the opposite. An ion must gain energy to get closer to the membrane, which means, at the same time, higher potential energy. This situation explains why the ions do not diffuse into the bulk region. At their present positions, the counterforces by the membrane (and the dipole layer on it) balance the attraction from the opposite side. The ion channels can work in continuous mode (the "resting ion channels") and impulse mode (gated by "caps" on the channels, in transient mode).

Notice that the resulting electrical field cannot accelerate an ion except when the ion is at the beginning of the open ion channel across the membrane. Here, the mechanical counterforce is missing, and the field drops to the value generated by the condenser inside the plates. Here, a vast electrical field accelerates the ions to a high speed (although a force needed to move against the viscosity decreases the electrostatic acceleration). The empty red circles show an ion that traverses through the channel. The figure shows their electrical fields on the two sides, at the beginning and end of the channel. (On the right side, the electrical field is negative, but so is the charge of the ions.)

D.4.2 Moving in the polarized electrolyte

The electrical acceleration sharply decreases after leaving the ion channel (see the right side). As shown by the difference of the vertical positions of the filled red and green balls, the difference in the electrical field values is huge across the membrane ($E_{accelerate}$) and considerable inside the dielectric electrolyte (E_{assist}) as shown by the difference in the electrical field values of the green balls. The field still accelerates, but significantly less intensely than the one in the gap. Suppose we assume that the ion quickly accelerates to its Stokes-Einstein speed. In that case, the travel speeds in those space regions are proportional to the difference of the fields in the shown positions and the reciprocals of their travel times.

The passed ions decrease the concentration, and the electrical field on the high-concentration side increases on the low-concentration side. Similarly, the gradients decrease and increase, changing locally the gradients. At the exit, the field continues to accelerate the ion. However, the increasing potential and concentration increasingly decelerate the passing ions, so they quickly brake them to the assisted speed. Then, the effect of the electrical field cancels, and the ions will move with their diffusion speed ("the ion stops").

D.4.3 Ion selectivity

As Eq.(D.5) states, the electrolyte separated by a permeable membrane is balanced when the electrical and thermodynamic forces are equal. The $E^{C_k}_{thermal}(d)$ "electrical field" depends on the chemical quality of the ion; the electrical field does not. That means the resulting driving forces acting on the different ions are different. While one ion experiences a resulting force and moves to the other segment, the other remains resting. Of course, the ions moving across the segments change the overall concentrations in both segments (and so the electrical field contribution), indirectly affecting the balanced state of the other ions. When a mixture of ions is present in the volume, all ions feel a different driving force, and the solution as a whole will be balanced when the resulting forces for all chemical ions are balanced. The permanently open ion channels only passively participate in the process. The driving forces of the ions are different, and they produce, depending on the actual concentrations, the selectivity of the channels.

When Na^+ ions rush into the intracellular layer, they roughly increase the overall concentration and the potential. All other ions, including K^+, feel a driving force. The targeted membrane potential is set electrically, and the driving gradients may behave controversially during the transient period. The actual voltage gradient may temporarily direct the chemical gradient in opposite directions. Phenomena such as 'over-relaxation', in this sense, similar to 'hyperpolarization', may happen.

However, the layer itself can also be modeled as having just a few ions under their mutual repulsion on the surface or a few atomic layers on top of each other, depending on the concentration and voltage in the bulk on the two sides. (The voltage gradient diagram line validates the plane crossing the membrane and the ion channel.)

Our results align with the observation (see caption of Fig. 11.22 in [**?**]). The amount of unbalanced ions is in the range of 10^7, and so is the amount of rush-in ions. In addition, those ions on the high-concentration side rush into the low-concentration side and cause a large change in the membrane potential. Their absolute amount is small compared to the total number of ions in the cell, but it is significant compared to the number of unbalanced ions.

Appendix E

Speculations

Contents

E.1 Cell pressure

The electrolyte contains dissociated molecules (free charges), which, due to the repulsion between the ions, are located on the surfaces of the membrane and exert a pressure on the membrane surface. We assume that the relationship between the amount of the charge on the membrane and the pressure is linear, i.e., we conclude a pressure change from a charge change, and extrapolate it to the neuron in the resting state.

Physics panel E.1.1:
Estimating the speed of ions in electric field

In **Physics Panel 4.1.4**, we estimated the acceleration and passage time of a Na^+ ion during a rush-in action. In this period the ion is accelerated to the speed $v = a * t$

$$v_{Na^+} = 0.42 \times 10^{14}\ [\frac{m}{s^2}]\ \times\ 1.54 \times 10^{-11}[s] = 600\ [\frac{m}{s}] \tag{E.1}$$

(This speed is just a speed component along the channel. The thermal speed is about an order of magnitude higher, but not necessarily in the direction of the channel. The speed due to acceleration may convert in direction and magnitude due to a single collision, as soon as the diaphragm effect of the channel is over. The accelerated ion 'stops' as soon as the accelerating voltage is over.) See our other estimation in **Physics Panel E.1.6**.

Physics panel E.1.2:
Estimating the current of ion channels

Based on the time caculated in **Physics Panel 4.1.4**, we can attribute to an ion channel for this short period, provided that 10^3 ions pass through a single ion channel, a current

$$I_{channel} = \frac{10^3 \times 1.60 \times 10^{-19}[C]}{1.54 \times 10^{-11}[s]} = 10^{-5} A \tag{E.2}$$

It is in the range of several μAs, for a short period. It is distributed in the spatiotemporal surface of a neuron, and contributes to a current of dozen of pAs.)

Physics panel E.1.3:
Estimating the pressure change due to rushin

Using the force from **Physics panel 2.0.1**, one can estimate how the pressure of the neural cell increases due to the rush-in changes at the beginning of the AP. As evidence shows, the local potential at the internal surface of the membrane is in the range of 100 mV in the resting state and increases by $\Delta U = 100\,\mathrm{mV}$ in the transition state. This increase means a change in the force acting on an ion (see Eq. (4.1)) by $1.6 * 10^{-12}$ $[N]$. When we assume 10^7 rush-in ions and unchanged cell size, the total force acting on the membrane increases by $1.6 * 10^{-5}$ $[N]$. This change in force means on the neuron's $8 * 10^{-9}$ $[m^2]$ surface a pressure change

$$\Delta P = \frac{1.6 * 10^{-5}\ [N]}{8 * 10^{-9}\ [m^2]} = 2 * 10^3 \quad \left[\frac{N}{m^2}\right] \tag{E.3}$$

Again, the pressure can be calculated using electricity.
(Here we used data from Johnston&Wu constants from **Numerics panel 2.0.1**)

Physics panel E.1.4:
Estimating the force measurable due to rushin

The pressure change calculated in **Physics panel E.1.3** enables to calculate the force measurable dur to rushin. That pressure on a 10×10 μm tooltip of the measuring device is converted to a force $2 * 10^3 * 100 * 10^{-12} = 200$ pN force. The measured force value [27] is about 600 pN, so our estimation is in the correct range.
The measured pressure value is [45] 5 $\frac{dyn}{cm^2} = 0.5$ $\frac{N}{m^2}$, which is an average value for some period. If we assume a 1 Hz frequency, and a 1 ms period for the duration of the AP peak, it means $1 * 10^3$ $\left[\frac{N}{m^2}\right]$. So, the measured mechanical change values [38] underpin our hypothesis that the increased pressure increases the neuron's size via its elasticity.

Physics panel E.1.5:
Comparing forces exerted on ions in cells

One can compare the force due to **Physics panel E.1.3**
phys:CellForceinField

Physics panel E.1.6:
Estimating the rush-in speed

A way to estimate the ions' rush-in speed is that one assumes that the Stokes-Einstein speed describes the ions' speed (Fig. 3 in [14], quantitatively underpins the hypothesis, see [46]) both in axons and ion channels. If so, the ratio of the potential gradients equals the ratio of the speeds of ions. One can estimate that the 100 mV AP on the 50 μm AIS generates a $2 * 10^3\ \frac{V}{m}$ electrical gradient, and [14] measured 20 $\frac{m}{s}$ for the speed of the AP. By assuming that the electrical field across the membrane is $10^7\ \frac{V}{m}$, one can estimate the speed at the exit of the ion channels as $10^3\ \frac{m}{s}$. See our other estimation in **Physics Panel E.1.1**

Physics panel E.1.7:
Estimating the rush-in mechanical work

By using the result derived in **Physics panel E.1.3**, one can also estimate the work done when the rush-in charge extends the neuron, as the $\Delta P \times \Delta V$. We consider that ΔP remains constant, and we calculate the change of the volume as neuron's surface $10^{-8}\ m^2$ multiplied by the change of the radius $\Delta r = 10^{-9}\ [m]$ multiplied by the pressure ΔP, that gives

$$E_{mech} = \Delta P \times \Delta V = 2 * 10^{-14}\ [J]$$

If one assumes that the pressure is of entirely mechanical origin [38], one must assume that the pressure is $10^6\ \frac{N}{m^2}$, which is well above the measured value. This underpins our conclusion that the overwhelming part of the pressure (force) is of electrostatic origin.

Physics panel E.1.8:
Estimating the rush-in electrical work

We can also estimate the electrical energy from reset to threshold, see also [47]

$$E_{electr} = \frac{1}{2} C * (\Delta V)^2 = 1.4 * 10^{-12}\ [J]$$

Physics panel E.1.9:
Estimating the rush-in time period

To estimate the time period required to change the pressure calculated in **Physics panel E.1.3**, one can assume that the collision force of a Na^+ ion provides the pressure (the shock wave), and this can be calculated as $F = m\frac{\Delta v}{\Delta t}$. Given that Eq. (4.1) provides the force, and one must assume that the ion entirely loses its 10^4 *to* $10^5 \frac{m}{s}$ speed, one concludes that it generates a shock wave in $\Delta t = 10^{-8}$ *to* 10^{-7} *seconds*.

Physics panel E.1.10:
Estimating the dissipated energy of thermodynamical cycle

We estimate the work done when the rush-in charge extends the neuron, as the $\Delta P \times \Delta V$. We consider that ΔP remains constant, and we calculate the change of the volume as neuron's surface 10^{-8} m^2 multiplied by the change of the radius $\Delta r = 10^{-9}$ $[m]$ multiplied by the pressure ΔP, that gives $\mathbf{E_{mech} = 2 * 10^{-14}}$ **[J]**, as we calculated in **Physics panel E.1.7**.
[33] reported that an AP dissipates $Q_e = 0.25\,\mu\mathrm{J\,m^{-2}}$, which on a $8 \times 10^{-9}\,\mathrm{m^{-2}}$ surface means $0.4 \times 10^{-14}\,\mathrm{J}$. As discussed in **Physics panel E.1.8**, the electrical work is $\mathbf{E_{electr} = 1.4 * 10^{-12}}$ **[J]**.

Physics panel E.1.11:
Estimating the energy cost of information transfer

The energy dissipated during issuing an AP, **Physics panel E.1.10**
The theoretically lowest possible energetic cost of information is $k_B T ln(2)$), numerically ca. 3×10^{-21} J i.e., compared to the optimal limit set by physics, this efficiency value is 10^6-10^8 times less energy efficient, in line with [47]. Nature protests: the phenomenon is different from tha one biology claim. In the neuron there is no single state which can be set or reset.

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